

# Group Theory in Condensed Matter

## A Practical Introduction

Companion Notes to the Video Lectures of Prof. Juan Luis Mañes

(UPV/EHU, recorded Winter 2022; published on the *Topological Matter School* channel)

These notes were written independently as a study companion. They follow the order and the examples of the eight video lectures, but the text, derivations, figures and exercises are our own. They are neither written nor endorsed by the lecturer; any error is ours. Timestamps such as [L2 @ 42:00](#) are clickable links to the corresponding moment in the video.

Playlist:

[https://www.youtube.com/playlist?list=PLyGH4pJ\\_4EhrbbYmYjEYlVRpaSrRRQTri](https://www.youtube.com/playlist?list=PLyGH4pJ_4EhrbbYmYjEYlVRpaSrRRQTri)



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# Preface

## What these notes are

The lecture series *Group Theory in Condensed Matter: A Practical Introduction* teaches group theory *backwards*: first you see what a group-theory computation produces, then you learn to perform it, and only at the end you learn why it works. The lecturer states the key theorems without proof and devotes the time to applications, from molecular vibrations to topological quantum chemistry.

These notes keep that order, so that each chapter can be read next to the corresponding lecture, and add three things:

1. a written, self-contained account of every lecture, with all computations carried out explicitly and checked numerically;
2. green boxes marked *Beyond the lectures*, and sections marked ★ that fill the gaps a theorist will need later: cosets and quotient groups (Sections 1.4 and 2.3), proofs of Schur's lemma and the orthogonality theorems together with projection operators (Sections 1.8, 2.6 and 3.6), the rotation group, SU(2) and Clebsch–Gordan series (Section 5.5), time reversal and Kramers degeneracy (Sections 5.8 and 6.8), and the method of invariants for  $\vec{k} \cdot \vec{p}$  Hamiltonians (Section 7.6);
3. exercises at the end of every chapter, each followed by its detailed solution.

Each starred section is placed where the lectures first need it; it can be skipped on a first reading without losing the thread.

## How the chapters map onto the lectures

| Chapter | Topic                                                  | Video                     | Length |
|---------|--------------------------------------------------------|---------------------------|--------|
| 1       | Symmetries, degeneracies and representations           | <a href="#">Lecture 1</a> | 89 min |
| 2       | Irreducible representations as building blocks         | <a href="#">Lecture 2</a> | 92 min |
| 3       | Operations with representations I: tensors, invariants | <a href="#">Lecture 3</a> | 91 min |
| 4       | Operations with representations II: spectroscopy       | <a href="#">Lecture 4</a> | 84 min |
| 5       | Spin, double groups, crystal-field splitting           | <a href="#">Lecture 5</a> | 96 min |
| 6       | Space groups and their representations                 | <a href="#">Lecture 6</a> | 90 min |
| 7       | The group theory of electron bands                     | <a href="#">Lecture 7</a> | 89 min |
| 8       | Group theory and topology                              | <a href="#">Lecture 8</a> | 88 min |

Lectures 5 and 8 have no captions; their timestamps in these notes were read off the slides and are approximate.

## Prerequisites

Linear algebra (matrices, eigenvalues, change of basis), quantum mechanics at the level of the hydrogen atom and angular momentum, and an introductory solid-state course (Bravais lattices, reciprocal lattice, Bloch's theorem, tight binding). No previous knowledge of group theory is assumed.

## Notation and typographical conventions

The typesetting follows ISO 80000-2:

- Variables are italic; mathematical constants and operators with fixed meaning are upright: the imaginary unit  $i$ , Euler's number  $e$ , the number  $\pi$ , the differential  $d$  (as in  $dx$ ), and named functions and operators ( $\exp$ ,  $\sin$ ,  $\text{tr}$ ,  $\det$ ).
- Vectors carry an arrow,  $\vec{r}$ ,  $\vec{k}$ ,  $\vec{E}$ ; their components are italic scalars,  $r_x$ ,  $k_i$ .
- Matrices are bold italic,  $\mathbf{D}(g)$ ; their elements are italic scalars,  $D_{ij}(g)$ . The transpose is  $\mathbf{A}^T$  and the hermitian conjugate  $\mathbf{A}^\dagger$ .
- Labels that are not variables are upright: the “v” in  $C_{3v}$ , the “d” in  $\sigma_d$ , the “g” in  $E_g$ . Irreducible representations carry upright Mulliken symbols,  $A_1$ ,  $E$ ,  $T_2$ , while the identity *operation* is the italic  $E$ .

| Symbol                               | Meaning                                                             |
|--------------------------------------|---------------------------------------------------------------------|
| $G,  G $                             | a group and its order (number of elements)                          |
| $g, h$                               | group elements                                                      |
| $\Gamma, \Gamma^{(\mu)}$             | a representation; the $\mu$ -th irreducible representation (irrep)  |
| $\mathbf{D}(g)$                      | matrix representing $g$                                             |
| $\chi(g) = \text{tr } \mathbf{D}(g)$ | character                                                           |
| $d_\mu$                              | dimension of irrep $\mu$                                            |
| $m_\mu$                              | multiplicity of irrep $\mu$ in a representation                     |
| $\Gamma_V, \Gamma_R$                 | vector and axial-vector (“rotational”) representations              |
| $\Gamma_M, \Gamma_{\text{vib}}$      | mechanical and vibrational representations of a molecule            |
| $\oplus, \otimes$                    | direct sum and (tensor/Kronecker) product                           |
| $[\Gamma^2], \{\Gamma^2\}$           | symmetric and antisymmetric squares                                 |
| $\mathcal{T}$                        | translation group of a crystal                                      |
| $\{R   \vec{v}\}$                    | Seitz symbol: point operation $R$ followed by translation $\vec{v}$ |
| $G_{\vec{k}}$                        | little group of $\vec{k}$                                           |
| $\Theta$                             | time-reversal operator                                              |

In the lectures, general representations are denoted by  $T$  and irreducible ones by  $\tau$  (following Lyubarskii). We use  $\Gamma$  throughout, because  $T$  is also needed for the translation group and for the tetrahedral group  $T_d$ .

Symmetry operations always act *actively*: they move the physical object (a vector, a wave function, a molecule) and leave the coordinate axes fixed (Section 3.1). A function transforms as  $(\hat{O}_g f)(\vec{r}) = f(g^{-1}\vec{r})$ .

## Recommended textbooks

The lecturer recommends: G. Ya. Lyubarskii, *The Application of Group Theory in Physics* (Pergamon, 1960), which he learned from; M. S. Dresselhaus, G. Dresselhaus and A. Jorio, *Group Theory: Application to the Physics of Condensed Matter* (Springer, 2008), rich in applications; and C. J. Bradley and A. P. Cracknell, *The Mathematical Theory of Symmetry in Solids* (Oxford, 1972), the reference for tables and for magnetic groups but not a book to learn from. For the topological part: J. Cano and B. Bradlyn, *Band representations and topological quantum chemistry*, Annu. Rev. Condens. Matter Phys. 12, 225 (2021), arXiv:2006.04890. The tables in these notes use the conventions of the Bilbao Crystallographic Server (<https://www.cryst.ehu.es>) wherever possible.



# Chapter 1

## Symmetries, degeneracies and representations

### Video: Lecture 1

Why group theory [L1 @ 0:51](#) · continuous vs discrete symmetries [L1 @ 3:13](#) · methane: what group theory predicts [L1 @ 13:44](#) · the idea of a representation [L1 @ 23:14](#) · Wigner's theorem [L1 @ 37:58](#) · the group  $C_{3v}$  and the vector representation [L1 @ 42:08](#) · definition of a representation [L1 @ 46:24](#) · reducible vs irreducible [L1 @ 71:42](#) · equivalent representations [L1 @ 83:10](#)

### 1.1 Why group theory?

Group theory is the systematic way of extracting *all* the consequences of the symmetry of a physical system, reliably and efficiently. Its computations are simple (mostly sums and products of integers) but the concepts need care.

In condensed matter we deal almost exclusively with *space-time* symmetries: rotations, reflections, inversion, lattice translations and time reversal. Internal symmetries (U(1), SU(2) isospin, SU(3) colour) belong mostly to high-energy physics. A second classification is more important for the method:

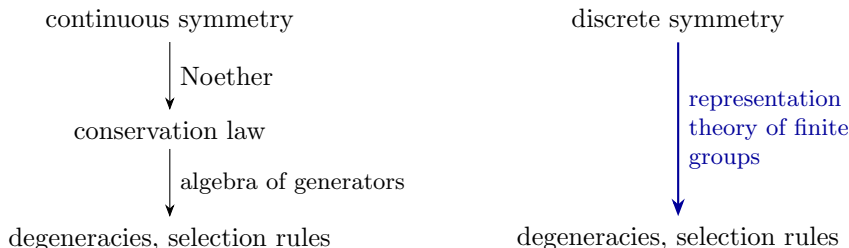
Continuous symmetries such as SO(3) are Lie groups. A finite rotation is the exponential of an infinitesimal one,

$$\hat{U}(\vec{\theta}) = \exp\left(-\frac{i}{\hbar} \vec{\theta} \cdot \hat{\vec{L}}\right), \quad (1.1)$$

so invariance of the Hamiltonian,  $\hat{U}\hat{H}\hat{U}^\dagger = \hat{H}$  for all angles, implies at first order  $[\hat{H}, \hat{\vec{L}}] = 0$ : a *conservation law*. Degeneracies and selection rules then follow from the conservation law and the algebra of the generators, without any group theory. The  $(2l + 1)$ -fold degeneracy of the levels of a central

potential and the dipole selection rule  $\Delta l = \pm 1$  are derived this way in every quantum-mechanics course.

Discrete symmetries such as the finite set of rotations and reflections that leave a molecule or a crystal invariant have no infinitesimal elements, hence no generators and no conservation laws. There is no Noether theorem for them. To go from symmetry to degeneracies and selection rules we must jump directly, and the bridge is the representation theory of finite groups: characters, their orthogonality, projection operators.



In this course we never need Lie-algebra techniques such as roots, Dynkin diagrams or Young tableaux. From the rotation group we only need its irreducible representations, which we obtain elementarily in [Chapter 5](#). All the groups we meet are discrete subgroups of  $O(3)$  (rotations, reflections, inversion), lattice translations, and time reversal.

## 1.2 What group theory predicts: the vibrations of methane

The methane molecule  $\text{CH}_4$  has its four hydrogen atoms at the vertices of a regular tetrahedron with the carbon atom at the centre. Its symmetry group, the tetrahedral group  $T_d$ , has 24 elements.

Without symmetry considerations we can only say the following. Each of the five atoms can move in three directions, so a distortion of the molecule is a vector with 15 components. In the harmonic approximation the frequencies are obtained by diagonalising a real symmetric  $15 \times 15$  matrix, which gives at most 15 distinct frequencies. Six of them vanish: three uniform translations and three rigid rotations do not stretch any bond (these are the *zero modes* of a non-linear molecule). So at most nine non-zero frequencies remain.

A group-theory computation, which after [Chapter 2](#) will take you five minutes with pencil and paper, says much more:

What group theory says about CH<sub>4</sub> and CH<sub>3</sub>D

- CH<sub>4</sub> has exactly four distinct vibrational frequencies, with degeneracies 1, 2, 3 and 3 (1 + 2 + 3 + 3 = 9).
- All four are Raman active; only the two triply degenerate ones are infrared active; the molecule has no microwave (pure rotational) spectrum.
- In deuterated methane CH<sub>3</sub>D each triply degenerate frequency splits into one non-degenerate and one doubly degenerate frequency: six distinct frequencies, three non-degenerate and three doubly degenerate, all of them Raman and infrared active; CH<sub>3</sub>D is also microwave active.

The result of the computation is written as

$$\begin{aligned}\Gamma_{\text{M}} &= \text{A}_1 \oplus \text{E} \oplus \text{T}_1 \oplus 3 \text{T}_2 & (\dim 15), \\ \Gamma_{\text{V}} &= \text{T}_2 & (\dim 3), \quad \Gamma_{\text{R}} = \text{T}_1 \quad (\dim 3), \quad (1.2) \\ \Gamma_{\text{vib}} &= \Gamma_{\text{M}} \ominus \Gamma_{\text{V}} \ominus \Gamma_{\text{R}} = \text{A}_1 \oplus \text{E} \oplus 2 \text{T}_2 & (\dim 9).\end{aligned}$$

Every symbol here is a *representation*: a rule telling how some kind of object transforms under the symmetry operations. Loosely:

- $\Gamma_{\text{V}}$ , the *vector representation*, tells how ordinary (polar) vectors transform: a set of  $3 \times 3$  matrices, one per symmetry operation.
- $\Gamma_{\text{R}}$ , the *rotational* or *axial-vector representation*, tells how pseudovectors (angular momentum, magnetic field) transform. It coincides with  $\Gamma_{\text{V}}$  on proper rotations and differs by a sign on improper operations (reflections, inversion), because  $\vec{L} = \vec{r} \times \vec{p}$  does not change sign under inversion while  $\vec{r}$  and  $\vec{p}$  do.
- $\Gamma_{\text{M}}$ , the *mechanical representation*, tells how an arbitrary distortion of the molecule (a 15-component vector) transforms:  $15 \times 15$  matrices.
- $\Gamma_{\text{vib}}$ , the *vibrational representation*, acts on the 9-dimensional space of pure vibrations, obtained by removing uniform translations (which transform as vectors) and rigid rotations (which transform as axial vectors).

The symbols A<sub>1</sub>, A<sub>2</sub>, E, T<sub>1</sub>, T<sub>2</sub> are the five *irreducible representations* (irreps) of  $T_d$ . Unlike  $\Gamma_{\text{V}}$  or  $\Gamma_{\text{M}}$  they do not describe any particular object; they are abstract building blocks. Their dimension is read off the letter: A and B are one-dimensional, E is two-dimensional, T is three-dimensional.

### How to read a decomposition (Wigner's theorem)

If the representation carried by the eigenvectors of a symmetric operator (here the dynamical matrix, later the Hamiltonian) decomposes as  $\Gamma = \bigoplus_{\mu} m_{\mu} \Gamma^{(\mu)}$ , then

- the multiplicity  $m_{\mu}$  is the number of distinct eigenvalues associated with the irrep  $\Gamma^{(\mu)}$ ;
- the dimension  $d_{\mu}$  of the irrep is the degeneracy of each of them.

For  $\text{CH}_4$ ,  $\Gamma_{\text{vib}} = \text{A}_1 \oplus \text{E} \oplus 2 \text{T}_2$  gives one non-degenerate, one doubly degenerate and two triply degenerate frequencies. For  $\text{CH}_3\text{D}$  the symmetry group is the smaller group  $C_{3v}$  and  $\Gamma_{\text{vib}} = 3 \text{A}_1 \oplus 3 \text{E}$  (Section 2.8). We prove Wigner's theorem in an explicit example in Section 3.5; the general proof rests on Schur's lemma (Section 2.6.1).

*Remark 1.1.* Group theory tells you which frequencies must be degenerate, how many distinct ones there are, and which matrix elements must vanish. It never tells you the numerical values of the frequencies: those need physical input (force constants).

## 1.3 Groups

Definition 1.2. A *group* is a set  $G$  with a composition law  $\circ$  such that

1.  $g_1 \circ g_2 \in G$  for all  $g_1, g_2 \in G$  (closure);
2.  $(g_1 \circ g_2) \circ g_3 = g_1 \circ (g_2 \circ g_3)$  (associativity);
3. there is a neutral element  $e$  with  $e \circ g = g \circ e = g$ ;
4. every  $g$  has an inverse  $g^{-1}$  with  $g \circ g^{-1} = g^{-1} \circ g = e$ .

The number of elements  $|G|$  is the *order* of the group.

For a group of symmetry operations,  $g_1 \circ g_2$  means “apply  $g_2$  first, then  $g_1$ ”. The composition law can be anything: for the translation group it is vector addition,  $\vec{t}_1 \circ \vec{t}_2 = \vec{t}_1 + \vec{t}_2$ . Symmetry operations are denoted as follows (Schoenflies notation):

- $E$ : identity (do nothing). Not to be confused with the irrep  $\text{E}$ .
- $C_n$ : rotation by  $2\pi/n$ ;  $C_n^+$  counterclockwise,  $C_n^-$  clockwise (seen from the tip of the axis).
- $\sigma$ : reflection in a plane;  $\sigma_v$  vertical (containing the main axis),  $\sigma_h$  horizontal,  $\sigma_d$  diagonal.
- $I$ : inversion,  $\vec{r} \mapsto -\vec{r}$ .

- $S_n$ : rotoreflection, a rotation  $C_n$  followed by reflection in the plane perpendicular to the axis (Section 2.7).

### 1.3.1 The group $C_{3v}$

Replacing one hydrogen of methane by deuterium leaves a triangular pyramid with the deuterium on top. Its symmetry group  $C_{3v}$  (crystallographic symbol  $3m$ ) has six elements (Fig. 1.1): the identity  $E$ , the rotations  $C_3^\pm$  by  $\pm 120^\circ$  about the vertical  $z$  axis, and three vertical mirror planes  $\sigma_{d1}, \sigma_{d2}, \sigma_{d3}$ , each containing the  $z$  axis and one hydrogen atom. We choose the  $x$  axis inside  $\sigma_{d1}$ .

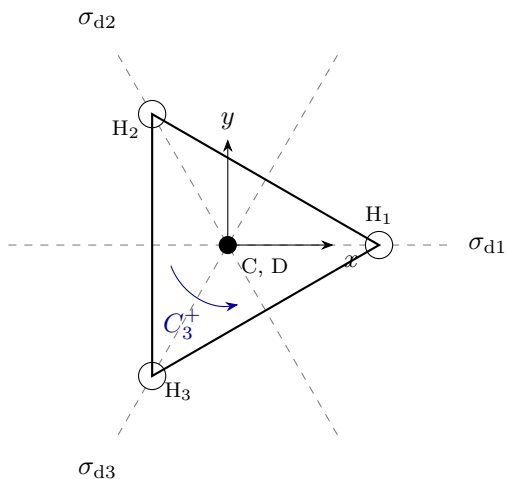


Figure 1.1: The group  $C_{3v}$  seen from above ( $z$  out of the page). The deuterium atom sits above the carbon on the  $z$  axis; the three hydrogens lie below. The mirror  $\sigma_{di}$  contains the  $z$  axis and hydrogen  $H_i$ .

Its multiplication table (row  $g$ , column  $h$ , entry  $gh$ , i.e.  $h$  applied first) is

|               | $E$           | $C_3^+$       | $C_3^-$       | $\sigma_{d1}$ | $\sigma_{d2}$ | $\sigma_{d3}$ |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| $E$           | $E$           | $C_3^+$       | $C_3^-$       | $\sigma_{d1}$ | $\sigma_{d2}$ | $\sigma_{d3}$ |
| $C_3^+$       | $C_3^+$       | $C_3^-$       | $E$           | $\sigma_{d3}$ | $\sigma_{d1}$ | $\sigma_{d2}$ |
| $C_3^-$       | $C_3^-$       | $E$           | $C_3^+$       | $\sigma_{d2}$ | $\sigma_{d3}$ | $\sigma_{d1}$ |
| $\sigma_{d1}$ | $\sigma_{d1}$ | $\sigma_{d2}$ | $\sigma_{d3}$ | $E$           | $C_3^+$       | $C_3^-$       |
| $\sigma_{d2}$ | $\sigma_{d2}$ | $\sigma_{d3}$ | $\sigma_{d1}$ | $C_3^-$       | $E$           | $C_3^+$       |
| $\sigma_{d3}$ | $\sigma_{d3}$ | $\sigma_{d1}$ | $\sigma_{d2}$ | $C_3^+$       | $C_3^-$       | $E$           |

(1.3)

The group is *not* commutative:  $C_3^+ \sigma_{d1} = \sigma_{d3}$  but  $\sigma_{d1} C_3^+ = \sigma_{d2}$ . Each row and each column contains every element exactly once (the *rearrangement theorem*).

## 1.4 Subgroups, cosets and Lagrange's theorem ★

The lectures skip the structural part of group theory. Space groups, little groups and induced representations use it constantly, so the starred sections supply it where it first becomes useful.

A subset  $H \subseteq G$  that is itself a group under the same law is a *subgroup*. For a finite subset it suffices to check closure.

**Definition 1.3.** For  $g \in G$  the *left coset* of  $H$  is  $gH = \{gh : h \in H\}$ ; the right coset is  $Hg$ .

Two left cosets are either identical or disjoint ( $g_1H \cap g_2H \neq \emptyset$  implies  $g_2^{-1}g_1 \in H$ , hence  $g_1H = g_2H$ ), and each contains  $|H|$  elements. So  $G$  is a disjoint union of cosets,

$$G = g_1H \cup g_2H \cup \cdots \cup g_nH, \quad n = |G|/|H|.$$

**Theorem 1.4 (Lagrange).** *The order of a subgroup divides the order of the group. The integer  $[G : H] = |G|/|H|$  is the index of  $H$ .*

Consequences: the order of every element (the smallest  $n$  with  $g^n = e$ ) divides  $|G|$ ; a group of prime order is cyclic.  $C_{3v}$  ( $|G| = 6$ ) can only have subgroups of order 1, 2, 3, 6:  $\{E\}$ , the three  $\{E, \sigma_{di}\}$ ,  $C_3$ , and  $C_{3v}$  itself.

## 1.5 Representations

### 1.5.1 The vector representation of $C_{3v}$

A rotation or reflection acts linearly on vectors, so it is described by a  $3 \times 3$  matrix acting on the column of components  $(v_x, v_y, v_z)^T$ . With the axes of [Fig. 1.1](#):

$$\begin{aligned} \mathbf{V}(E) &= \mathbf{1}, & \mathbf{V}(\sigma_{d1}) &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \\ \mathbf{V}(C_3^+) &= \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, & \mathbf{V}(C_3^-) &= \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, & (1.4) \\ \mathbf{V}(\sigma_{d2}) &= \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} & 0 \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}, & \mathbf{V}(\sigma_{d3}) &= \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} & 0 \\ \frac{\sqrt{3}}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \end{aligned}$$

The mirror  $\sigma_{d1}$  contains the  $x$  and  $z$  axes, so it only reverses  $v_y$ . How to construct the other matrices without memorising formulas is explained in [Section 3.1](#). The

matrices reproduce the multiplication table; for example  $\mathbf{V}(C_3^+)\mathbf{V}(C_3^+) = \mathbf{V}(C_3^-)$ , because two rotations by  $120^\circ$  make a rotation by  $240^\circ = -120^\circ$ .

### 1.5.2 Definition

Definition 1.5. A (linear) *representation*  $\Gamma$  of a group  $G$  on a vector space  $L$  assigns to each  $g \in G$  an invertible linear operator  $\mathbf{D}(g)$  on  $L$  such that

$$\mathbf{D}(g_1)\mathbf{D}(g_2) = \mathbf{D}(g_1g_2) \quad \text{for all } g_1, g_2 \in G. \quad (1.5)$$

The dimension of  $L$  is the *dimension* of the representation. Choosing a basis of  $L$ , each  $\mathbf{D}(g)$  becomes a matrix.

Whatever the composition law of the group (vector addition for translations), the operators of a representation are always composed by *matrix multiplication*. From (1.5) it follows that  $\mathbf{D}(e) = \mathbf{1}$  and  $\mathbf{D}(g^{-1}) = \mathbf{D}(g)^{-1}$ .

- The assignment  $\mathbf{D}(g) = \mathbf{1}$  (a  $1 \times 1$  matrix) for every  $g$  satisfies (1.5). It is the *identity* or *trivial* representation, called  $A_1$  in  $C_{3v}$  and  $T_d$ . It is not useless: it describes invariant quantities.
- A representation is *faithful* if different elements get different matrices. The vector representation of  $C_{3v}$  is faithful; the trivial one is not. Faithfulness is not required.
- There are infinitely many representations (any dimension, any basis). The central task of the theory is to classify them and to reduce any given one to a few standard pieces.

## 1.6 Reducible and irreducible representations

All six matrices (1.4) have the same block-diagonal structure: a  $2 \times 2$  block acting on  $(v_x, v_y)$  and a  $1 \times 1$  block acting on  $v_z$ . Because matrix multiplication preserves this structure, each set of blocks is itself a representation. The  $1 \times 1$  blocks are all equal to 1: that is  $A_1$ . The  $2 \times 2$  blocks form a two-dimensional representation called  $E$ . We write

$$\Gamma_V = A_1 \oplus E. \quad (1.6)$$

The symbol  $\oplus$  (*direct sum*) does not add matrices of different sizes; it means that in a suitable basis all matrices take block-diagonal form with the given blocks.

Geometrically: the  $z$  axis and the  $xy$  plane are *invariant subspaces*. No operation of the group takes a vector on the  $z$  axis off the axis, nor a vector in the plane out of the plane.

Definition 1.6. A representation on  $L$  is *reducible* if  $L$  has a proper invariant subspace ( $\neq \{0\}, L$ ) and *irreducible* otherwise.

The representation  $E$  is irreducible: every line in the  $xy$  plane is moved by some  $C_3$  rotation or some mirror.

For finite groups every representation is equivalent to a unitary one (Section 1.8), and for unitary representations the orthogonal complement of an invariant subspace is also invariant. Hence a reducible representation always splits completely into a direct sum of irreducible ones. This is why irreps are *building blocks*.

## 1.7 Equivalent representations

The block structure depends on the basis. Rotate the  $y$  and  $z$  axes by  $45^\circ$  about  $x$ , i.e. change basis with

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{W}(g) = \mathbf{A} \mathbf{V}(g) \mathbf{A}^{-1}.$$

Then, for instance,

$$\mathbf{W}(C_3^+) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{6}}{4} & -\frac{\sqrt{6}}{4} \\ \frac{\sqrt{6}}{4} & \frac{1}{4} & -\frac{3}{4} \\ \frac{\sqrt{6}}{4} & -\frac{3}{4} & \frac{1}{4} \end{pmatrix}, \quad \mathbf{W}(\sigma_{d1}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & -1 & 0 \end{pmatrix},$$

and no block structure is visible, although  $\mathbf{W}$  describes exactly the same physics in rotated axes.

**Definition 1.7.** Two representations  $\mathbf{D}_1$  and  $\mathbf{D}_2$  of  $G$  are *equivalent* if there is an invertible matrix  $\mathbf{A}$  such that  $\mathbf{D}_2(g) = \mathbf{A} \mathbf{D}_1(g) \mathbf{A}^{-1}$  for all  $g \in G$ .

Equivalent representations are the same representation written in different bases. This raises two practical questions: given a set of matrices, how do we know whether it is reducible, and how do we find its irreducible components, without trying infinitely many changes of basis? The answer is the theory of characters (Chapter 2). It also shows that a finite group has only finitely many inequivalent irreps, as many as it has conjugacy classes:  $C_{3v}$  has three and  $T_d$  has five.

## 1.8 Unitarity and complete reducibility ★

The lectures state the main theorems of representation theory without proof. The starred sections supply the proofs, starting here with the two facts used in Section 1.6.  $G$  is a finite group throughout; sums over  $g$  run over all elements.

Theorem 1.8. *Every representation of a finite group is equivalent to a unitary one.*

*Proof.* Given  $\mathbf{D}(g)$  on  $\mathbb{C}^n$  with the standard inner product  $(x, y) = x^\dagger y$ , define the averaged inner product

$$\{x, y\} = \sum_g (\mathbf{D}(g)x, \mathbf{D}(g)y).$$

It is positive definite and invariant:  $\{\mathbf{D}(h)x, \mathbf{D}(h)y\} = \sum_g (\mathbf{D}(gh)x, \mathbf{D}(gh)y) = \{x, y\}$  by the rearrangement theorem. In a basis orthonormal for  $\{\cdot, \cdot\}$  the matrices are unitary.  $\square$

Theorem 1.9 (Maschke). *If  $W$  is an invariant subspace of a unitary representation, so is its orthogonal complement  $W^\perp$ . Hence every representation is a direct sum of irreps.*

*Proof.* If  $y \in W^\perp$  and  $x \in W$ , then  $(x, \mathbf{D}(g)y) = (\mathbf{D}(g)^{-1}x, y) = 0$ , because  $\mathbf{D}(g)^{-1}x = \mathbf{D}(g^{-1})x \in W$ . Iterate on  $W$  and  $W^\perp$  until the pieces are irreducible (the dimension decreases at each step).  $\square$

## Exercises and solutions

Exercise 1.1. Verify three entries of the multiplication table (1.3) by multiplying the matrices (1.4):  $C_3^+ \sigma_{d1}$ ,  $\sigma_{d1} C_3^+$  and  $\sigma_{d2} \sigma_{d1}$ . Interpret the last one geometrically.

*Solution.* With the matrices (1.4) (only the upper-left  $2 \times 2$  blocks matter; the  $zz$  entry is always 1):

$$\mathbf{V}(C_3^+) \mathbf{V}(\sigma_{d1}) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \mathbf{V}(\sigma_{d3}),$$

$$\mathbf{V}(\sigma_{d1}) \mathbf{V}(C_3^+) = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \mathbf{V}(\sigma_{d2}),$$

$$\mathbf{V}(\sigma_{d2}) \mathbf{V}(\sigma_{d1}) = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix} = \mathbf{V}(C_3^-).$$

Geometrically, reflecting first in a line and then in a second line making the angle  $\alpha$  with it is a rotation by  $2\alpha$ . From  $\sigma_{d1}$  ( $0^\circ$ ) to  $\sigma_{d2}$  ( $120^\circ$ ) the angle is  $120^\circ$ , so the product is a rotation by  $240^\circ = -120^\circ$ , i.e.  $C_3^-$ .

Exercise 1.2. List all subgroups of  $C_{3v}$ . Which of them are commutative?

*Solution.* By Lagrange's theorem the orders can be 1, 2, 3, 6:  $\{E\}$ ; the three groups  $\{E, \sigma_{di}\}$ ;  $C_3 = \{E, C_3^+, C_3^-\}$ ; and  $C_{3v}$ . All proper subgroups have prime or unit order, hence are cyclic and commutative;  $C_{3v}$  itself is not commutative.

Exercise 1.3. Show that assigning  $+1$  to  $E, C_3^\pm$  and  $-1$  to the three mirrors is a representation of  $C_{3v}$  (called  $A_2$ ). Is it faithful? Give a physical quantity that transforms according to  $A_2$ .

*Solution.* The assignment is  $g \mapsto \det \mathbf{V}(g)$ , and  $\det$  is multiplicative:  $\det(\mathbf{V}(g_1)\mathbf{V}(g_2)) = \det \mathbf{V}(g_1) \det \mathbf{V}(g_2)$ . (Equivalently: rotation times rotation and mirror times mirror are rotations, rotation times mirror is a mirror, as the table (1.3) confirms.) It is not faithful: its kernel is  $C_3$ . Quantities transforming as  $A_2$ : the  $z$  component of an axial vector ( $R_z, L_z, B_z$ ), which is invariant under the rotations about  $z$  and reversed by vertical mirrors.

Exercise 1.4. A planar molecule  $\text{BF}_3$  has four atoms. How many vibrational degrees of freedom does it have? What about the linear molecule  $\text{CO}_2$ ? (For a linear molecule, rotation about the molecular axis does not move the atoms.)

*Solution.*  $\text{BF}_3$ :  $3 \times 4 = 12$  degrees of freedom, minus 3 translations and 3 rotations: 6 vibrational.  $\text{CO}_2$  (linear):  $9 - 3 - 2 = 4$ .

Exercise 1.5. Show that the matrices  $\mathbf{W}(g)$  of Section 1.7 have the same traces as the matrices  $\mathbf{V}(g)$ . Then prove that  $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{A}^{-1}) = \text{tr} \mathbf{B}$  for any invertible  $\mathbf{A}$ .

*Solution.*  $\text{tr} \mathbf{W}(C_3^+) = -\frac{1}{2} + \frac{1}{4} + \frac{1}{4} = 0 = \text{tr} \mathbf{V}(C_3^+)$  and  $\text{tr} \mathbf{W}(\sigma_{\text{d1}}) = 1 + 0 + 0 = 1 = \text{tr} \mathbf{V}(\sigma_{\text{d1}})$ . In general  $\text{tr}(\mathbf{A}\mathbf{B}\mathbf{A}^{-1}) = \text{tr}(\mathbf{A}^{-1}\mathbf{A}\mathbf{B}) = \text{tr} \mathbf{B}$  by the cyclic property  $\text{tr}(\mathbf{X}\mathbf{Y}) = \sum_{ij} X_{ij}Y_{ji} = \text{tr}(\mathbf{Y}\mathbf{X})$ .

## Chapter 2

# Irreducible representations as building blocks

### Video: Lecture 2

Recap and the problem of finding invariant subspaces L2 @ 8:18 · traces and characters L2 @ 18:49 · classes L2 @ 24:07 · number of irreps and  $\sum d_\mu^2 = |G|$  L2 @ 33:37 · orthogonality of characters L2 @ 36:44 · the “magic formula” L2 @ 41:01 · computing characters without matrices L2 @ 57:21 · roto reflections and rotoinversions L2 @ 61:31 · axial vectors L2 @ 72:10 · the mechanical representation L2 @ 74:13 · CH<sub>3</sub>D worked out L2 @ 86:55

The aim of this chapter is to derive, with nothing but a character table and a few multiplications, the decomposition  $\Gamma_{\text{vib}}(\text{CH}_3\text{D}) = 3\text{A}_1 \oplus 3\text{E}$ .

## 2.1 Characters

Matrices change under a change of basis, but traces do not:  $\text{tr}(\mathbf{A}\mathbf{D}\mathbf{A}^{-1}) = \text{tr} \mathbf{D}$ . Hence equivalent representations have the same traces. The converse is far from obvious, and is one of the central results of the theory (proved in [Section 2.6](#)):

**Theorem 2.1.** *Two representations of a finite group are equivalent if and only if  $\text{tr} \mathbf{D}_1(g) = \text{tr} \mathbf{D}_2(g)$  for every  $g \in G$ .*

**Definition 2.2.** The *character* of a representation  $\Gamma$  is the function  $\chi_\Gamma: G \rightarrow \mathbb{C}$ ,  $\chi_\Gamma(g) = \text{tr} \mathbf{D}(g)$ .

A character is the whole list of traces, one per group element; the trace of a single element is not enough to characterise a representation. Two consequences:

- $\chi(E) = \dim \Gamma$ , because  $\mathbf{D}(E)$  is the unit matrix.
- If  $\Gamma = \Gamma_1 \oplus \Gamma_2$  then  $\chi_\Gamma = \chi_{\Gamma_1} + \chi_{\Gamma_2}$ : the trace of a block-diagonal matrix is the sum of the traces of the blocks.

For the vector representation of  $C_{3v}$  (1.4), the traces of the full matrices, of the  $2 \times 2$  blocks and of the  $1 \times 1$  blocks are

|              | $E$ | $C_3^+$ | $C_3^-$ | $\sigma_{d1}$ | $\sigma_{d2}$ | $\sigma_{d3}$ |
|--------------|-----|---------|---------|---------------|---------------|---------------|
| $\chi_V$     | 3   | 0       | 0       | 1             | 1             | 1             |
| $\chi_E$     | 2   | -1      | -1      | 0             | 0             | 0             |
| $\chi_{A_1}$ | 1   | 1       | 1       | 1             | 1             | 1             |

and indeed  $\chi_V = \chi_{A_1} + \chi_E$ .

## 2.2 Conjugacy classes

In the table above  $C_3^+$  and  $C_3^-$  have equal traces, and so do the three mirrors. This is no accident.

**Definition 2.3.** Two elements  $a, b \in G$  are *conjugate* if  $b = gag^{-1}$  for some  $g \in G$ . Conjugacy is an equivalence relation; its equivalence classes are the (*conjugacy*) *classes* of  $G$ .

Since  $\text{tr } \mathbf{D}(gag^{-1}) = \text{tr}(\mathbf{D}(g)\mathbf{D}(a)\mathbf{D}(g)^{-1}) = \text{tr } \mathbf{D}(a)$ , *characters are class functions*: constant on each class.

Intuitively, a class collects the elements “of the same kind” that are related by some symmetry of the group. In  $C_{3v}$ :

- $\{E\}$ : the identity is always a class by itself.
- $\{C_3^+, C_3^-\}$ : a vertical mirror turns a counterclockwise rotation into a clockwise one (a rotation seen in a mirror parallel to its axis reverses its sense),  $\sigma_{d1}C_3^+\sigma_{d1}^{-1} = C_3^-$ .
- $\{\sigma_{d1}, \sigma_{d2}, \sigma_{d3}\}$ : a  $C_3$  rotation carries each mirror plane into another,  $C_3^+\sigma_{d1}C_3^- = \sigma_{d2}$ .

Being “of the same kind” is not enough: there must be a group element relating them. In a group containing only rotations about  $z$  (such as  $C_3$ ),  $C_3^+$  and  $C_3^-$  are *not* conjugate, since nothing reverses the sense of rotation. In a commutative (Abelian) group every element is a class by itself.

### 2.2.1 Class sizes and rules of thumb ★

Conjugation  $a \mapsto gag^{-1}$  is the change of “point of view” by the symmetry  $g$ . The number of elements in the class of  $a$  is  $|G|/|Z(a)|$ , where  $Z(a) = \{g : ga = ag\}$  is the centraliser; in particular class sizes divide  $|G|$ . Rules of thumb for point groups:

- rotations by the same angle about axes related by a group operation are conjugate;

- $C_n^+$  and  $C_n^-$  about the same axis are conjugate if and only if the group contains a  $C_2$  perpendicular to the axis or a mirror containing the axis;
- mirrors related by a group operation are conjugate.

## 2.3 Normal subgroups, quotient groups and homomorphisms



Definition 2.4. A subgroup  $N$  is *normal* ( $N \trianglelefteq G$ ) if  $gNg^{-1} = N$  for all  $g$ , i.e. if it is a union of whole classes. Equivalently, left and right cosets coincide,  $gN = Ng$ .

For a normal subgroup the product of cosets (Section 1.4) is well defined,  $(g_1N)(g_2N) = g_1g_2N$ , and the cosets form the *quotient group*  $G/N$  of order  $|G|/|N|$ .

Example 2.5.  $C_3 = \{E, C_3^+, C_3^-\}$  is normal in  $C_{3v}$  (it is the union of the classes  $\{E\}$  and  $\{C_3^\pm\}$ ). The quotient  $C_{3v}/C_3$  has two elements, “rotations” and “reflections”. It is isomorphic to  $\{+1, -1\}$  under multiplication, which is precisely the irrep  $A_2$  viewed as a homomorphism.

The most important example in this course is the translation subgroup of a space group, whose quotient is the point group (Section 6.5.1).

### 2.3.1 Homomorphisms

A map  $f: G \rightarrow G'$  with  $f(g_1g_2) = f(g_1)f(g_2)$  is a *homomorphism*. Its kernel  $\text{Ker } f = \{g : f(g) = e'\}$  is a normal subgroup, and

Theorem 2.6 (First isomorphism theorem).  $G/\text{Ker } f \cong f(G)$ .

A representation is a homomorphism  $G \rightarrow \text{GL}(n, \mathbb{C})$ . It is faithful exactly when its kernel is  $\{e\}$ . A non-faithful representation of  $G$  is a faithful one of  $G/\text{Ker}$ : the irrep  $A_2$  of  $C_{3v}$  “lives” on  $C_{3v}/C_3$ . The map  $\text{SU}(2) \rightarrow \text{SO}(3)$  is a homomorphism with kernel  $\{\mathbf{1}, -\mathbf{1}\}$ , so  $\text{SO}(3) \cong \text{SU}(2)/\{\pm\mathbf{1}\}$  (Section 5.5).

## 2.4 Character tables

Because characters are class functions, they are tabulated by class, with the number of elements of each class written in front of its symbol:

| $C_{3v}$ | $E$ | $2C_3$ | $3\sigma_d$ | basis functions                                 | (2.1) |
|----------|-----|--------|-------------|-------------------------------------------------|-------|
| $A_1$    | 1   | 1      | 1           | $z, x^2 + y^2, z^2$                             |       |
| $A_2$    | 1   | 1      | -1          | $R_z$                                           |       |
| $E$      | 2   | -1     | 0           | $(x, y), (R_x, R_y), (x^2 - y^2, xy), (xz, yz)$ |       |

The last column lists functions or quantities that transform according to each irrep;  $R_i$  denote components of an axial vector.

Theorem 2.7. *For a finite group  $G$ :*

1. *the number of inequivalent irreps equals the number of conjugacy classes;*
2.  $\sum_{\mu} d_{\mu}^2 = |G|$  (Burnside).

For  $C_{3v}$ : three classes, hence three irreps, with  $d_1^2 + d_2^2 + d_3^2 = 6$ . The only solution in positive integers is 1, 1, 2. Infinitely many representations of any dimension, and all of them are built from just these three blocks. For  $T_d$  (24 elements, five classes):  $1 + 1 + 4 + 9 + 9 = 24$ , dimensions 1, 1, 2, 3, 3.

You never need to construct character tables yourself: they are tabulated in Bradley and Cracknell and in the POINT program of the Bilbao Crystallographic Server. The table of  $T_d$  is

| $T_d$ | $E$ | $8C_3$ | $3C_2$ | $6S_4$ | $6\sigma_d$ | basis functions                 |
|-------|-----|--------|--------|--------|-------------|---------------------------------|
| $A_1$ | 1   | 1      | 1      | 1      | 1           | $x^2 + y^2 + z^2$               |
| $A_2$ | 1   | 1      | 1      | -1     | -1          |                                 |
| $E$   | 2   | -1     | 2      | 0      | 0           | $(2z^2 - x^2 - y^2, x^2 - y^2)$ |
| $T_1$ | 3   | 0      | -1     | 1      | -1          | $(R_x, R_y, R_z)$               |
| $T_2$ | 3   | 0      | -1     | -1     | 1           | $(x, y, z), (yz, xz, xy)$       |

(2.2)

## 2.5 Orthogonality and the reduction formula

Characters of a group with  $|G|$  elements can be regarded as vectors with  $|G|$  components, and we give them the natural inner product

$$\langle \chi, \chi' \rangle = \frac{1}{|G|} \sum_{g \in G} \chi(g)^* \chi'(g). \quad (2.3)$$

Theorem 2.8 (Orthogonality of characters). *If  $\chi^{(\mu)}$  and  $\chi^{(\nu)}$  are characters of irreps, then  $\langle \chi^{(\mu)}, \chi^{(\nu)} \rangle = \delta_{\mu\nu}$ .*

The irreducible characters are therefore an orthonormal set, and since their number equals the number of classes they are a basis of the space of class functions. If  $\Gamma = \bigoplus_{\nu} m_{\nu} \Gamma^{(\nu)}$ , then  $\chi_{\Gamma} = \sum_{\nu} m_{\nu} \chi^{(\nu)}$ , and projecting on  $\chi^{(\mu)}$  gives the multiplicities:

Reduction formula (the “magic formula”)

$$m_\mu = \langle \chi^{(\mu)}, \chi_\Gamma \rangle = \frac{1}{|G|} \sum_{g \in G} \chi^{(\mu)}(g)^* \chi_\Gamma(g) = \frac{1}{|G|} \sum_{\text{classes } \mathcal{C}} N_{\mathcal{C}} \chi^{(\mu)}(\mathcal{C})^* \chi_\Gamma(\mathcal{C}), \quad (2.4)$$

where  $N_{\mathcal{C}}$  is the number of elements of class  $\mathcal{C}$ . A useful corollary:  $\Gamma$  is irreducible if and only if  $\langle \chi_\Gamma, \chi_\Gamma \rangle = 1$ ; in general  $\langle \chi_\Gamma, \chi_\Gamma \rangle = \sum_\mu m_\mu^2$ .

The multiplicities come out as non-negative integers:  $m_\mu$  counts diagonal blocks, and there is no such thing as half a block or a negative number of blocks. A fractional result is a sure sign of an arithmetic mistake (or, as in [Section 5.4](#), of using the wrong group).

*Remark 2.9* (The most common mistake). The sum in (2.4) runs over *elements*, not classes. When using a table organised by classes, multiply each term by the class size  $N_{\mathcal{C}}$ . A practical trick: write the class sizes above the columns before starting.

*Example 2.10* (Decomposing  $\Gamma_V$  of  $C_{3V}$ ). With  $\chi_V = (3, 0, 1)$  on the classes  $(E, 2C_3, 3\sigma_d)$  and the table (2.1):

$$\begin{aligned} m_{A_1} &= \frac{1}{6} (1 \cdot 3 \cdot 1 + 2 \cdot 0 \cdot 1 + 3 \cdot 1 \cdot 1) = 1, \\ m_{A_2} &= \frac{1}{6} (1 \cdot 3 \cdot 1 + 2 \cdot 0 \cdot 1 + 3 \cdot 1 \cdot (-1)) = 0, \\ m_E &= \frac{1}{6} (1 \cdot 3 \cdot 2 + 2 \cdot 0 \cdot (-1) + 3 \cdot 1 \cdot 0) = 1. \end{aligned}$$

So  $\Gamma_V = A_1 \oplus E$ , without ever looking for a good basis. Checks:  $\langle \chi_V, \chi_V \rangle = \frac{1}{6}(9 + 0 + 3) = 2 = 1^2 + 1^2$ .

## 2.6 Schur's lemma and the orthogonality theorems ★

The lecture states [Theorems 2.1](#), [2.7](#) and [2.8](#) without proof. All of them rest on Schur's lemma. By [Section 1.8](#) we may take every representation unitary.

### 2.6.1 Schur's lemma

**Lemma 2.11** (Schur I). *Let  $\mathbf{D}^{(1)}$  ( $d_1$ -dimensional) and  $\mathbf{D}^{(2)}$  ( $d_2$ -dimensional) be irreps and let  $\mathbf{A}$  be a  $d_1 \times d_2$  matrix with  $\mathbf{D}^{(1)}(g)\mathbf{A} = \mathbf{A}\mathbf{D}^{(2)}(g)$  for all  $g$  (an intertwiner). Then either  $\mathbf{A} = 0$ , or  $\mathbf{A}$  is invertible and the irreps are equivalent.*

*Proof.*  $\text{Ker } \mathbf{A}$  is invariant under  $\mathbf{D}^{(2)}$  (if  $\mathbf{A}x = 0$  then  $\mathbf{A}\mathbf{D}^{(2)}x = \mathbf{D}^{(1)}\mathbf{A}x = 0$ ). By irreducibility it is  $\{0\}$  or everything. Likewise the image of  $\mathbf{A}$  is invariant under  $\mathbf{D}^{(1)}$ , so it is  $\{0\}$  or everything. Either  $\mathbf{A} = 0$ , or  $\mathbf{A}$  is injective and surjective.  $\square$

**Lemma 2.12** (Schur II). *A matrix commuting with all matrices of an irrep (over  $\mathbb{C}$ ) is a multiple of the identity.*

*Proof.* Let  $\lambda$  be an eigenvalue of  $\mathbf{A}$ . Then  $\mathbf{A} - \lambda \mathbf{1}$  commutes with the irrep and is not invertible, so by Schur I it vanishes.  $\square$

Schur II is the heart of Wigner's theorem: an invariant Hamiltonian restricted to one copy of an irrep is proportional to the identity, so all its  $d_\mu$  partner states have the same energy.

### 2.6.2 The great orthogonality theorem

Theorem 2.13. For unitary irreps  $\mathbf{D}^{(\mu)}$ ,  $\mathbf{D}^{(\nu)}$  (chosen identical if equivalent),

$$\sum_{g \in G} D_{ij}^{(\mu)}(g)^* D_{kl}^{(\nu)}(g) = \frac{|G|}{d_\mu} \delta_{\mu\nu} \delta_{ik} \delta_{jl}. \quad (2.5)$$

*Proof.* For any  $d_\nu \times d_\mu$  matrix  $\mathbf{X}$ , the average  $\mathbf{A} = \sum_g \mathbf{D}^{(\nu)}(g) \mathbf{X} \mathbf{D}^{(\mu)}(g)^{-1}$  is an intertwiner:  $\mathbf{D}^{(\nu)}(h) \mathbf{A} = \mathbf{A} \mathbf{D}^{(\mu)}(h)$  by rearrangement. For  $\mu \neq \nu$ ,  $\mathbf{A} = 0$  (Schur I). For  $\mu = \nu$ ,  $\mathbf{A} = \lambda \mathbf{1}$  (Schur II), with  $\lambda$  fixed by the trace:  $d_\mu \lambda = \text{tr } \mathbf{A} = |G| \text{tr } \mathbf{X}$ . Now choose  $\mathbf{X}$  with a single entry 1 in position  $(l, j)$ . Using unitarity,  $[\mathbf{D}^{(\mu)}(g)^{-1}]_{ji} = D_{ij}^{(\mu)}(g)^*$ , the  $(k, i)$  element of  $\mathbf{A}$  is  $\sum_g D_{kl}^{(\nu)}(g) D_{ij}^{(\mu)}(g)^*$ . It vanishes for  $\mu \neq \nu$  and equals  $\lambda \delta_{ki} = (|G|/d_\mu) \delta_{lj} \delta_{ki}$  for  $\mu = \nu$ , which is (2.5).  $\square$

Interpretation: the  $\sum_\mu d_\mu^2$  functions  $g \mapsto \sqrt{d_\mu/|G|} D_{ij}^{(\mu)}(g)$  are orthonormal in the  $|G|$ -dimensional space of functions on  $G$ . Hence  $\sum_\mu d_\mu^2 \leq |G|$ .

### 2.6.3 Proofs of the character theorems

Setting  $i = j$ ,  $k = l$  in (2.5) and summing gives the orthogonality of characters, Theorem 2.8:  $\frac{1}{|G|} \sum_g \chi^{(\mu)}(g)^* \chi^{(\nu)}(g) = \delta_{\mu\nu}$ . From it follow the reduction formula (2.4) and the irreducibility test  $\langle \chi, \chi \rangle = 1$ .

*Proof of Theorem 2.1.* By complete reducibility every representation is  $\bigoplus m_\mu \Gamma^{(\mu)}$ , and the multiplicities are determined by the character through (2.4). Two representations with the same character have the same multiplicities, hence are both equivalent to the same direct sum.  $\square$

The regular representation. Let  $G$  act on the  $|G|$ -dimensional space with basis  $\{e_h\}_{h \in G}$  by  $g e_h = e_{gh}$ . Since  $gh \neq h$  unless  $g = e$ , its character is  $\chi_{\text{reg}}(e) = |G|$  and  $\chi_{\text{reg}}(g) = 0$  otherwise. By (2.4),  $m_\mu = \frac{1}{|G|} |G| d_\mu = d_\mu$ : every irrep occurs in the regular representation as many times as its dimension. Counting dimensions,  $\sum_\mu d_\mu^2 = |G|$  (Burnside), which proves part 2 of Theorem 2.7.

Number of irreps. The characters are class functions, orthonormal, so their number is at most the number of classes  $n_c$ . Completeness of the matrix elements (they span all functions on  $G$ , by Burnside) implies that the characters span all class functions, so the number of irreps is exactly  $n_c$ . An equivalent statement is the *second orthogonality relation*,

$$\sum_{\mu} \chi^{(\mu)}(\mathcal{C})^* \chi^{(\mu)}(\mathcal{C}') = \frac{|G|}{N_{\mathcal{C}}} \delta_{\mathcal{C}\mathcal{C}'}, \quad (2.6)$$

orthogonality of the *columns* of a character table. It is a quick check of a table: for  $C_{3v}$  the column of  $\sigma_d$  gives  $1 + 1 + 0 = 2 = 6/3$ .

## 2.7 Computing characters without matrices

To use the reduction formula we still need the character of the representation of interest. Two observations make this almost free:

1. Characters are class functions: compute one element per class, choosing the easiest.
2. Traces do not depend on the basis: for each element, use the most convenient axes.

### 2.7.1 The operations of a point group

Besides the identity, a point group contains four kinds of operations:

- proper rotations  $C_n^{\pm}$  by  $\theta = \pm 2\pi/n$ ;
- reflections  $\sigma$ ;
- the inversion  $I$ ;
- *rotoreflections*  $S_n^{\pm} = \sigma_h C_n^{\pm}$ : rotate by  $\pm 2\pi/n$ , then reflect in the plane perpendicular to the axis.

Crystallographers (and the Bilbao server) use *rotoinversions*  $\bar{n}$  instead: rotate by  $2\pi/n$ , then invert. The two are related through  $\sigma_h = IC_2$  (reflection in a plane equals inversion times a half-turn about the normal):

$$S_n^{\pm} = \sigma_h C_n^{\pm} = I C_2 C_n^{\pm} = I C(\pi \pm 2\pi/n).$$

For example  $S_3^+ = IC_6^- = \bar{6}^-$  and  $S_4^+ = \bar{4}^-$ : a rotoreflection corresponds to a rotoinversion about the same axis but with a different angle and the *opposite* sense. Tables in Schoenflies notation (Bradley–Cracknell) use  $S_n$ ; tables in international notation use  $\bar{n}$ . Check which convention a table uses.

Every improper operation (determinant  $-1$ ) is the inversion times a proper rotation.

### 2.7.2 Character of the vector representation

Take the rotation axis as  $z$ . A rotation by  $\theta$  has diagonal elements  $\cos \theta, \cos \theta, 1$ ; a rotoreflection  $S(\theta) = \sigma_h C(\theta)$  has  $\cos \theta, \cos \theta, -1$ . Hence

$$\boxed{\chi_V(C(\theta)) = 1 + 2 \cos \theta, \quad \chi_V(S(\theta)) = -1 + 2 \cos \theta} \quad (2.7)$$

with the special cases  $\chi_V(E) = 3$ ,  $\chi_V(C_2) = -1$ ,  $\chi_V(C_3) = 0$ ,  $\chi_V(C_4) = 1$ ,  $\chi_V(C_6) = 2$ ,  $\chi_V(\sigma) = \chi_V(S(0)) = 1$ ,  $\chi_V(I) = \chi_V(S(\pi)) = -3$ ,  $\chi_V(S_4) = -1$ ,  $\chi_V(S_6) = 0$ ,  $\chi_V(S_3) = -2$ .

### 2.7.3 Axial vectors

An axial vector (pseudovector) transforms like a vector under proper rotations but does not change sign under inversion,  $\vec{L} = \vec{r} \times \vec{p} \mapsto (-\vec{r}) \times (-\vec{p}) = \vec{L}$ . Since every improper operation is  $I$  times a rotation, the axial-vector matrices are  $\mathbf{R}(g) = \det(\mathbf{V}(g)) \mathbf{V}(g)$  and

$$\chi_R(g) = \det(\mathbf{V}(g)) \chi_V(g). \quad (2.8)$$

A counter-intuitive consequence: a mirror *reverses* the components of a pseudovector parallel to the mirror and *preserves* the perpendicular one, exactly the opposite of an ordinary vector. Think of a current loop and its mirror image: a magnetic field lying in the mirror plane flips, while the component normal to the mirror does not.

For  $C_{3v}$ :  $\chi_R = (3, 0, -1)$ , so  $\Gamma_R = A_2 \oplus E$  ( $R_z$  spans  $A_2$ , which explains the entry in (2.1)).

## 2.8 The mechanical representation

Label the atoms of a molecule  $a = 1, \dots, N$  and collect their displacements in the  $3N$ -component vector  $\mathbf{u} = (\vec{u}_1, \dots, \vec{u}_N)$ . A symmetry operation  $g$  does two things to a distortion:

1. it *permutes* atoms of the same species: atom  $a$  is carried to the position of atom  $g(a)$ ;
2. it *rotates* each displacement vector by  $\mathbf{V}(g)$ .

Hence  $\vec{u}'_{g(a)} = \mathbf{V}(g) \vec{u}_a$ , and the  $3N \times 3N$  matrix  $\mathbf{M}(g)$  consists of  $3 \times 3$  blocks equal to  $\mathbf{V}(g)$ , placed in block-row  $g(a)$  and block-column  $a$ . For  $\text{CH}_3\text{D}$  with atoms ordered  $(\text{H}_1, \text{H}_2, \text{H}_3, \text{D}, \text{C})$ , the rotation  $C_3^+$  sends  $\text{H}_1 \rightarrow \text{H}_2 \rightarrow \text{H}_3 \rightarrow \text{H}_1$  and fixes

D and C:

$$\mathbf{M}(C_3^+) = \begin{pmatrix} 0 & 0 & \mathbf{V} & 0 & 0 \\ \mathbf{V} & 0 & 0 & 0 & 0 \\ 0 & \mathbf{V} & 0 & 0 & 0 \\ 0 & 0 & 0 & \mathbf{V} & 0 \\ 0 & 0 & 0 & 0 & \mathbf{V} \end{pmatrix}, \quad \mathbf{V} \equiv \mathbf{V}(C_3^+).$$

Only blocks on the diagonal contribute to the trace, and a diagonal block appears exactly when an atom is left in place. Therefore

Key result

$$\chi_{\mathbf{M}}(g) = N_{\text{inv}}(g) \chi_{\mathbf{V}}(g), \quad (2.9)$$

where  $N_{\text{inv}}(g)$  is the number of atoms left invariant by  $g$ .

There is no need to construct any  $15 \times 15$  matrix: look at the molecule and count.

### 2.8.1 Removing translations and rotations

Uniform translations  $\vec{u}_a = \vec{t}$  span a 3-dimensional invariant subspace that transforms as  $\Gamma_{\mathbf{V}}$ . Rigid rotations about the centre of mass span another one that transforms as  $\Gamma_{\mathbf{R}}$ . What is left is the space of pure vibrations:

$$\Gamma_{\text{vib}} = \Gamma_{\mathbf{M}} \ominus \Gamma_{\mathbf{V}} \ominus \Gamma_{\mathbf{R}}, \quad (2.10)$$

a formal subtraction of multiplicities ([Chapter 3](#) shows that the three subspaces are orthogonal with respect to the mass-weighted inner product, so the subtraction is legitimate).

### 2.8.2 Worked example: CH<sub>3</sub>D

| $C_{3v}$                                                                        | $E$ | $2C_3$ | $3\sigma_d$ |
|---------------------------------------------------------------------------------|-----|--------|-------------|
| $N_{\text{inv}}$                                                                | 5   | 2      | 3           |
| $\chi_{\mathbf{V}}$                                                             | 3   | 0      | 1           |
| $\chi_{\mathbf{M}} = N_{\text{inv}}\chi_{\mathbf{V}}$                           | 15  | 0      | 3           |
| $\chi_{\mathbf{R}}$                                                             | 3   | 0      | -1          |
| $\chi_{\text{vib}} = \chi_{\mathbf{M}} - \chi_{\mathbf{V}} - \chi_{\mathbf{R}}$ | 9   | 0      | 3           |

The identity leaves all five atoms in place; a  $C_3$  rotation leaves C and D; each mirror contains C, D and one H. The reduction formula gives

$$m_{A_1} = \frac{1}{6}(9 + 0 + 9) = 3, \quad m_{A_2} = \frac{1}{6}(9 + 0 - 9) = 0, \quad m_E = \frac{1}{6}(18 + 0 + 0) = 3,$$

so  $\Gamma_{\text{vib}} = 3A_1 \oplus 3E$ : three non-degenerate and three doubly degenerate frequencies. (Alternatively,  $\Gamma_M = 4A_1 \oplus A_2 \oplus 5E$ , and subtracting  $\Gamma_V = A_1 \oplus E$  and  $\Gamma_R = A_2 \oplus E$  gives the same result.)

Summary: the recipe for molecular vibrations

1. Find the point group and its character table.
2. For one element per class, write  $\chi_V$  from (2.7) and count  $N_{\text{inv}}$ .
3.  $\chi_M = N_{\text{inv}}\chi_V$ ,  $\chi_R = \det \cdot \chi_V$ ,  $\chi_{\text{vib}} = \chi_M - \chi_V - \chi_R$ .
4. Reduce with (2.4); read degeneracies from the dimensions and the number of distinct frequencies from the multiplicities.

## Exercises and solutions

Exercise 2.1 (*methane*). Using the table (2.2), derive  $\Gamma_M(\text{CH}_4) = A_1 \oplus E \oplus T_1 \oplus 3T_2$  and  $\Gamma_{\text{vib}} = A_1 \oplus E \oplus 2T_2$ . You will need the number of atoms left invariant by each class of  $T_d$ .

*Solution.* Classes  $E, 8C_3, 3C_2, 6S_4, 6\sigma_d$ . The  $C_3$  axes pass through C and one H; the  $C_2$  and  $S_4$  axes pass only through C; each  $\sigma_d$  contains C and two H. Hence

|                  | $E$ | $8C_3$ | $3C_2$ | $6S_4$ | $6\sigma_d$ |
|------------------|-----|--------|--------|--------|-------------|
| $N_{\text{inv}}$ | 5   | 2      | 1      | 1      | 3           |
| $\chi_V$         | 3   | 0      | -1     | -1     | 1           |
| $\chi_M$         | 15  | 0      | -1     | -1     | 3           |
| $\chi_R$         | 3   | 0      | -1     | 1      | -1          |

with  $\chi_V$  from (2.7) ( $S_4$ :  $-1 + 2\cos 90^\circ = -1$ ). The reduction formula gives

$$\begin{aligned}
 m_{A_1} &= \frac{1}{24}(15 + 0 - 3 - 6 + 18) = 1, & m_{A_2} &= \frac{1}{24}(15 + 0 - 3 + 6 - 18) = 0, \\
 m_E &= \frac{1}{24}(30 + 0 - 6 + 0 + 0) = 1, & m_{T_1} &= \frac{1}{24}(45 + 0 + 3 - 6 - 18) = 1, \\
 m_{T_2} &= \frac{1}{24}(45 + 0 + 3 + 6 + 18) = 3,
 \end{aligned}$$

so  $\Gamma_M = A_1 \oplus E \oplus T_1 \oplus 3T_2$ . Since  $\chi_V = \chi_{T_2}$  and  $\chi_R = \chi_{T_1}$ ,  $\Gamma_{\text{vib}} = A_1 \oplus E \oplus 2T_2$ .

Exercise 2.2 (*ammonia*). Find  $\Gamma_{\text{vib}}$  for the pyramidal molecule  $\text{NH}_3$  (point group  $C_{3v}$ ). How many distinct vibrational frequencies are there?

*Solution.*  $N_{\text{inv}} = (4, 1, 2)$  (each mirror contains N and one H),  $\chi_M = (12, 0, 2)$ , so  $\Gamma_M = 3A_1 \oplus A_2 \oplus 4E$ . Subtracting  $\Gamma_V = A_1 \oplus E$  and  $\Gamma_R = A_2 \oplus E$ :  $\Gamma_{\text{vib}} = 2A_1 \oplus 2E$ . Four distinct frequencies: two non-degenerate, two doubly degenerate ( $2 + 4 = 6 = 3 \cdot 4 - 6$ ).

Exercise 2.3 (*water*). The water molecule lies in the  $yz$  plane and has point group  $C_{2v}$ , with elements  $E$ ,  $C_2$ ,  $\sigma_v(xz)$  and  $\sigma'_v(yz)$ . Its character table is

| $C_{2v}$ | $E$ | $C_2$ | $\sigma_v(xz)$ | $\sigma'_v(yz)$ |
|----------|-----|-------|----------------|-----------------|
| $A_1$    | 1   | 1     | 1              | 1               |
| $A_2$    | 1   | 1     | -1             | -1              |
| $B_1$    | 1   | -1    | 1              | -1              |
| $B_2$    | 1   | -1    | -1             | 1               |

Show that  $\Gamma_{\text{vib}} = 2A_1 \oplus B_2$ .

*Solution.*  $\chi_V = (3, -1, 1, 1)$ ,  $N_{\text{inv}} = (3, 1, 1, 3)$  (the  $xz$  mirror contains only O),  $\chi_M = (9, -1, 1, 3)$ ,  $\chi_R = \det \cdot \chi_V = (3, -1, -1, -1)$ ,  $\chi_{\text{vib}} = \chi_M - \chi_V - \chi_R = (3, 1, 1, 3)$ . Then  $m_{A_1} = \frac{1}{4}(3 + 1 + 1 + 3) = 2$ ,  $m_{A_2} = \frac{1}{4}(3 + 1 - 1 - 3) = 0$ ,  $m_{B_1} = \frac{1}{4}(3 - 1 + 1 - 3) = 0$ ,  $m_{B_2} = \frac{1}{4}(3 - 1 - 1 + 3) = 1$ :  $\Gamma_{\text{vib}} = 2A_1 \oplus B_2$  (symmetric stretch, bend, antisymmetric stretch).

Exercise 2.4 (*boron trifluoride*). The planar molecule  $\text{BF}_3$  has point group  $D_{3h}$  with classes  $E$ ,  $2C_3$ ,  $3C'_2$ ,  $\sigma_h$ ,  $2S_3$ ,  $3\sigma_v$  and character table

| $D_{3h}$ | $E$ | $2C_3$ | $3C'_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|----------|-----|--------|---------|------------|--------|-------------|
| $A'_1$   | 1   | 1      | 1       | 1          | 1      | 1           |
| $A'_2$   | 1   | 1      | -1      | 1          | 1      | -1          |
| $E'$     | 2   | -1     | 0       | 2          | -1     | 0           |
| $A''_1$  | 1   | 1      | 1       | -1         | -1     | -1          |
| $A''_2$  | 1   | 1      | -1      | -1         | -1     | 1           |
| $E''$    | 2   | -1     | 0       | -2         | 1      | 0           |

Compute  $\Gamma_V$ ,  $\Gamma_R$ ,  $\Gamma_M$  and  $\Gamma_{\text{vib}}$ .

*Solution.* With  $\chi_V$  from (2.7) ( $S_3$ :  $-1 + 2\cos 120^\circ = -2$ ) and the invariant atoms (each  $C'_2$  and each  $\sigma_v$  contains B and one F;  $\sigma_h$  contains all atoms):

|                  | $E$ | $2C_3$ | $3C'_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|------------------|-----|--------|---------|------------|--------|-------------|
| $N_{\text{inv}}$ | 4   | 1      | 2       | 4          | 1      | 2           |
| $\chi_V$         | 3   | 0      | -1      | 1          | -2     | 1           |
| $\chi_M$         | 12  | 0      | -2      | 4          | -2     | 2           |
| $\chi_R$         | 3   | 0      | -1      | -1         | 2      | -1          |

Reduction:  $\Gamma_V = E' \oplus A''_2$ ,  $\Gamma_R = A'_2 \oplus E''$ ,  $\Gamma_M = A'_1 \oplus A'_2 \oplus 3E' \oplus 2A''_2 \oplus E''$ , and therefore  $\Gamma_{\text{vib}} = A'_1 \oplus 2E' \oplus A''_2$  ( $1 + 4 + 1 = 6$ ). For instance  $m_{E'}(\Gamma_M) = \frac{1}{12}(24 + 0 + 0 + 8 + 4 + 0) = 3$ .

Exercise 2.5. Use  $\langle \chi, \chi \rangle$  to show that the vector representation of  $T_d$  is irreducible, while that of  $C_{3v}$  is not.

$\left| \begin{array}{l} \text{Solution. } T_d: \langle \chi_V, \chi_V \rangle = \frac{1}{24}(9 + 0 + 3 \cdot 1 + 6 \cdot 1 + 6 \cdot 1) = 1: \text{ irreducible. } C_{3v}: \\ \frac{1}{6}(9 + 0 + 3) = 2 = 1^2 + 1^2: \text{ the sum of two different irreps.} \end{array} \right.$

## Chapter 3

# Operations with representations I: tensors and invariants

### Video: Lecture 3

Components vs vectors, active vs passive [L3 @ 4:19](#) · the column recipe for matrices [L3 @ 11:50](#) · pseudovectors and mirrors [L3 @ 16:09](#) · molecular polarizability [L3 @ 31:59](#) · invariants from irreps [L3 @ 39:19](#) · molecular vibrations: Lagrangian and dynamical matrix [L3 @ 54:06](#) · the potential energy of CH<sub>4</sub> in symmetry-adapted coordinates [L3 @ 69:52](#) · Wigner's theorem seen explicitly [L3 @ 83:37](#) · product representations [L3 @ 86:45](#)

## 3.1 How to build representation matrices

### 3.1.1 Active and passive

Two opposite conventions exist. In the *active* view, which we always use, the symmetry operation moves the object (vector, molecule, wave function) and the coordinate axes stay fixed. In the *passive* view the axes are moved and the object stays, which is equivalent to moving the object by the inverse operation. Some classic books (for example Bradley–Cracknell for translations) use the passive convention; mixing the two is a common source of sign errors.

Do not confuse the *coordinate system* (the axes, fixed) with the *coordinates* of a vector (its components, which change when the vector is moved).

### 3.1.2 The column recipe

Representation matrices act on columns of components,

$$(v'_x, v'_y, v'_z)^T = \mathbf{V}(g) (v_x, v_y, v_z)^T.$$

Applying  $\mathbf{V}(g)$  to the basis vector  $\vec{e}_1 = (1, 0, 0)^T$  returns the first column. Hence:

### Column recipe

The  $j$ -th column of  $\mathbf{D}(g)$  contains the components of the transformed basis vector  $g\vec{e}_j$ .

This works for every representation and every basis, and avoids general formulas for rotation or reflection matrices. As an example, construct  $\mathbf{V}(\sigma_{d2})$  for  $C_{3v}$  (Fig. 1.1); the mirror line makes  $120^\circ$  with the  $x$  axis.

- $\vec{e}_x$  (angle  $0^\circ$ ) is reflected to the direction at  $240^\circ$ :  $g\vec{e}_x = (-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0)$ .
- $\vec{e}_y$  (angle  $90^\circ$ ) is reflected to  $150^\circ$ :  $g\vec{e}_y = (-\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ .
- $\vec{e}_z$  lies in the mirror and is unchanged.

Placing these as columns reproduces  $\mathbf{V}(\sigma_{d2})$  in (1.4).

### 3.1.3 Pseudovectors, again

The axial-vector matrices are  $\mathbf{R}(g) = \det \mathbf{V}(g) \mathbf{V}(g)$ : identical to  $\mathbf{V}(g)$  for proper rotations, opposite for improper operations. For instance  $\mathbf{R}(\sigma_{d1}) = \text{diag}(-1, 1, -1)$  reverses the components of a pseudovector lying in the mirror ( $x, z$ ) and keeps the normal one ( $y$ ).

## 3.2 Tensors and their invariance

A physical property of a molecule or crystal in equilibrium is typically a tensor. Our first example is the electric *polarizability*  $\alpha_{ij}$ , which relates the induced dipole to an applied field:

$$p_i = \alpha_{ij} E_j \quad (\text{sum over repeated indices}). \quad (3.1)$$

A rank-2 tensor transforms like a product of coordinates,  $x_i x_j$ : one factor of  $\mathbf{V}$  for each index,

$$\alpha'_{ij} = V_{ik} V_{jl} \alpha_{kl} \quad \Longleftrightarrow \quad \boldsymbol{\alpha}' = \mathbf{V} \boldsymbol{\alpha} \mathbf{V}^T = \mathbf{V} \boldsymbol{\alpha} \mathbf{V}^{-1}, \quad (3.2)$$

where we used that  $\mathbf{V}(g)$  is orthogonal. If  $g$  is a symmetry, the molecule looks the same after the operation and its polarizability cannot change,  $\boldsymbol{\alpha}' = \boldsymbol{\alpha}$ . Equivalently  $[\boldsymbol{\alpha}, \mathbf{V}(g)] = 0$  for all  $g \in G$ .

The brute-force approach writes the most general symmetric  $3 \times 3$  matrix and imposes these commutation relations. That is feasible for  $\boldsymbol{\alpha}$ , tedious for the  $6 \times 6$  elastic tensor, and hopeless for the  $15 \times 15$  dynamical matrix of methane. Group theory offers a shortcut.

### 3.3 Invariants

Contract the tensor into a scalar. The energy of the molecule in a static field is

$$U = -\frac{1}{2} \alpha_{ij} E_i E_j. \quad (3.3)$$

Being a property of the molecule,  $U$  must be invariant under  $G$ ; being quadratic in  $\vec{E}$ , it must be a linear combination of the independent *quadratic invariants* that can be built from the components of  $\vec{E}$ . Group theory tells us all of them.

#### Quadratic invariants

Decompose the vector space into irreps with symmetry-adapted coordinates. Let all irreps be unitary and *real* (orthogonal matrices). Then:

1. within each irrep the squared norm  $\sum_i x_i^2$  is invariant, and it is the only quadratic invariant built from that irrep alone;
2. if an irrep occurs twice, with coordinates  $x_i$  and  $y_i$  written *in the same basis* (same matrices), the scalar product  $\sum_i x_i y_i$  is an additional invariant;
3. products of coordinates belonging to inequivalent irreps are never invariant.

Point 1 follows from unitarity: an orthogonal transformation preserves the norm. That the norm is the *only* invariant, and that inequivalent irreps give none, is a consequence of Schur's lemma (Section 2.6.1), as shown in Section 3.8. For complex irreps the recipe changes slightly (Section 6.1).

Example 3.1 (Polarizability of  $\text{CH}_3\text{D}$ ). In  $C_{3v}$ ,  $\Gamma_V = A_1 \oplus E$  with coordinates  $E_z$  ( $A_1$ ) and  $(E_x, E_y)$  ( $E$ ). The quadratic invariants are  $E_z^2$  and  $E_x^2 + E_y^2$ , so

$$U = -\frac{1}{2} [a(E_x^2 + E_y^2) + bE_z^2], \quad \alpha = \begin{pmatrix} a & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{pmatrix}.$$

Any molecule or crystal with symmetry  $C_{3v}$  has a polarizability determined by two constants. Group theory cannot give their values.

Example 3.2 (Polarizability of  $\text{CH}_4$ ). In  $T_d$ ,  $\Gamma_V = T_2$  is irreducible. The only quadratic invariant is  $E_x^2 + E_y^2 + E_z^2$ , so  $\alpha = a\mathbf{1}$ : the polarizability of methane is isotropic, although the molecule is not spherical. (This fact decides its rotational Raman spectrum, Section 4.5.)

### 3.4 Molecular vibrations

Let  $\vec{R}_a^0$  be the equilibrium position of atom  $a$  (measured from the centre of mass) and  $\vec{u}_a$  its displacement. Collect the displacements in the  $3N$ -vector  $\mathbf{u}$ . In the

harmonic approximation

$$L = \frac{1}{2} \dot{\mathbf{u}}^T \mathbf{M} \dot{\mathbf{u}} - \frac{1}{2} \mathbf{u}^T \Phi \mathbf{u}, \quad (3.4)$$

with the diagonal mass matrix  $\mathbf{M} = \text{diag}(m_1, m_1, m_1, m_2, \dots)$  and the matrix of force constants  $\Phi_{ai,bj} = \partial^2 V / \partial u_{ai} \partial u_{bj}$  at equilibrium. The equations of motion  $\mathbf{M} \ddot{\mathbf{u}} + \Phi \mathbf{u} = 0$  with  $\mathbf{u}(t) = \text{Re}(\mathbf{n} e^{-i\omega t})$  give the generalised eigenvalue problem  $\Phi \mathbf{n} = \omega^2 \mathbf{M} \mathbf{n}$ . With mass-weighted coordinates  $\mathbf{w} = \mathbf{M}^{1/2} \mathbf{u}$  it becomes an ordinary one,

$$\mathbf{D} \mathbf{w} = \omega^2 \mathbf{w}, \quad \mathbf{D} = \mathbf{M}^{-1/2} \Phi \mathbf{M}^{-1/2}. \quad (3.5)$$

The real symmetric *dynamical matrix*  $\mathbf{D}$  has  $3N$  eigenvalues  $\omega^2$  and orthonormal eigenvectors (the *normal modes*). In the original variables the modes are orthogonal with respect to the metric  $\mathbf{M}$ .

The  $3N$ -dimensional space splits into three  $\mathbf{M}$ -orthogonal invariant subspaces:

- translations,  $\vec{u}_a = \vec{t}$  for all  $a$  (transform as  $\Gamma_V$ );
- rigid rotations,  $\vec{u}_a = \vec{\varphi} \times \vec{R}_a^0$  (transform as  $\Gamma_R$ );
- pure vibrations, defined by the six constraints  $\sum_a m_a \vec{u}_a = 0$  (no displacement of the centre of mass) and  $\sum_a m_a \vec{R}_a^0 \times \vec{u}_a = 0$  (no angular momentum at linear order).

The first two cost no potential energy,  $\omega = 0$ . The third is awkward to describe explicitly, and group theory never needs to: it only uses its character  $\chi_M - \chi_V - \chi_R$ .

A symmetry operation only exchanges atoms of the same mass, so  $\mathbf{M}(g)$  commutes with the mass matrix,  $\mathbf{M}(g)\mathbf{M} = \mathbf{M}\mathbf{M}(g)$ . By the same argument as for the polarizability,  $\mathbf{D} = \mathbf{M}(g)\mathbf{D}\mathbf{M}(g)^T$ : the dynamical matrix is an invariant rank-2 tensor on the space of distortions.

### 3.5 Wigner's theorem, seen explicitly

Take methane,  $\Gamma_{\text{vib}} = A_1 \oplus E \oplus 2T_2$ , and assume we use symmetry-adapted vibrational coordinates:

$$\underbrace{q_1}_{A_1}, \quad \underbrace{(q_2, q_3)}_E, \quad \underbrace{(q_4, q_5, q_6)}_{T_2}, \quad \underbrace{(q_7, q_8, q_9)}_{T_2},$$

where the two  $T_2$  triplets transform with *identical* matrices. (How to find such coordinates with projection operators is explained in [Section 3.6](#).) The potential energy is an invariant quadratic form, so by [Section 3.3](#) it must be

$$2V = a q_1^2 + b (q_2^2 + q_3^2) + c (q_4^2 + q_5^2 + q_6^2) + d (q_7^2 + q_8^2 + q_9^2) + 2e (q_4 q_7 + q_5 q_8 + q_6 q_9). \quad (3.6)$$

Five constants, instead of the 45 independent entries of a general symmetric  $9 \times 9$  matrix. Now reorder the coordinates, pairing the first component of the first triplet

with the first component of the second, and so on:  $(q_1; q_2; q_3; q_4, q_7; q_5, q_8; q_6, q_9)$ . In this order the matrix of the quadratic form is block diagonal,

$$\mathbf{D}_{\text{vib}} = \text{diag}\left(a, b, b, \begin{pmatrix} c & e \\ e & d \end{pmatrix}, \begin{pmatrix} c & e \\ e & d \end{pmatrix}, \begin{pmatrix} c & e \\ e & d \end{pmatrix}\right). \quad (3.7)$$

Its eigenvalues are  $a$  (once),  $b$  (twice) and the two eigenvalues  $\omega_{\pm}^2$  of the  $2 \times 2$  block (three times each): four distinct frequencies with degeneracies 1, 2, 3, 3. The mechanism is general:

#### Wigner's theorem, block form

If an invariant hermitian operator acts on a space carrying  $\bigoplus_{\mu} m_{\mu} \Gamma^{(\mu)}$ , then in a symmetry-adapted basis (ordered as above) it consists, for each irrep  $\mu$ , of one  $m_{\mu} \times m_{\mu}$  block repeated  $d_{\mu}$  times. Hence  $m_{\mu}$  distinct eigenvalues, each  $d_{\mu}$ -fold degenerate.

The same statement with “Hamiltonian” for “dynamical matrix” is the basis of all quantum-mechanical applications: energy levels are labelled by irreps, and the degeneracy of a level equals the dimension of its irrep, unless two levels coincide *accidentally*.

*Remark 3.3* (Symmetry-adapted vs normal coordinates). In (3.6),  $q_1, q_2, q_3$  are already normal coordinates (their blocks are  $1 \times 1$ ). The  $T_2$  normal coordinates are linear combinations such as  $Q_4 = \cos \phi q_4 + \sin \phi q_7$  that diagonalise the  $2 \times 2$  block; symmetry alone cannot fix the mixing angle  $\phi$ .

## 3.6 Projection operators ★

Wigner's theorem is used in a symmetry-adapted basis. Projection operators construct such a basis from any starting vector. Let  $\Gamma$  act on a space  $L$  with operators  $\hat{O}_g$ , and define

$$\hat{P}^{(\mu)} = \frac{d_{\mu}}{|G|} \sum_g \chi^{(\mu)}(g)^* \hat{O}_g, \quad \hat{P}_{ij}^{(\mu)} = \frac{d_{\mu}}{|G|} \sum_g D_{ij}^{(\mu)}(g)^* \hat{O}_g. \quad (3.8)$$

**Theorem 3.4.**  $\hat{P}^{(\mu)}$  is the projector onto the subspace of  $L$  that transforms as  $\Gamma^{(\mu)}$  (all its copies). If  $\phi$  is any vector,  $\phi_i = \hat{P}_{ii}^{(\mu)} \phi$  (no sum) is either zero or the  $i$ -th partner of a basis transforming exactly with the matrices  $\mathbf{D}^{(\mu)}$ , and the other partners are  $\phi_j = \hat{P}_{ji}^{(\mu)} \phi$ .

The proof applies  $\hat{P}_{ij}^{(\mu)}$  to a symmetry-adapted basis and uses the great orthogonality theorem (2.5). In practice:

1. Take any trial function (an atomic orbital, a displacement of one atom).

2. Apply  $\hat{P}^{(\mu)}$  to get its  $\mu$  component. Repeat with other trial functions until you have  $m_\mu d_\mu$  independent results.
3. For multidimensional irreps, use  $\hat{P}_{ji}^{(\mu)}$  (needs the matrices) to generate partners transforming with the tabulated matrices. This is required for the invariant constructions of [Section 3.3](#).

Example 3.5 (Three hydrogen orbitals in  $C_{3v}$ ). Let  $h_1, h_2, h_3$  be the  $1s$  orbitals of the three hydrogens of  $\text{CH}_3\text{D}$ . The operations permute them:  $C_3^+ : h_1 \rightarrow h_2 \rightarrow h_3 \rightarrow h_1$ , and  $\sigma_{d1}$  fixes  $h_1$  and swaps  $h_2 \leftrightarrow h_3$ . The permutation character is  $(3, 0, 1)$ , so the span is  $A_1 \oplus E$ . Projecting  $h_1$ :

$$\begin{aligned}\hat{P}^{(A_1)} h_1 &= \frac{1}{6} (h_1 + h_2 + h_3 + h_1 + h_3 + h_2) = \frac{1}{3} (h_1 + h_2 + h_3), \\ \hat{P}^{(E)} h_1 &= \frac{2}{6} (2h_1 - h_2 - h_3 + 0) = \frac{1}{3} (2h_1 - h_2 - h_3).\end{aligned}$$

The second  $E$  partner is the orthogonal combination  $h_2 - h_3$  (odd under  $\sigma_{d1}$ , like  $y$ ). Normalised symmetry-adapted combinations:  $\frac{1}{\sqrt{3}}(h_1 + h_2 + h_3)$  for  $A_1$ , and  $(\frac{1}{\sqrt{6}}(2h_1 - h_2 - h_3), \frac{1}{\sqrt{2}}(h_2 - h_3))$  for  $E$ , which transform like  $(x, y)$ .

### 3.7 Product representations

The nine products  $x_i x'_j$  of the components of two vectors span the tensor product space  $\mathbb{R}^3 \otimes \mathbb{R}^3$ , and transform as  $(x_i x'_j)' = V_{ik} V_{jl} x_k x'_l$ . Treating the pair  $(ij)$  as a single index, the  $9 \times 9$  matrix

$$[\mathbf{V} \otimes \mathbf{V}]_{(ij),(kl)} = V_{ik} V_{jl} \quad (3.9)$$

(the *Kronecker product*) defines the *product representation*  $\Gamma_V \otimes \Gamma_V$ . More generally, if  $\Gamma_1$  acts on  $L_1$  and  $\Gamma_2$  on  $L_2$ , then  $\Gamma_1 \otimes \Gamma_2$  acts on  $L_1 \otimes L_2$  with matrices  $\mathbf{D}_1(g) \otimes \mathbf{D}_2(g)$ , of dimension  $d_1 d_2$ . Rank-2 tensors “belong to”  $\Gamma_V \otimes \Gamma_V$ . As with the direct sum, the matrices of the product representation are not products of the original matrices. Their traces, however, are products of the traces ([Chapter 4](#)):

$$\chi_{\Gamma_1 \otimes \Gamma_2}(g) = \chi_{\Gamma_1}(g) \chi_{\Gamma_2}(g). \quad (3.10)$$

### 3.8 Bilinear invariants: the general statement

Theorem 3.6. Let  $\Gamma^{(\mu)}$  and  $\Gamma^{(\nu)}$  be real orthogonal irreps with coordinates  $x_i$  and  $y_j$ . A bilinear form  $B = \sum_{ij} b_{ij} x_i y_j$  is invariant under  $G$  if and only if  $\mathbf{b} \mathbf{D}^{(\nu)}(g) = \mathbf{D}^{(\mu)}(g) \mathbf{b}$  for all  $g$ . By Schur’s lemma ([Section 2.6.1](#)):

1. if  $\Gamma^{(\mu)}$  and  $\Gamma^{(\nu)}$  are inequivalent, then  $\mathbf{b} = 0$ : no invariant;
2. if they are identical (same matrices), then  $\mathbf{b} = \lambda \mathbf{1}$ : the only invariant is the scalar product  $\sum_i x_i y_i$ .

*Proof.* Invariance of  $B$  under  $x \mapsto D^{(\mu)}x$ ,  $y \mapsto D^{(\nu)}y$  means  $D^{(\mu)\text{T}}\mathbf{b}D^{(\nu)} = \mathbf{b}$ , which for orthogonal matrices is the intertwining condition above. By Schur I,  $\mathbf{b} = 0$  for inequivalent irreps. For identical irreps Schur II gives  $\mathbf{b} = \lambda\mathbf{1}$ . For real irreps Schur II holds over  $\mathbb{R}$  as well, because a real matrix commuting with a real, absolutely irreducible representation is still a multiple of the identity.  $\square$

If the two irreps are equivalent but written in different bases,  $D^{(\nu)} = \mathbf{A}D^{(\mu)}\mathbf{A}^{-1}$ , there is still one invariant, but it is  $x^{\text{T}}\mathbf{A}^{-1}y$  rather than the scalar product. Always write repeated irreps with the same matrices.

The number of independent bilinear invariants in  $\Gamma_1 \otimes \Gamma_2$  is the multiplicity of the identity irrep,  $m_{A_1}(\Gamma_1 \otimes \Gamma_2) = \langle \chi_{\Gamma_1}^*, \chi_{\Gamma_2} \rangle$ , which by orthogonality equals 1 if  $\Gamma_2 \simeq \Gamma_1^*$  and 0 otherwise (Chapter 4).

## Exercises and solutions

Exercise 3.1. Use the column recipe to construct  $\mathbf{V}(\sigma_{d3})$  and  $\mathbf{V}(C_3^-)$  for  $C_{3v}$ , and check them against (1.4).

*Solution.*  $\sigma_{d3}$  is the mirror line at  $240^\circ$  (equivalently  $60^\circ$ ). A direction at angle  $\theta$  is reflected to  $2 \cdot 240^\circ - \theta$ :  $\vec{e}_x$  ( $0^\circ$ ) goes to  $480^\circ \equiv 120^\circ$ ,  $(-\frac{1}{2}, \frac{\sqrt{3}}{2}, 0)$ ;  $\vec{e}_y$  ( $90^\circ$ ) goes to  $390^\circ \equiv 30^\circ$ ,  $(\frac{\sqrt{3}}{2}, \frac{1}{2}, 0)$ ;  $\vec{e}_z$  is fixed. For  $C_3^-$ :  $\vec{e}_x \rightarrow (-\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0)$  (angle  $-120^\circ$ ),  $\vec{e}_y \rightarrow (\frac{\sqrt{3}}{2}, -\frac{1}{2}, 0)$  (angle  $-30^\circ$ ). Placing these as columns reproduces (1.4).

Exercise 3.2. Show that uniform translations and rigid rotations are orthogonal to each other and to all pure vibrations with respect to the inner product  $\langle \mathbf{u}, \mathbf{v} \rangle_M = \mathbf{u}^{\text{T}}\mathbf{M}\mathbf{v}$ .

*Solution.* Translation  $\vec{t}$  versus any  $\mathbf{v}$ :  $\sum_a m_a \vec{t} \cdot \vec{v}_a = \vec{t} \cdot \sum_a m_a \vec{v}_a$ , which vanishes for rotations ( $\sum_a m_a \vec{\varphi} \times \vec{R}_a^0 = \vec{\varphi} \times \sum_a m_a \vec{R}_a^0 = 0$ , the centre of mass being the origin) and for vibrations (first constraint). Rotation versus vibration:  $\sum_a m_a (\vec{\varphi} \times \vec{R}_a^0) \cdot \vec{u}_a = \vec{\varphi} \cdot \sum_a m_a \vec{R}_a^0 \times \vec{u}_a = 0$  (second constraint).

Exercise 3.3 (*conductivity tensor*). A crystal has point group  $C_{4v}$  (a four-fold axis along  $z$  and four vertical mirrors). Using quadratic invariants, find the most general symmetric conductivity tensor  $\sigma_{ij}$ . Compare with  $C_{3v}$  and  $T_d$ .

*Solution.* In  $C_{4v}$ ,  $\Gamma_V = A_1(z) \oplus E(x, y)$ , both real. The quadratic invariants are  $E_z^2$  and  $E_x^2 + E_y^2$ , hence  $\boldsymbol{\sigma} = \text{diag}(\sigma_\perp, \sigma_\perp, \sigma_\parallel)$  with two constants, exactly as for  $C_{3v}$  (uniaxial crystals). For  $T_d$  (cubic),  $\boldsymbol{\sigma} = \sigma\mathbf{1}$  is isotropic.

Exercise 3.4. For  $\text{NH}_3$ ,  $\Gamma_{\text{vib}} = 2A_1 \oplus 2E$  (Exercise 2.2). How many independent constants does the most general harmonic potential energy contain? Describe the block structure of the dynamical matrix in symmetry-adapted coordinates.

*Solution.*  $\Gamma_{\text{vib}} = 2A_1 \oplus 2E$ . The  $A_1$  part gives a  $2 \times 2$  symmetric block (3 constants: two squares and one cross term); the  $E$  part gives a  $2 \times 2$  block repeated twice (3 constants). Six constants in total, instead of the 21 of a general symmetric  $6 \times 6$  matrix. Two non-degenerate and two doubly degenerate frequencies.

Exercise 3.5. For  $C_{3v}$  compute the character of  $E \otimes E$  and decompose it. How many independent bilinear invariants can be built from two  $E$  doublets  $(x_1, x_2)$  and  $(y_1, y_2)$ ? Write them down.

*Solution.*  $\chi_{E \otimes E} = (4, 1, 0)$ , so  $m_{A_1} = \frac{1}{6}(4 + 2) = 1$ ,  $m_{A_2} = \frac{1}{6}(4 + 2) = 1$ ,  $m_E = \frac{1}{6}(8 - 2) = 1$ :  $E \otimes E = A_1 \oplus A_2 \oplus E$ . There is one bilinear invariant,  $x_1y_1 + x_2y_2$  (both doublets written with the same real matrices). The  $A_2$  combination  $x_1y_2 - x_2y_1$  is invariant under  $C_3$  but changes sign under the mirrors. The  $E$  combination is  $(x_1y_1 - x_2y_2, -x_1y_2 - x_2y_1)$ .

## Chapter 4

# Operations with representations II: spectroscopy and selection rules

### Video: Lecture 4

Characters of product representations [L4 @ 2:03](#) · microwave and infrared spectroscopy (blackboard) [L4 @ 10:32](#) · counting bilinear invariants [L4 @ 23:44](#) · microwave activity [L4 @ 28:55](#) · infrared activity [L4 @ 32:02](#) · symmetry-adapted vs normal coordinates [L4 @ 42:37](#) · symmetric tensors and the symmetric square [L4 @ 46:56](#) · Raman scattering (blackboard) [L4 @ 55:04](#) · Raman activity [L4 @ 76:10](#)

### 4.1 Characters of product representations

Let  $\Gamma_1$  and  $\Gamma_2$  have matrices  $\mathbf{D}_1(g)$  and  $\mathbf{D}_2(g)$ . The product representation acts on objects  $t_{ij}$  with one index of each kind:

$$t'_{ij} = [\mathbf{D}_1(g)]_{ik} [\mathbf{D}_2(g)]_{jl} t_{kl} \equiv [\mathbf{D}_1 \otimes \mathbf{D}_2]_{(ij),(kl)} t_{kl}.$$

The trace sets the double index  $(kl)$  equal to  $(ij)$  and sums:

$$\chi_{\Gamma_1 \otimes \Gamma_2}(g) = \sum_{i,j} [\mathbf{D}_1(g)]_{ii} [\mathbf{D}_2(g)]_{jj} = \chi_1(g) \chi_2(g).$$

So, just as characters of direct sums add, characters of products multiply.

Example 4.1 ( $\Gamma_V \otimes \Gamma_V$  for  $C_{3v}$ ).  $\chi_V = (3, 0, 1)$ , hence  $\chi_{\Gamma_V \otimes \Gamma_V} = (9, 0, 1)$  and

$$m_{A_1} = \frac{1}{6}(9 + 0 + 3) = 2, \quad m_{A_2} = \frac{1}{6}(9 + 0 - 3) = 1, \quad m_E = \frac{1}{6}(18 + 0 + 0) = 3,$$

so  $\Gamma_V \otimes \Gamma_V = 2A_1 \oplus A_2 \oplus 3E$  ( $2 + 1 + 6 = 9$ ).

## 4.2 Counting bilinear invariants

A bilinear quantity  $\sum b_{ij}x_iy_j$  built from  $x \in \Gamma_1$  and  $y \in \Gamma_2$  belongs to  $\Gamma_1 \otimes \Gamma_2$ . It is invariant if it lies in a copy of the identity representation  $A_1$  (all characters 1). Therefore

$$\begin{aligned} \#\{\text{independent bilinear invariants}\} &= m_{A_1}(\Gamma_1 \otimes \Gamma_2) \\ &= \frac{1}{|G|} \sum_g \chi_1(g)\chi_2(g) = \langle \chi_1^*, \chi_2 \rangle. \end{aligned} \quad (4.1)$$

If  $\Gamma_1 = \Gamma^{(\mu)}$  and  $\Gamma_2 = \Gamma^{(\nu)}$  are irreps, orthogonality gives  $m_{A_1} = 1$  when  $\Gamma^{(\nu)}$  is equivalent to the complex conjugate of  $\Gamma^{(\mu)}$  and 0 otherwise. For real irreps ( $\Gamma^{(\mu)*} = \Gamma^{(\mu)}$ ): *an invariant can only be formed from two copies of the same irrep*, and then there is exactly one, the scalar product. This proves the recipe of [Section 3.3](#).

## 4.3 Symmetric and antisymmetric squares

A *symmetric* rank-2 tensor ( $t_{ij} = t_{ji}$ , like the polarizability) lives in the 6-dimensional subspace of  $\mathbb{R}^3 \otimes \mathbb{R}^3$  spanned by the symmetric products  $x^2, y^2, z^2, xy, yz, zx$ . It carries the *symmetric square*  $[\Gamma^2]$ . The antisymmetric tensors ( $t_{ij} = -t_{ji}$ , 3 components) carry the *antisymmetric square*  $\{\Gamma^2\}$ . Their characters are

$$\chi_{[\Gamma^2]}(g) = \frac{1}{2}[\chi(g)^2 + \chi(g^2)], \quad \chi_{\{\Gamma^2\}}(g) = \frac{1}{2}[\chi(g)^2 - \chi(g^2)]. \quad (4.2)$$

*Proof.* Diagonalise  $\mathbf{D}(g)$  (possible for a unitary matrix) with eigenvalues  $\lambda_i$ . The symmetric products  $e_i e_j$  ( $i \leq j$ ) are eigenvectors with eigenvalues  $\lambda_i \lambda_j$ , so  $\chi_{[\Gamma^2]} = \sum_{i \leq j} \lambda_i \lambda_j = \frac{1}{2}[(\sum \lambda_i)^2 + \sum \lambda_i^2]$ , and  $\sum_i \lambda_i^2 = \text{tr } \mathbf{D}(g)^2 = \chi(g^2)$ . The antisymmetric case is the same with  $i < j$ .  $\square$

To use (4.2) you need, for each class, the class of  $g^2$ . For  $C_{3v}$ :  $E^2 = E$ ,  $(C_3^\pm)^2 = C_3^\mp$  (same class),  $\sigma^2 = E$ . Hence

|                         |     |        |             |            |                                                                       |
|-------------------------|-----|--------|-------------|------------|-----------------------------------------------------------------------|
| $C_{3v}$                | $E$ | $2C_3$ | $3\sigma_d$ |            |                                                                       |
| $\chi_V(g)$             | 3   | 0      | 1           |            |                                                                       |
| $\chi_V(g^2)$           | 3   | 0      | 3           | $\implies$ | $[\Gamma_V^2] = 2A_1 \oplus 2E, \quad \{\Gamma_V^2\} = A_2 \oplus E.$ |
| $\chi_{[\Gamma_V^2]}$   | 6   | 0      | 2           |            |                                                                       |
| $\chi_{\{\Gamma_V^2\}}$ | 3   | 0      | -1          |            |                                                                       |

Notice that  $\{\Gamma_V^2\} = \Gamma_R$ : an antisymmetric tensor  $t_{ij}$  is equivalent to the axial vector  $\frac{1}{2}\varepsilon_{ijk}t_{jk}$ , just as  $\vec{r} \times \vec{p}$  is built from the antisymmetric products  $x_i p_j - x_j p_i$  ([Exercise 4.1](#)).

For  $T_d$  ( $C_2^2 = E$ ,  $S_4^2 = C_2$ ,  $\sigma_d^2 = E$ ) one finds  $[\Gamma_V^2] = A_1 \oplus E \oplus T_2$  and  $\{\Gamma_V^2\} = T_1 = \Gamma_R$ .

## 4.4 A little spectroscopy

We need only a qualitative picture of three kinds of molecular spectroscopy.

**Electric-dipole coupling.** A molecule is a neutral cloud of charges. To lowest order in the multipole expansion it couples to a (long-wavelength) electromagnetic field through its dipole moment,  $H_{\text{int}} = -\vec{p} \cdot \vec{E}$ .

**Microwave (rotational) spectroscopy.** A molecule with a *permanent* dipole  $\vec{p}_0$  that rotates produces an oscillating dipole and can absorb or emit at its rotation frequencies. For a diatomic molecule  $H = \hat{L}^2/2I$  gives levels  $\hbar^2 l(l+1)/2I$  and the selection rule  $\Delta l = \pm 1$  produces equally spaced lines in the microwave range. HCl is active; H<sub>2</sub> has no dipole and is inactive.

**Infrared (vibrational) spectroscopy.** A vibration that makes the dipole oscillate,  $\delta\vec{p}(t) \neq 0$ , absorbs infrared light at its frequency. The breathing mode of BF<sub>3</sub> keeps the dipole zero and is inactive; a mode that pushes the boron out of the plane against the fluorines creates an oscillating dipole and is active.

**Raman scattering.** Light of frequency  $\omega_0$  far from any electronic resonance induces a dipole  $\vec{p} = \alpha\vec{E}$  in the electron cloud, which re-radiates at  $\omega_0$  (Rayleigh scattering, the blue of the sky). By the Born–Oppenheimer approximation the electron cloud follows the nuclei. If the molecule rotates or vibrates,  $\alpha(t)$  oscillates at the rotation or vibration frequency  $\Omega$ , and the product  $\alpha(t)\vec{E}_0 \cos \omega_0 t$  contains the frequencies  $\omega_0 \pm \Omega$ : weak side lines. Quantum mechanically the photon loses energy to the molecule (Stokes line,  $\omega_0 - \Omega$ ) or gains it (anti-Stokes,  $\omega_0 + \Omega$ ). The anti-Stokes line needs a thermally excited vibration and is much weaker at room temperature.

## 4.5 Selection rules from invariants

In each case we write the most general interaction energy allowed by symmetry and ask whether it can be non-zero. The strategy is always the same: the energy must be an *invariant*, and (4.1) tells when an invariant exists.

**Microwave activity.** A permanent dipole is a property of the molecule at rest, hence invariant under  $G$ . A non-zero invariant vector exists only if some component of a vector is invariant:

## Key result

A molecule is microwave active  $\iff A_1 \subset \Gamma_V$ . The permanent dipole lies in the  $A_1$  subspace.

CH<sub>4</sub>:  $\Gamma_V = T_2$ , inactive, with no need to know anything about its charge distribution. CH<sub>3</sub>D:  $\Gamma_V = A_1 \oplus E$ , active, with the dipole along the three-fold axis.

Infrared activity. Expand the dipole in the normal coordinates,  $\delta p_i = \sum_{\alpha} c_{i\alpha} Q_{\alpha}$ . The interaction  $H_{IR} = -\sum_{i,\alpha} c_{i\alpha} E_i Q_{\alpha}$  is a bilinear invariant in  $\Gamma_V \otimes \Gamma_{vib}$ . A mode transforming as  $\Gamma^{(\mu)}$  appears in  $H_{IR}$  only if  $\Gamma^{(\mu)}$  is also contained in  $\Gamma_V$  (for real irreps):

## Key result

A vibrational mode of symmetry  $\Gamma^{(\mu)}$  is infrared active  $\iff \Gamma^{(\mu)} \subset \Gamma_V$ .

Rotational Raman activity. A rotating molecule modulates  $\alpha$  only if  $\alpha$  is anisotropic,  $\alpha \neq a\mathbf{1}$ . The number of independent invariant symmetric tensors is  $m_{A_1}([\Gamma_V^2])$ ; one of them is always  $\mathbf{1}$ . Hence:

## Key result

Rotational Raman activity  $\iff m_{A_1}([\Gamma_V^2]) \geq 2$ . For point groups this happens exactly when  $\Gamma_V$  is reducible.

Vibrational Raman activity. To first order in the normal coordinates,  $\delta\alpha_{ij} = \sum_{\alpha} a_{ij\alpha} Q_{\alpha}$ , so the Raman Hamiltonian  $H_R = -\frac{1}{2} \sum a_{ij\alpha} E_i E_j Q_{\alpha}$  is a trilinear invariant in  $[\Gamma_V^2] \otimes \Gamma_{vib}$ :

## Key result

A vibrational mode of symmetry  $\Gamma^{(\mu)}$  is Raman active  $\iff \Gamma^{(\mu)} \subset [\Gamma_V^2]$ .

## 4.5.1 Methane and deuterated methane

|                                | $\Gamma_{vib}$             | $\Gamma_V$     | $[\Gamma_V^2]$            |
|--------------------------------|----------------------------|----------------|---------------------------|
| CH <sub>4</sub> ( $T_d$ )      | $A_1 \oplus E \oplus 2T_2$ | $T_2$          | $A_1 \oplus E \oplus T_2$ |
| CH <sub>3</sub> D ( $C_{3v}$ ) | $3A_1 \oplus 3E$           | $A_1 \oplus E$ | $2A_1 \oplus 2E$          |

- CH<sub>4</sub>: microwave inactive ( $A_1 \not\subset \Gamma_V$ ); only the two  $T_2$  frequencies are infrared active; all four frequencies ( $A_1, E, T_2, T_2$ ) are Raman active; no rotational

Raman spectrum ( $\Gamma_V$  irreducible,  $\alpha$  isotropic).

- $\text{CH}_3\text{D}$ : microwave active; all six frequencies are infrared and Raman active; rotational Raman active.

This completes the list of predictions announced in [Section 1.2](#).

### Beyond the lectures: the general selection rule

In quantum mechanics the same reasoning is phrased with matrix elements. Let  $|\psi_i\rangle$  belong to  $\Gamma_i$ , let the components of an operator  $\hat{O}$  form a set transforming as  $\Gamma_O$ , and let  $|\psi_f\rangle$  belong to  $\Gamma_f$ . Then

$$\langle\psi_f|\hat{O}|\psi_i\rangle \neq 0 \quad \text{only if} \quad A_1 \subset \Gamma_f^* \otimes \Gamma_O \otimes \Gamma_i \quad \Longleftrightarrow \quad \Gamma_f \subset \Gamma_O \otimes \Gamma_i.$$

The matrix element is invariant under the group (it is a number), so it can only pick up the identity component of the triple product. For electric-dipole transitions  $\Gamma_O = \Gamma_V$ ; for magnetic-dipole transitions  $\Gamma_O = \Gamma_R$ ; for Raman and quadrupole transitions  $\Gamma_O = [\Gamma_V^2]$ . The Wigner–Eckart theorem ([Section 5.5.5](#)) refines the statement: all non-zero matrix elements within a pair of multiplets are fixed up to one constant per copy of  $\Gamma_f$  in  $\Gamma_O \otimes \Gamma_i$ .

## Exercises and solutions

**Exercise 4.1.** Show from (4.2) and (2.7) that  $\chi_{\{\Gamma_V^2\}}(g) = \chi_R(g)$  for every rotation  $C(\theta)$  and every rotoreflection  $S(\theta)$ . Conclude that antisymmetric rank-2 tensors transform as axial vectors in every point group.

*Solution.* Rotation:  $\chi = 1 + 2c$  ( $c = \cos \theta$ ) and  $g^2 = C(2\theta)$  has  $\chi(g^2) = 1 + 2 \cos 2\theta = 4c^2 - 1$ . Then  $\frac{1}{2}[(1+2c)^2 - (4c^2 - 1)] = 1 + 2c = \chi_V = \chi_R$ . Rotoreflexion:  $\chi = -1 + 2c$  and  $S(\theta)^2 = C(2\theta)$ , so  $\frac{1}{2}[(2c-1)^2 - (4c^2 - 1)] = 1 - 2c = -\chi_V = \chi_R$ . The characters agree on every element, so  $\{\Gamma_V^2\} \cong \Gamma_R$  by [Theorem 2.1](#).

**Exercise 4.2** ( $\text{BF}_3$ ). Using [Exercise 2.4](#), decide which vibrational frequencies of  $\text{BF}_3$  are infrared active and which are Raman active. Is the molecule microwave active? Does it show a rotational Raman spectrum?

*Solution.* The squares in  $D_{3h}$ :  $C_2'^2 = \sigma_h^2 = \sigma_v^2 = E$ ,  $C_3^2 \in \{2C_3\}$ ,  $S_3^2 \in \{2C_3\}$ . So  $\chi_V(g^2) = (3, 0, 3, 3, 0, 3)$ ,  $\chi_V^2 = (9, 0, 1, 1, 4, 1)$  and  $\chi_{[\Gamma_V^2]} = (6, 0, 2, 2, 2, 2)$ , giving  $[\Gamma_V^2] = 2A_1' \oplus E' \oplus E''$ . With  $\Gamma_{\text{vib}} = A_1' \oplus 2E' \oplus A_2''$  and  $\Gamma_V = E' \oplus A_2''$ :

- $A_1'$  (breathing): Raman only;
- $A_2''$  (out-of-plane): infrared only;
- the two  $E'$  modes: infrared and Raman.

Microwave inactive ( $A_1' \not\subset \Gamma_V$ : no permanent dipole). Rotational Raman active ( $A_1'$  appears twice in  $[\Gamma_V^2]$ : anisotropic polarizability).

Exercise 4.3 ( $\text{H}_2\text{O}$ ). With the axes of [Exercise 2.3](#), find the irreps of  $x$ ,  $y$ ,  $z$  and of the six quadratic functions. Which vibrations of water are infrared active, which are Raman active?

*Solution.*  $z \in A_1$ ,  $x \in B_1$ ,  $y \in B_2$ ;  $x^2, y^2, z^2 \in A_1$ ,  $xy \in A_2$ ,  $xz \in B_1$ ,  $yz \in B_2$ . So  $\Gamma_V = A_1 \oplus B_1 \oplus B_2$  and  $[\Gamma_V^2] = 3A_1 \oplus A_2 \oplus B_1 \oplus B_2$ . All three modes ( $2A_1 \oplus B_2$ ) are both infrared and Raman active.

Exercise 4.4 (*mutual exclusion*). Suppose the point group contains the inversion  $I$ . Irreps are then even (g,  $\chi(I) = +d$ ) or odd (u,  $\chi(I) = -d$ ). Show that  $\Gamma_V$  contains only odd irreps and  $[\Gamma_V^2]$  only even ones. Conclude that no vibration of a centrosymmetric molecule is both infrared and Raman active.

*Solution.*  $V(I) = -1$ , so every irrep contained in  $\Gamma_V$  has  $D(I) = -1$ : odd. On symmetric tensors  $I$  acts as  $(-1)(-1) = +1$ , so every irrep in  $[\Gamma_V^2]$  is even. An infrared-active mode must be odd, a Raman-active one even; no mode is both. ( $\text{CO}_2$  is the textbook example.)

Exercise 4.5 (*electronic transitions*). An electronic state of a  $C_{3v}$  molecule has symmetry  $A_1$ . To which states ( $A_1$ ,  $A_2$ ,  $E$ ) can it be excited by electric-dipole absorption, and with which light polarisation? Answer the same question for an initial state of symmetry  $E$ .

*Solution.* From  $A_1$ :  $\Gamma_f \subset \Gamma_V \otimes A_1 = \Gamma_V = A_1 \oplus E$ . Transitions to  $A_1$  need  $z$ -polarised light, to  $E$  in-plane polarisation;  $A_1 \rightarrow A_2$  is forbidden. From  $E$ :  $z$  polarisation gives  $A_1 \otimes E = E$  (only  $E \rightarrow E$ ), in-plane polarisation gives  $E \otimes E = A_1 \oplus A_2 \oplus E$  (all final states allowed).

# Chapter 5

## Spin, double groups and crystal-field splitting

Video: Lecture 5 (no captions; times read off the slides)

Subduced representations,  $\text{CH}_4 \rightarrow \text{CH}_3\text{D}$ : first part of the lecture · irreps of  $\text{SO}(3)$  [L5 @ 49:00](#) · splitting of atomic orbitals, weak spin-orbit [L5 @ 63:10](#) · splitting of multiplets, strong spin-orbit [L5 @ 74:50](#) · double groups [L5 @ 81:50](#) · reality of a representation [L5 @ 93:40](#)

### 5.1 Subgroups and subduction

Deuterating one hydrogen of methane lowers the symmetry from  $T_d$  to its subgroup  $C_{3v}$ . A representation  $\Gamma$  of  $G$  automatically gives a representation of any subgroup  $H \subset G$ : just keep the matrices of the elements of  $H$ . This is the *subduced* (or restricted) representation  $\Gamma \downarrow H$ . Its character is the restriction of  $\chi_\Gamma$  to  $H$ . An irrep of  $G$  is in general reducible over  $H$ , and its decomposition is given by the reduction formula in [H](#).

Take  $C_{3v} \subset T_d$  with the three-fold axis along  $[111]$  and one of the six mirrors of  $T_d$  containing that axis. The classes  $E$ ,  $2C_3$ ,  $3\sigma_d$  of  $C_{3v}$  lie in the classes  $E$ ,  $8C_3$ ,  $6\sigma_d$  of  $T_d$ . Reading the columns of [\(2.2\)](#):

| $T_d \downarrow C_{3v}$ | $E$ | $2C_3$ | $3\sigma_d$ |                |
|-------------------------|-----|--------|-------------|----------------|
| $A_1$                   | 1   | 1      | 1           | $A_1$          |
| $A_2$                   | 1   | 1      | -1          | $A_2$          |
| $E$                     | 2   | -1     | 0           | $E$            |
| $T_1$                   | 3   | 0      | -1          | $A_2 \oplus E$ |
| $T_2$                   | 3   | 0      | 1           | $A_1 \oplus E$ |

(5.1)

Such tables are called *correlation* or *subduction* tables. Applied to methane:

$$\Gamma_{\text{vib}}(\text{CH}_4) \downarrow C_{3v} = A_1 \oplus E \oplus 2(A_1 \oplus E) = 3A_1 \oplus 3E = \Gamma_{\text{vib}}(\text{CH}_3\text{D}).$$

Each triply degenerate frequency splits into one non-degenerate and one doubly degenerate frequency, with no new computation. The general lesson: *lowering the symmetry can only split levels*, and subduction tells how.

## 5.2 The rotation group

### 5.2.1 Irreps of $\text{SO}(3)$

The spherical harmonics  $Y_l^m$ ,  $m = -l, \dots, l$ , span a  $(2l+1)$ -dimensional irrep  $D_l$  of  $\text{SO}(3)$  ( $l = 0, 1, 2, \dots$ ). Every rotation is conjugate to a rotation by the same angle about  $z$ , so the character depends only on the angle  $\theta$ . Using  $\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$ ,

$$\chi_l(\theta) = \sum_{m=-l}^l e^{-im\theta} = \frac{e^{i(l+1/2)\theta} - e^{-i(l+1/2)\theta}}{e^{i\theta/2} - e^{-i\theta/2}} = \frac{\sin[(l + \frac{1}{2})\theta]}{\sin(\theta/2)}, \quad (5.2)$$

with  $\chi_l(0) = 2l + 1$ . (The sign convention of the exponent does not matter: the sum is symmetric in  $m \rightarrow -m$ .) Useful values:  $\chi_l(\pi) = (-1)^l$ ;  $\chi_l(2\pi/3) = 1, 0, -1$  for  $l \equiv 0, 1, 2 \pmod{3}$ .

### 5.2.2 $\text{O}(3)$ and parity

The full orthogonal group is  $\text{O}(3) = \text{SO}(3) \times \{E, I\}$ , and the inversion commutes with everything. Each irrep of  $\text{SO}(3)$  gives two irreps of  $\text{O}(3)$ ,  $D_l^+$  and  $D_l^-$ , with  $D(I) = \pm 1$ . An improper element  $IC(\theta)$  has character  $\pm\chi_l(\theta)$ . Atomic orbitals have the parity of  $Y_l^m$ ,  $(-1)^l$ :  $s = D_0^+$ ,  $p = D_1^-$ ,  $d = D_2^+$ ,  $f = D_3^-$ . A mirror is  $\sigma = IC_2$ , so  $\chi_{D_l^{(-1)^l}}(\sigma) = (-1)^l \chi_l(\pi) = 1$  for every orbital.

### 5.2.3 Direct products ★

$\text{O}(3) = \text{SO}(3) \times \{E, I\}$  is an instance of a general construction. If  $G$  has two normal subgroups (Section 2.3)  $H, K$  with  $H \cap K = \{e\}$  and  $HK = G$ , then  $G \cong H \times K$  and elements of  $H$  commute with those of  $K$ . The irreps of  $H \times K$  are products of irreps of the factors,  $\Gamma_H^{(\mu)} \otimes \Gamma_K^{(\nu)}$ , with characters  $\chi^{(\mu)}(h)\chi^{(\nu)}(k)$ . Other examples:  $O_h = O \times C_i$  (Exercise 5.3) and  $D_{6h} = C_{6v} \times \{E, \sigma_h\}$  (Chapter 7). Tables for such groups are often given only for one factor; the irreps of the full group are the g (even) and u (odd) copies.

### 5.3 Crystal-field splitting (weak spin-orbit coupling)

An atom in a molecule or crystal no longer feels a spherically symmetric potential, only one invariant under its *local* point group  $G$  (its site-symmetry group, [Section 7.4](#)). If spin-orbit coupling is weak, an  $l$  level splits according to  $D_l^\pm \downarrow G$ . For  $C_{3v}$ :

| $C_{3v}$ | $E$ | $2C_3$ | $3\sigma_d$ |                                           |
|----------|-----|--------|-------------|-------------------------------------------|
| $D_0^+$  | 1   | 1      | 1           | $s \rightarrow a_1$                       |
| $D_1^-$  | 3   | 0      | 1           | $p \rightarrow a_1 \oplus e$              |
| $D_2^+$  | 5   | -1     | 1           | $d \rightarrow a_1 \oplus 2e$             |
| $D_3^-$  | 7   | 1      | 1           | $f \rightarrow 2a_1 \oplus a_2 \oplus 2e$ |

(5.3)

Lower-case labels are customary for one-electron orbitals. The most famous example is a  $d$  ion in an octahedral environment ( $O_h$ ):  $d \rightarrow e_g \oplus t_{2g}$ , the splitting  $\Delta_o$  of ligand-field theory ([Exercise 5.3](#)). In a tetrahedral environment  $d \rightarrow e \oplus t_2$ .

**Orbital mixing.** Once the symmetry is lowered, orbitals from different  $l$  that belong to the same irrep of  $G$  may mix. In  $C_{3v}$ ,  $s$ ,  $p_z$  and  $d_{z^2}$  are all  $a_1$ . Strictly speaking there is no “ $p_z$  orbital” in a crystal, only an  $a_1$  orbital that resembles one. This matters when one builds tight-binding models ([Section 7.4](#)).

### 5.4 Half-integer angular momentum

With spin, the relevant angular momenta  $j$  can be half-integers. Formula (5.2) still gives the character of  $D_j$  with  $l \rightarrow j$ , but for half-integer  $j$  it has a disturbing property:

$$\chi_j(\theta + 2\pi) = -\chi_j(\theta).$$

A rotation by  $2\pi$  multiplies a spinor by  $-1$ . These are the irreps of  $SU(2)$ , which covers  $SO(3)$  twice ([Section 5.5](#)); as “representations” of  $SO(3)$  they are *double valued*.

Try anyway to split a  $j = \frac{1}{2}$  or  $j = \frac{3}{2}$  level in a  $C_{3v}$  environment (strong spin-orbit coupling, so that  $j$  is a good quantum number of the free atom). For improper elements, the inversion acts trivially on spin and  $\sigma = IC_2$ , so  $\chi(\sigma) = \pm\chi_j(\pi) = 0$  for half-integer  $j$ :

| $C_{3v}$    | $E$ | $2C_3$ | $3\sigma_d$ |                                                                            |
|-------------|-----|--------|-------------|----------------------------------------------------------------------------|
| $D_{1/2}^-$ | 2   | 1      | 0           | $D_{1/2}^- : m_{A_1} = m_{A_2} = \frac{2}{3}, m_E = \frac{1}{3} \quad (!)$ |
| $D_{3/2}^-$ | 4   | -1     | 0           | $D_{3/2}^- : m_{A_1} = m_{A_2} = \frac{1}{3}, m_E = \frac{5}{3} \quad (!)$ |

Fractional multiplicities: the characters of spinors are not combinations of the characters of  $C_{3v}$ . The group is too small.

## 5.5 The rotation group and SU(2) ★

The lecture only needs the characters of the rotation group. A theorist also needs the relation between SO(3) and SU(2), the Clebsch–Gordan series and the Wigner–Eckart theorem; they are summarised here.

### 5.5.1 Rotations

A rotation  $R(\vec{n}, \theta)$  by the angle  $\theta \in [0, \pi]$  about the unit axis  $\vec{n}$  acts on vectors as  $R\vec{v} = \cos \theta \vec{v} + \sin \theta \vec{n} \times \vec{v} + (1 - \cos \theta)(\vec{n} \cdot \vec{v})\vec{n}$ . Since  $R'R(\vec{n}, \theta)R'^{-1} = R(R'\vec{n}, \theta)$ , all rotations by the same angle are conjugate: the classes of SO(3) are labelled by  $\theta$ , and characters are functions of  $\theta$  only. On states, rotations act as  $\hat{U}(\vec{n}, \theta) = \exp(-i\theta \vec{n} \cdot \hat{\vec{J}}/\hbar)$ , and the generators satisfy  $[\hat{J}_x, \hat{J}_y] = i\hbar \hat{J}_z$  and cyclic permutations.

### 5.5.2 SU(2) covers SO(3) twice

Every  $U \in \text{SU}(2)$  can be written

$$U(\vec{n}, \theta) = \exp(-i\frac{\theta}{2} \vec{n} \cdot \vec{\sigma}) = \cos \frac{\theta}{2} \mathbf{1} - i \sin \frac{\theta}{2} \vec{n} \cdot \vec{\sigma}, \quad 0 \leq \theta \leq 2\pi. \quad (5.4)$$

Acting by conjugation on the traceless hermitian matrices  $\vec{v} \cdot \vec{\sigma}$  (which form a copy of  $\mathbb{R}^3$ ),  $U(\vec{v} \cdot \vec{\sigma})U^\dagger = (R\vec{v}) \cdot \vec{\sigma}$  defines a rotation  $R(U)$ , with  $R_{ij} = \frac{1}{2} \text{tr}(\sigma_i U \sigma_j U^\dagger)$ . The map  $U \mapsto R(U)$  is a homomorphism onto SO(3) with kernel  $\{\mathbf{1}, -\mathbf{1}\}$  (Section 2.3):  $U(\vec{n}, \theta)$  and  $U(\vec{n}, \theta + 2\pi) = -U(\vec{n}, \theta)$  give the same rotation. In particular

$$U(\vec{n}, 2\pi) = -\mathbf{1} : \quad (5.5)$$

a rotation by  $2\pi$  multiplies a spinor by  $-1$ , and only a rotation by  $4\pi$  is the identity. Topologically, SU(2) is the 3-sphere (simply connected), and  $\text{SO}(3) = \text{SU}(2)/\{\pm \mathbf{1}\}$  is doubly connected. A closed path of rotations from 0 to  $2\pi$  cannot be shrunk to a point; the same path traversed twice can. This is the “belt trick”.

### 5.5.3 Irreps and characters

The irreps of SU(2) are  $D_j$ ,  $j = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ , of dimension  $2j+1$ , with basis  $|j, m\rangle$  and  $D_j(U(\vec{e}_z, \theta))|j, m\rangle = e^{-im\theta}|j, m\rangle$ . Their characters are (5.2) with  $l \rightarrow j$ , and

$$\chi_j(\theta + 2\pi) = (-1)^{2j} \chi_j(\theta). \quad (5.6)$$

For integer  $j$ ,  $D_j(-\mathbf{1}) = +\mathbf{1}$  and  $D_j$  is a genuine representation of SO(3) (orbital angular momentum,  $j = l$ ). For half-integer  $j$ ,  $D_j(-\mathbf{1}) = -\mathbf{1}$ : a representation of SU(2) but only a “double-valued” one of SO(3).

### 5.5.4 Clebsch–Gordan series

The product  $D_{j_1} \otimes D_{j_2}$  has eigenvalues  $e^{-i(m_1+m_2)\theta}$  under  $z$  rotations. Collecting the values of  $m_1 + m_2$  gives

$$\chi_{j_1} \chi_{j_2} = \sum_{j=|j_1-j_2|}^{j_1+j_2} \chi_j, \quad D_{j_1} \otimes D_{j_2} = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} D_j. \quad (5.7)$$

For example, a  $p$  electron with spin:  $D_1 \otimes D_{1/2} = D_{3/2} \oplus D_{1/2}$ . This is the origin of the  $j = \frac{3}{2}$ ,  $j = \frac{1}{2}$  multiplets split by spin–orbit coupling. The change of basis from  $|j_1 m_1\rangle |j_2 m_2\rangle$  to  $|j m\rangle$  is given by the Clebsch–Gordan coefficients  $\langle j_1 m_1; j_2 m_2 | j m \rangle$ .

### 5.5.5 The Wigner–Eckart theorem

A set of operators  $T_q^{(k)}$ ,  $q = -k, \dots, k$ , transforming like  $|k, q\rangle$  under rotations is an *irreducible tensor operator* of rank  $k$ . Then

$$\langle j, m | T_q^{(k)} | j', m' \rangle = \langle j' m'; k q | j m \rangle \frac{\langle j || T^{(k)} || j' \rangle}{\sqrt{2j+1}}. \quad (5.8)$$

All  $(2j+1)(2k+1)(2j'+1)$  matrix elements are fixed by a single reduced matrix element. The Clebsch–Gordan coefficient gives the selection rules  $m = m' + q$  and  $|j - j'| \leq k \leq j + j'$ . The dipole rule  $\Delta l = \pm 1$  follows from  $k = 1$  together with parity.

For a finite group the statement is the same: matrix elements  $\langle \Gamma_f, i | \hat{O}_a^\Gamma | \Gamma_i, j \rangle$  are products of coupling coefficients of  $G$  and reduced matrix elements, one reduced element for each copy of  $\Gamma_f$  in  $\Gamma_O \otimes \Gamma_i$  (Section 4.5).

## 5.6 Double groups

The cure is to treat the rotation by  $2\pi$  as a group element of its own,  $\bar{E} \equiv C(2\pi) \neq E$ , with  $\bar{E}^2 = E$ . The *double group*  $G^D$  contains, for each  $g \in G$ , the two elements  $g$  and  $\bar{E}g$ , so  $|G^D| = 2|G|$ . Concretely,  $G^D$  is the set of  $SU(2)$  matrices (times the trivial action of  $I$ ) that project onto  $G$ . In it:

- $C_n^n = \bar{E}$  rather than  $E$  (a rotation by  $2\pi$ );
- $\sigma^2 = \bar{E}$ , since  $\sigma = IC_2$  and  $C_2^2 = \bar{E}$ ;
- $g$  and  $\bar{E}g$  usually fall in different classes (the precise rules, due to Opechowski, are in Bradley–Cracknell).

Its irreps are of two kinds: the *single-valued* irreps of  $G$  (with  $\mathbf{D}(\bar{E}) = +\mathbf{1}$ ) and new *double-valued* irreps with  $\mathbf{D}(\bar{E}) = -\mathbf{1}$ , which describe systems with an odd number of electrons (half-integer spin).

Example 5.1 (The double group of  $C_{3v}$ ). Order 12, six classes, six irreps ( $1 + 1 + 4 + 1 + 1 + 4 = 12$ ):

| $C_{3v}^D$    | $E$ | $\bar{E}$ | $2C_3$ | $2\bar{C}_3$ | $3\sigma_d$ | $3\bar{\sigma}_d$ | type |
|---------------|-----|-----------|--------|--------------|-------------|-------------------|------|
| $A_1$         | 1   | 1         | 1      | 1            | 1           | 1                 | 1    |
| $A_2$         | 1   | 1         | 1      | 1            | -1          | -1                | 1    |
| $E$           | 2   | 2         | -1     | -1           | 0           | 0                 | 1    |
| ${}^1\bar{E}$ | 1   | -1        | -1     | 1            | i           | -i                | 2    |
| ${}^2\bar{E}$ | 1   | -1        | -1     | 1            | -i          | i                 | 2    |
| $\bar{E}_1$   | 2   | -2        | 1      | -1           | 0           | 0                 | 3    |
| $D_{1/2}$     | 2   | -2        | 1      | -1           | 0           | 0                 |      |
| $D_{3/2}$     | 4   | -4        | -1     | 1            | 0           | 0                 |      |

(5.9)

Now the reduction works:  $D_{1/2} \downarrow C_{3v}^D = \bar{E}_1$  and  $D_{3/2} \downarrow C_{3v}^D = {}^1\bar{E} \oplus {}^2\bar{E} \oplus \bar{E}_1$ . The last column is explained in the next section.

The one-dimensional characters are forced by the group structure: in the double group  $C_3^3 = \bar{E}$ , so  $\chi(C_3)^3 = \chi(\bar{E}) = -1$ ; and  $\sigma^2 = \bar{E}$  gives  $\chi(\sigma)^2 = -1$ ,  $\chi(\sigma) = \pm i$ .

### 5.6.1 Construction, classes and irreps ★

For a point group  $G \subset O(3)$ , the double group  $G^D$  is the set of pairs  $(g, \pm \mathbf{U}_g)$ , where  $\mathbf{U}_g \in SU(2)$  is a lift of the rotational part of  $g$  (Section 5.5). The inversion acts trivially on spin, so a mirror  $\sigma$  with normal  $\vec{n}$  is represented by the lift of  $C_2(\vec{n})$ ,  $-i \vec{n} \cdot \vec{\sigma}$ , and  $\sigma^2 = \bar{E}$ . The order is  $2|G|$ , and  $\bar{E} = (E, -1)$  is central. Useful facts:

- *Classes (Opechowski's rules).*  $g$  and  $\bar{E}g$  always lie in different classes, except when  $g$  is a two-fold rotation or a reflection and  $G$  contains a two-fold rotation or a reflection whose axis (or normal) is perpendicular to that of  $g$ . For example, in  $C_{3v}$  the mirrors  $\sigma_d$  and  $\bar{\sigma}_d$  are in different classes, while in  $C_{2v}$  and  $C_{6v}$  they are not.
- *Irreps.* Those with  $D(\bar{E}) = +1$  are the ordinary irreps of  $G$ . The others, with  $D(\bar{E}) = -1$ , are the double-valued irreps, and  $\sum_{\text{double-valued}} d^2 = |G|$ .
- *Electrons.* With spin-orbit coupling the states of an odd number of electrons carry double-valued irreps only. For weak spin-orbit coupling, obtain them as  $\Gamma_{\text{orb}} \otimes D_{1/2}$  (Exercise 5.4).

## 5.7 Reality of a representation

Given a representation  $D$ , the complex conjugate matrices  $D^*$  also form a representation, with character  $\chi^*$ . For an irrep there are three possibilities:

1. *real* (type 1):  $\mathbf{D}$  is equivalent to a representation by real matrices;
2. *complex* (type 2):  $\mathbf{D}^*$  is not equivalent to  $\mathbf{D}$  ( $\chi$  is not real);
3. *pseudoreal* (type 3):  $\mathbf{D}^*$  is equivalent to  $\mathbf{D}$  ( $\chi$  is real), yet no basis makes all matrices real.

The *Frobenius–Schur indicator* distinguishes them from the character alone:

$$\frac{1}{|G|} \sum_{g \in G} \chi(g^2) = \begin{cases} +1 & \text{real,} \\ 0 & \text{complex,} \\ -1 & \text{pseudoreal.} \end{cases} \quad (5.10)$$

In the double group of  $C_{3v}$ :  $A_1, A_2, E$  are real;  ${}^1\bar{E}$  and  ${}^2\bar{E}$  are complex and conjugate to each other;  $\bar{E}_1$  is pseudoreal ([Exercise 5.5](#)). The spin- $\frac{1}{2}$  representation of  $SU(2)$  itself is the prototypical pseudoreal representation:  $\sigma_y \mathbf{U}^* \sigma_y = \mathbf{U}$  for every  $\mathbf{U} \in SU(2)$ .

Why this matters: time reversal is antiunitary and effectively relates a representation to its complex conjugate. It glues complex-conjugate pairs of irreps into degenerate *physically irreducible* representations. For  ${}^1\bar{E} \oplus {}^2\bar{E}$  this gives the Kramers degeneracy of the  $j = \frac{3}{2}$  level, which in a  $C_{3v}$  field splits only into two Kramers doublets. It also decides when the bilinear invariants of [Section 3.3](#) come in pairs ([Section 6.1](#)). See [Section 5.8](#).

## 5.8 Time reversal ★

Time reversal appears in the lectures in two places: in the pairing of complex-conjugate irreps ([Section 5.7](#), and [Section 6.1](#) in the next chapter) and in band topology ([Chapter 8](#)). This section gives the minimum theory for an atom or a molecule; [Section 6.8](#) adds what changes in a crystal.

### 5.8.1 An antiunitary symmetry

By Wigner’s theorem on symmetries, a symmetry of quantum mechanics is represented by an operator that is either unitary or *antiunitary*: antilinear,  $\Theta(a\psi + b\phi) = a^*\Theta\psi + b^*\Theta\phi$ , with  $\langle \Theta\psi | \Theta\phi \rangle = \langle \psi | \phi \rangle^*$ . Time reversal must reverse momenta and angular momenta but not positions, and it must preserve  $[\hat{x}, \hat{p}] = i\hbar$ . This forces antiunitarity. Every antiunitary operator can be written  $\Theta = \hat{U}K$  with  $\hat{U}$  unitary and  $K$  complex conjugation in a fixed basis.

- *Spinless particles*:  $\Theta = K$  in the position representation,  $(\Theta\psi)(\vec{r}) = \psi(\vec{r})^*$ , and  $\Theta^2 = +1$ .
- *Spin  $\frac{1}{2}$* :  $\Theta = -i\sigma_y K$ , which reverses  $\vec{\sigma}$  ( $\Theta\vec{\sigma}\Theta^{-1} = -\vec{\sigma}$ ), and  $\Theta^2 = -1$ .
- $N$  electrons:  $\Theta^2 = (-1)^N$ .

## 5.8.2 Kramers' theorem

Theorem 5.2 (Kramers). *If  $[\hat{H}, \Theta] = 0$  and  $\Theta^2 = -1$ , every energy level is at least doubly degenerate.*

*Proof.*  $\Theta\psi$  has the same energy as  $\psi$ . If they were the same state,  $\Theta\psi = c\psi$ , then  $\Theta^2\psi = \Theta(c\psi) = c^*\Theta\psi = |c|^2\psi = \psi$ , contradicting  $\Theta^2 = -1$ . In fact the two states are orthogonal: antiunitarity gives  $\langle a|b \rangle = \langle \Theta b|\Theta a \rangle$ , and with  $a = \psi$ ,  $b = \Theta\psi$ ,  $\langle \psi|\Theta\psi \rangle = \langle \Theta^2\psi|\Theta\psi \rangle = -\langle \psi|\Theta\psi \rangle$ , so  $\langle \psi|\Theta\psi \rangle = 0$ .  $\square$

## 5.8.3 Extra degeneracies

Time reversal commutes with the point-group operations, so it maps the states of a level onto states of the same energy that transform under the complex-conjugate representation. It can therefore force two irreps to stick together, or double a single one. For a point group  $G$  the answer depends on the reality type of the irrep (Section 5.7), measured by the Frobenius–Schur indicator  $s = \frac{1}{|G|} \sum_g \chi(g^2)$ , and on  $\Theta^2$ :

| reality type of $\Gamma$ | $\Theta^2 = +1$ (no spin)           | $\Theta^2 = -1$ (spin $\frac{1}{2}$ ) |        |
|--------------------------|-------------------------------------|---------------------------------------|--------|
| real ( $s = +1$ )        | no extra degeneracy                 | doubled: $\Gamma \oplus \Gamma$       | (5.11) |
| complex ( $s = 0$ )      | $\Gamma \oplus \Gamma^*$ degenerate | $\Gamma \oplus \Gamma^*$ degenerate   |        |
| pseudoreal ( $s = -1$ )  | doubled: $\Gamma \oplus \Gamma$     | no extra degeneracy                   |        |

Examples. For spinless electrons in  $C_4$  (Section 6.1), the complex pair  ${}^1E$ ,  ${}^2E$  becomes a degenerate doublet, the physically irreducible representation  $E$ . For spin- $\frac{1}{2}$  electrons in  $C_{3v}$ , the complex pair  ${}^1\bar{E}$ ,  ${}^2\bar{E}$  becomes a Kramers doublet, while the pseudoreal  $\bar{E}_1$  is already a doublet and is not doubled again. So a  $j = \frac{3}{2}$  level at a  $C_{3v}$  site splits into two Kramers doublets.

## Exercises and solutions

Exercise 5.1. Let  $C_s = \{E, \sigma\}$  be the subgroup of  $C_{3v}$  generated by one mirror, with irreps  $A'$  (even) and  $A''$  (odd). Construct the subduction table  $C_{3v} \rightarrow C_s$ .

*Solution.* Keep the columns  $E$  and  $\sigma$ :  $A_1 \rightarrow (1, 1) = A'$ ,  $A_2 \rightarrow (1, -1) = A''$ ,  $E \rightarrow (2, 0) = A' \oplus A''$ .

Exercise 5.2. Show that in a tetrahedral environment  $p \rightarrow t_2$ ,  $d \rightarrow e \oplus t_2$  and  $f \rightarrow a_1 \oplus t_1 \oplus t_2$ . Remember that  $S_4 = \sigma_h C_4$  is improper and use  $\chi(IC(\theta)) = \pm\chi_I(\theta)$ .

*Solution.* Write  $S_4 = \sigma_h C_4 = IC(270^\circ)$  and  $\sigma_d = IC_2$ . Then  $\chi(S_4) = \pm\chi_l(90^\circ)$  and  $\chi(\sigma_d) = \pm\chi_l(180^\circ)$ , with the parity sign  $(-1)^l$ . Using (5.2):

|             | $E$ | $8C_3$ | $3C_2$ | $6S_4$ | $6\sigma_d$ |                             |
|-------------|-----|--------|--------|--------|-------------|-----------------------------|
| $p \ (l=1)$ | 3   | 0      | -1     | -1     | 1           | $T_2$                       |
| $d \ (l=2)$ | 5   | -1     | 1      | -1     | 1           | $E \oplus T_2$              |
| $f \ (l=3)$ | 7   | 1      | -1     | 1      | 1           | $A_1 \oplus T_1 \oplus T_2$ |

For instance for  $d$ :  $m_E = \frac{1}{24}(10+8+6+0+0) = 1$  and  $m_{T_2} = \frac{1}{24}(15+0-3+6+6) = 1$ .

Exercise 5.3 (*octahedral field*). Since  $d$  orbitals are even under inversion, their splitting in  $O_h = O \times C_i$  is decided by the rotation group  $O$ , with classes  $E$ ,  $8C_3$ ,  $3C_2$ ,  $6C_4$ ,  $6C'_2$  and character table

| $O$   | $E$ | $8C_3$ | $3C_2$ | $6C_4$ | $6C'_2$ |
|-------|-----|--------|--------|--------|---------|
| $A_1$ | 1   | 1      | 1      | 1      | 1       |
| $A_2$ | 1   | 1      | 1      | -1     | -1      |
| $E$   | 2   | -1     | 2      | 0      | 0       |
| $T_1$ | 3   | 0      | -1     | 1      | -1      |
| $T_2$ | 3   | 0      | -1     | -1     | 1       |

Show that  $d \rightarrow e \oplus t_2$  (in  $O_h$ :  $e_g \oplus t_{2g}$ ). Which of the two is spanned by  $d_{xy}, d_{yz}, d_{zx}$ ?

*Solution.*  $\chi_2$ :  $E \rightarrow 5$ ,  $C_3 \rightarrow -1$ ,  $C_2 \rightarrow 1$ ,  $C_4 \rightarrow \sin(5\pi/4)/\sin(\pi/4) = -1$ ,  $C'_2 \rightarrow 1$ . Then  $m_E = \frac{1}{24}(10+8+6) = 1$ ,  $m_{T_2} = \frac{1}{24}(15-3+6+6) = 1$ , all others 0:  $d \rightarrow e \oplus t_2$ , which are  $e_g \oplus t_{2g}$  in  $O_h$ . The three orbitals  $xy, yz, zx$  are permuted among themselves by the cubic rotations and span  $t_{2g}$ ;  $x^2 - y^2$  and  $3z^2 - r^2$  span  $e_g$ .

Exercise 5.4 (*weak vs strong spin-orbit*). A  $p$  electron sits at a  $C_{3v}$  site. (a) Weak spin-orbit coupling: split the orbitals first,  $p \rightarrow a_1 \oplus e$ , then multiply by the spin representation  $D_{1/2} = \bar{E}_1$ . (b) Strong spin-orbit coupling: split  $j = \frac{1}{2}$  and  $j = \frac{3}{2}$  directly. Show that both orders give the same set of double-group irreps,  $2\bar{E}_1 \oplus {}^1\bar{E} \oplus {}^2\bar{E}$ .

*Solution.* (a)  $A_1 \otimes \bar{E}_1 = \bar{E}_1$ . The product  $E \otimes \bar{E}_1$  has characters

$$(2 \cdot 2, 2 \cdot (-2), (-1)(1), (-1)(-1), 0, 0) = (4, -4, -1, 1, 0, 0),$$

equal to  ${}^1\bar{E} \oplus {}^2\bar{E} \oplus \bar{E}_1$ . Total:  $2\bar{E}_1 \oplus {}^1\bar{E} \oplus {}^2\bar{E}$ . (b)  $D_{1/2} \rightarrow \bar{E}_1$  and  $D_{3/2} \rightarrow {}^1\bar{E} \oplus {}^2\bar{E} \oplus \bar{E}_1$ : the same set. The final labels do not depend on the order in which the perturbations are applied; only the energies do.

Exercise 5.5. Verify the “type” column of (5.9) with the Frobenius–Schur indicator. You need the classes of the squares:  $(C_3)^2 \in \{2\bar{C}_3\}$ ,  $(\bar{C}_3)^2 \in \{2\bar{C}_3\}$ ,  $\sigma^2 = \bar{\sigma}^2 = \bar{E}$ .

*Solution.* Sum  $\chi(g^2)$  over the 12 elements, grouped by class (sizes 1, 1, 2, 2, 3, 3), with  $g^2$  equal to  $E$ ,  $E$ ,  $\bar{C}_3$ ,  $\bar{C}_3$ ,  $\bar{E}$ ,  $\bar{E}$  respectively:

$$E : 2 + 2 + 2(-1) + 2(-1) + 3 \cdot 2 + 3 \cdot 2 = 12 \quad \Rightarrow s = +1,$$

$${}^1\bar{E} : 1 + 1 + 2 \cdot 1 + 2 \cdot 1 + 3(-1) + 3(-1) = 0 \quad \Rightarrow s = 0,$$

$$\bar{E}_1 : 2 + 2 + 2(-1) + 2(-1) + 3(-2) + 3(-2) = -12 \quad \Rightarrow s = -1.$$

$A_1$  and  $A_2$  give  $s = +1$  trivially.

# Chapter 6

## Space groups and their representations

### Video: Lecture 6

Complex irreps and bilinear invariants [L6 @ 0:03](#) ·  $C_{4v}$  versus  $C_4$  [L6 @ 5:14](#) · circular polarisation and time reversal [L6 @ 18:01](#) · from atoms to crystals; the bands of graphene [L6 @ 25:19](#) · symmetry-enforced vs accidental degeneracies [L6 @ 29:35](#) · irreps of the translation group [L6 @ 45:24](#) · the Brillouin zone as the set of irreps [L6 @ 53:52](#) · Bloch's theorem as Wigner's theorem [L6 @ 54:55](#) · Bloch sums from orbitals [L6 @ 63:52](#) · space groups, symmorphic and non-symmorphic [L6 @ 70:17](#) · star and little group [L6 @ 77:41](#) · small representations [L6 @ 86:15](#)

### 6.1 Complex representations and bilinear invariants

So far all irreps were real. When an irrep is complex, the invariant scalar product of two copies must be the *hermitian* one,  $\sum_i z_i^* w_i$ , because the matrices are unitary rather than orthogonal. Two consequences:

- a bilinear invariant pairs an irrep with its complex conjugate, as in (4.1);
- $z^* w$  is complex: its real and imaginary parts are two independent real invariants.

Compare  $C_{4v}$  with its rotation subgroup  $C_4 = \{E, C_4^+, C_2, C_4^-\}$  (rotations about  $z$  only).  $C_4$  is Abelian, every element is a class, so there are four irreps, all one-dimensional (the only solution of  $\sum d_\mu^2 = 4$  with four terms):

| $C_4$   | $E$ | $C_4^+$ | $C_2$ | $C_4^-$ | components         |
|---------|-----|---------|-------|---------|--------------------|
| A       | 1   | 1       | 1     | 1       | $E_z$              |
| B       | 1   | -1      | 1     | -1      |                    |
| ${}^1E$ | 1   | i       | -1    | -i      | $E_+ = E_x + iE_y$ |
| ${}^2E$ | 1   | -i      | -1    | i       | $E_- = E_x - iE_y$ |

(6.1)

*Convention.* Here we list vector *components*: the rotation  $C_4^+$  maps  $(E_x, E_y) \mapsto (-E_y, E_x)$ , hence  $E_+ \mapsto iE_+$ . If instead one lists *basis functions* with  $(\hat{O}_g f)(\vec{r}) = f(g^{-1}\vec{r})$ , the function  $x + iy$  picks up  $-i$ . The two conventions are complex conjugates of each other; tables differ on this point, so check before using one.

Every irrep of an Abelian group is one-dimensional. This will be crucial for the translation group.

Physically,  $E_\pm$  are circular polarisations. A rotation by  $90^\circ$  shifts their phase by  $\pm 90^\circ$ . Time reversal reverses the sense of rotation and exchanges them. The pair  ${}^1E \oplus {}^2E$  is therefore glued by time reversal into a two-dimensional *physically irreducible* representation, which is why it carries the letter E.

Invariants. In  $C_{4v}$ ,  $\Gamma_V = A_1 \oplus E$  as in  $C_{3v}$ , and the bilinear invariants of two vectors  $\vec{E}, \vec{P}$  are  $E_z P_z$  and  $E_x P_x + E_y P_y$ . In  $C_4$ ,  $\Gamma_V = A \oplus {}^1E \oplus {}^2E$ , and

$$E_+^* P_+ = (E_x P_x + E_y P_y) + i(E_x P_y - E_y P_x)$$

provides, besides  $E_z P_z$ , *two* invariants: the in-plane scalar product and  $(\vec{E} \times \vec{P})_z$ . The second is allowed because  $C_4$  has no mirrors. Groups made only of proper rotations are called *chiral*; they allow such “vector-product” couplings, with physical consequences such as optical activity.

Checking symmetry-adapted coordinates. To verify that given coordinates belong to an irrep, transform them with the *generators* only (for  $C_4$ , just  $C_4^+$ ; for  $C_{4v}$ ,  $C_4^+$  and one mirror); every other element is a product of generators. For a multi-dimensional irrep compare with the tabulated matrices, since the character is not enough.

## 6.2 From molecules to crystals

A crystal is a “molecule” (a set of atoms, the basis) repeated at the points of a Bravais lattice

$$\vec{t} = n_1 \vec{a}_1 + n_2 \vec{a}_2 + n_3 \vec{a}_3, \quad n_i \in \mathbb{Z}.$$

For graphene the basis consists of two carbon atoms, A and B, identical but with different environments (Fig. 6.1). The crystal is invariant under all lattice translations, which form the Abelian *translation group*  $\mathcal{T}$ . Taken literally, a perfect crystal has infinitely many degrees of freedom and its representations are infinite dimensional. These are the *band representations* of Chapter 8. The translation group lets us break them into finite pieces.

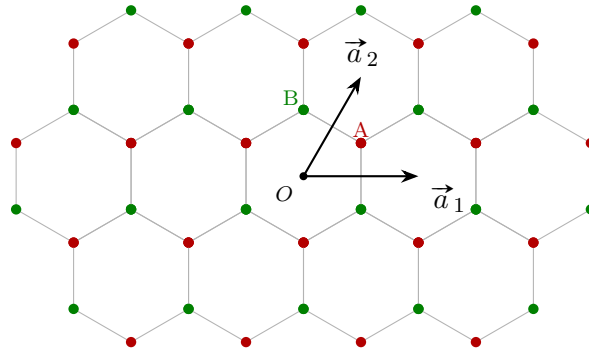


Figure 6.1: The honeycomb lattice. The origin  $O$  is the centre of a hexagon (Wyckoff position  $1a$ , site symmetry  $C_{6v}$ ). The atoms of sublattice A (red) sit at  $\vec{q}_A = \frac{1}{3}(\vec{a}_1 + \vec{a}_2)$  and those of sublattice B (green) at  $\vec{q}_B = \frac{2}{3}(\vec{a}_1 + \vec{a}_2)$  plus lattice vectors (Wyckoff position  $2b$ , site symmetry  $C_{3v}$ ). Lattice constant  $a = |\vec{a}_1|$ .

### 6.3 Irreducible representations of the translation group

Since  $\mathcal{T}$  is Abelian, its irreps are one-dimensional: to each translation a unimodular complex number (a phase), with  $D(\vec{t}_1)D(\vec{t}_2) = D(\vec{t}_1 + \vec{t}_2)$ . The only continuous functions turning sums into products are exponentials, so

$$D^{\vec{k}}(\vec{t}) = e^{-i\vec{k} \cdot \vec{t}} \quad (6.2)$$

for some wave vector  $\vec{k}$ . (Imposing Born–von Kármán periodic boundary conditions on a crystal of  $N$  cells makes  $\vec{k}$  discrete, with  $N$  allowed values; nothing else changes.)

Introduce the reciprocal basis,  $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ , and write  $\vec{k} = \sum_i k_i \vec{b}_i$ . Then  $\vec{k} \cdot \vec{t} = 2\pi \sum_i k_i n_i$ . If  $\vec{G} = \sum_i m_i \vec{b}_i$  with integer  $m_i$  (a reciprocal-lattice vector),  $e^{-i\vec{G} \cdot \vec{t}} = 1$  for every  $\vec{t}$ . Hence  $\vec{k}$  and  $\vec{k} + \vec{G}$  label the same irrep: they are *equivalent*, written  $\vec{k} \equiv \vec{k} + \vec{G}$ .

#### The Brillouin zone

The inequivalent irreps of  $\mathcal{T}$  are in one-to-one correspondence with the points of any primitive cell of the reciprocal lattice. The first Brillouin zone is the most symmetric such cell. *The Brillouin zone is the set of irreps of the translation group*, and  $\vec{k}$  (“crystal momentum”) is a label of an irrep, just as  $l$  labels an irrep of  $\text{SO}(3)$ .

### 6.4 Bloch’s theorem is Wigner’s theorem

Let the translation operators act on functions as  $(\hat{T}_{\vec{t}}\psi)(\vec{r}) = \psi(\vec{r} - \vec{t})$  (active convention: the function is moved by  $+\vec{t}$ ). A function belongs to the irrep  $\vec{k}$  if

$\hat{T}_{\vec{t}}\psi = e^{-i\vec{k}\cdot\vec{t}}\psi$ , i.e.  $\psi(\vec{r} + \vec{t}) = e^{i\vec{k}\cdot\vec{t}}\psi(\vec{r})$ . The general solution is

$$\psi_{\vec{k}}(\vec{r}) = e^{i\vec{k}\cdot\vec{r}} u_{\vec{k}}(\vec{r}), \quad u_{\vec{k}}(\vec{r} + \vec{t}) = u_{\vec{k}}(\vec{r}). \quad (6.3)$$

Since the Hamiltonian of a crystal commutes with  $\mathcal{T}$ , its eigenfunctions can be chosen to belong to irreps of  $\mathcal{T}$  (Wigner). That is all Bloch's theorem says. (Bradley and Cracknell use the passive convention, with the opposite sign of the exponent in (6.2).)

#### 6.4.1 Bloch sums

Place an orbital  $\varphi_{\alpha}$  at position  $\vec{q}_{\alpha}$  of every cell and form

$$\Phi_{\alpha\vec{k}}(\vec{r}) = \sum_{\vec{t}} e^{i\vec{k}\cdot\vec{t}} \varphi_{\alpha}(\vec{r} - \vec{t} - \vec{q}_{\alpha}). \quad (6.4)$$

Relabelling the sum shows  $\hat{T}_{\vec{t}}\Phi_{\alpha\vec{k}} = e^{-i\vec{k}\cdot\vec{t}}\Phi_{\alpha\vec{k}}$ : the Bloch sum belongs to the irrep  $\vec{k}$ . With this convention  $\Phi_{\alpha,\vec{k}+\vec{G}} = \Phi_{\alpha\vec{k}}$ . Bloch sums are “Bloch-like” waves: eigenfunctions of translations but not in general of  $\hat{H}$ .

Because  $\hat{H}$  is invariant, it has no matrix elements between inequivalent irreps of  $\mathcal{T}$  (Section 4.2). With  $N_o$  orbitals per cell, the  $\infty \times \infty$  tight-binding problem in real space splits into one  $N_o \times N_o$  problem at each  $\vec{k}$ , giving  $N_o$  bands. Different points of the Brillouin zone do not talk to each other. This is the only use of symmetry you can make without group theory. To go further we need the rest of the space group.

## 6.5 Space groups

A space-group element combines a point operation  $R$  (rotation, reflection, rotoinversion) with a translation  $\vec{v}$ . In Seitz notation

$$\{R | \vec{v}\} \vec{r} = R\vec{r} + \vec{v}, \quad (6.5)$$

first  $R$ , then the translation. From this,

$$\{R_1 | \vec{v}_1\}\{R_2 | \vec{v}_2\} = \{R_1R_2 | \vec{v}_1 + R_1\vec{v}_2\}, \quad \{R | \vec{v}\}^{-1} = \{R^{-1} | -R^{-1}\vec{v}\}. \quad (6.6)$$

The pure translations  $\{E | \vec{t}\}$  form a normal subgroup  $\mathcal{T}$ , and the set of rotational parts forms the *point group*  $\bar{G} \cong G/\mathcal{T}$  (Sections 2.3 and 6.5.1).

**Definition 6.1.** A space group is *symmorphic* if, for a suitable origin, every element has the form  $\{R | \vec{t}\}$  with  $\vec{t} \in \mathcal{T}$ ; equivalently,  $G$  contains the point group  $\bar{G}$  as a subgroup,  $G = \mathcal{T} \rtimes \bar{G}$ . Otherwise it is *non-symmorphic*: some operations come only with a fractional translation that no choice of origin removes. These are *glide planes* (reflection plus a half translation parallel to the plane) and *screw axes* (rotation  $C_n$  plus a translation  $\frac{p}{n}\vec{t}$  along the axis).

In three dimensions there are 230 space groups, of which 73 are symmorphic. In two dimensions there are 17 plane (wallpaper) groups, 13 of them symmorphic; the honeycomb lattice has the symmorphic plane group  $p6mm$ . Its three-dimensional analogue, used in the lecture and in the Bilbao tables, is  $P6mm$  (No. 183). For symmorphic groups the representation theory reduces to that of point groups; for non-symmorphic groups it is subtler but fully tabulated.

### 6.5.1 Semidirect products ★

The translation subgroup  $\mathcal{T}$  is normal (Section 2.3):  $\{R|\vec{v}\}\{E|\vec{t}\}\{R|\vec{v}\}^{-1} = \{E|R\vec{t}\}$ , and  $R\vec{t}$  is again a lattice vector (Exercise 6.3). The quotient  $G/\mathcal{T}$  is isomorphic to the point group  $\bar{G}$ : the coset of  $\{R|\vec{v}\}$  is determined by  $R$  alone.

If  $G$  has a normal subgroup  $N$  and a subgroup  $H$  with  $N \cap H = \{e\}$  and  $NH = G$ , then  $G = N \rtimes H$  is a *semidirect product*:  $H$  acts on  $N$  by conjugation. (When  $H$  is normal too, this is the direct product of Section 5.2.3.) Examples:  $C_{3v} = C_3 \rtimes \{E, \sigma_d\}$ ; a symmorphic space group is  $\mathcal{T} \rtimes \bar{G}$ , the point group acting on the lattice by rotation.

Non-symmorphic space groups are *not* semidirect products:  $\bar{G}$  is still the quotient  $G/\mathcal{T}$  but is not a subgroup of  $G$ , because some rotations only occur combined with fractional translations. This is the structural reason why their representation theory needs projective representations (see the box on non-symmorphic groups in Section 6.6).

### 6.5.2 Space-group operations move $\vec{k}$

Let  $g = \{R|\vec{v}\}$  act on functions by  $(\hat{O}_g\psi)(\vec{r}) = \psi(g^{-1}\vec{r})$ . From (6.6),  $g^{-1}\{E|\vec{t}\}g = \{E|R^{-1}\vec{t}\}$ . Hence, if  $\psi$  belongs to  $\vec{k}$ ,

$$\hat{T}_{\vec{t}} \hat{O}_g \psi = \hat{O}_g \hat{T}_{R^{-1}\vec{t}} \psi = e^{-i\vec{k} \cdot R^{-1}\vec{t}} \hat{O}_g \psi = e^{-i(R\vec{k}) \cdot \vec{t}} \hat{O}_g \psi :$$

$\hat{O}_g$  maps the irrep  $\vec{k}$  of  $\mathcal{T}$  onto the irrep  $R\vec{k}$ .

## 6.6 Star, little group and small representations

Definition 6.2. • The *star* of  $\vec{k}$  is the set of inequivalent vectors  $\{R\vec{k} : R \in \bar{G}\}$ .

- The *little group*  $G_{\vec{k}}$  is the set of  $g = \{R|\vec{v}\} \in G$  with  $R\vec{k} \equiv \vec{k}$ , i.e.  $R\vec{k} = \vec{k} + \vec{G}$  for some reciprocal vector  $\vec{G}$ . Its point parts form the *little co-group*  $\bar{G}_{\vec{k}} \subset \bar{G}$ .
- A *small representation* is an irrep of  $G_{\vec{k}}$  in which the translations are represented by  $e^{-i\vec{k} \cdot \vec{t}} \mathbf{1}$ .

The equivalence “up to  $\vec{G}$ ” is essential: for graphene the corner  $K$  of the hexagonal zone is carried by  $C_3$  to another corner that differs from it by a reciprocal-lattice vector, so  $C_3$  belongs to the little co-group of  $K$  (Chapter 7).

Why the little group matters: the Bloch states at a given  $\vec{k}$  are mapped onto themselves by  $G_{\vec{k}}$ , and the  $N_o \times N_o$  Hamiltonian  $\mathbf{H}(\vec{k})$  commutes with the small representation. By Wigner’s theorem the energy levels at  $\vec{k}$  are labelled by small irreps, and their degeneracies are the dimensions of those irreps. For bilinear quantities (Hamiltonians, optical matrix elements between states at the same  $\vec{k}$ ) only small representations are needed. Full space-group irreps are needed for electron–phonon processes, superstructures and similar problems. They are obtained by *induction* from small irreps and have dimension (size of the star)  $\times$  (dimension of the small irrep).

#### Small irreps of symmorphic groups

If  $G$  is symmorphic, the small irreps at  $\vec{k}$  are

$$D^{\vec{k},\mu}(\{R | \vec{t}\}) = e^{-i\vec{k} \cdot \vec{t}} \mathbf{D}^{(\mu)}(R), \quad R \in \bar{G}_{\vec{k}}, \quad \vec{t} \in \mathcal{T}, \quad (6.7)$$

where  $\mathbf{D}^{(\mu)}$  runs over the irreps of the little co-group (a point group). They are labelled like point-group irreps, or as  $\Gamma_1, \Gamma_2, \dots, K_1, K_2, \dots$  in the Bilbao convention.

#### Beyond the lectures: non-symmorphic groups

For  $g = \{R | \vec{v}\} \in G_{\vec{k}}$  with fractional  $\vec{v}$ , try  $\Delta(g) = e^{-i\vec{k} \cdot \vec{v}} \mathbf{D}(R)$ . Since  $g_1 g_2 = \{R_1 R_2 | \vec{v}_1 + R_1 \vec{v}_2\}$ , the product  $\Delta(g_1) \Delta(g_2) = e^{-i\vec{k} \cdot (\vec{v}_1 + \vec{v}_2)} \mathbf{D}(R_1) \mathbf{D}(R_2)$  must equal  $e^{-i\vec{k} \cdot (\vec{v}_1 + R_1 \vec{v}_2)} \mathbf{D}(R_1 R_2)$ . The two phases differ by  $e^{-i\vec{k} \cdot \vec{v}_2 + i\vec{k} \cdot R_1 \vec{v}_2} = e^{i(R_1^{-1} \vec{k} - \vec{k}) \cdot \vec{v}_2}$ . If every  $R \in \bar{G}_{\vec{k}}$  fixes  $\vec{k}$  exactly (always the case inside the zone), this factor is 1 and  $\mathbf{D}$  is an ordinary irrep of  $\bar{G}_{\vec{k}}$ . On the zone boundary,  $R_1^{-1} \vec{k} = \vec{k} + \vec{G}_1$  and the factor  $e^{i\vec{G}_1 \cdot \vec{v}_2}$  can differ from 1 when  $\vec{v}_2$  is fractional. Then  $\mathbf{D}$  must be a *projective* representation of the little co-group. This is how glides and screws force bands to stick together on the zone boundary (Exercise 6.5).

## 6.7 Degeneracies: enforced or accidental?

Look at a first-principles band structure of graphene (Fig. 7.2 shows the  $\pi$  bands). Its bands carry labels of small irreps at  $\Gamma, K, M$  and along the lines joining them. Two kinds of degeneracy appear:

- *Symmetry-enforced*: at  $K$  two bands meet because they carry a two-dimensional small irrep. Any Hamiltonian with the symmetry of graphene must have a degeneracy there: the Dirac point.

- *Accidental*: along a symmetry line two bands with *different* one-dimensional irreps cross. Group theory permits the crossing (the Hamiltonian cannot couple different irreps) but does not require it. Moving the bands changes where they cross, or removes the crossing.

Two further observations. At an isolated high-symmetry point an accidental degeneracy requires fine-tuning of the energies to infinite precision, so it has probability zero. If you find one numerically, either the precision is insufficient or you have missed a symmetry. Along a line,  $\vec{k}$  is a continuous parameter, and crossings of different irreps happen generically. Two bands of the *same* irrep generically repel (avoided crossing), because they can be coupled.

Near a degeneracy the bands disperse linearly unless a symmetry forbids the linear term. Group theory also decides this, through the method of invariants (Section 7.6).

## 6.8 Time reversal in crystals ★

Section 5.8 treated time reversal for an atom or a molecule. In a crystal it also acts on the wave vector.

### 6.8.1 Time reversal and Bloch states

Complex conjugation turns  $e^{i\vec{k}\cdot\vec{r}}u_{\vec{k}}$  into  $e^{-i\vec{k}\cdot\vec{r}}u_{\vec{k}}^*$ : time reversal maps  $\vec{k}$  to  $-\vec{k}$ . Hence

- $E_n(\vec{k}) = E_n(-\vec{k})$  (without spin), or  $E_{n\uparrow}(\vec{k}) = E_{n\downarrow}(-\vec{k})$  and in general Kramers pairs at  $\vec{k}$  and  $-\vec{k}$  (with spin);
- at the *time-reversal-invariant momenta* (TRIM),  $\vec{k} \equiv -\vec{k}$  (in graphene:  $\Gamma$  and the three  $M$  points), time reversal maps the states at  $\vec{k}$  onto themselves. With  $\Theta^2 = -1$  every band is then Kramers degenerate; in the presence of inversion symmetry as well, every band is doubly degenerate at every  $\vec{k}$ .

$K$  and  $K' = -K$  in graphene are exchanged by time reversal;  $K$  is not a TRIM.

At a TRIM of a symmorphic space group, the test of Section 5.8.3 applies to the little co-group. For general  $\vec{k}$  and non-symmorphic groups the same question is answered by Herring's criterion, a sum of  $\chi((\Theta g)^2)$  over the elements  $g$  that map  $\vec{k}$  to  $-\vec{k}$ ; see Bradley–Cracknell.

### 6.8.2 Magnetic groups

In a magnetically ordered crystal time reversal alone is broken, but a combination  $\Theta g$  with a spatial operation  $g$  may survive. For example, an antiferromagnet can be invariant under  $\Theta$  combined with a translation or with inversion. The groups

that contain such antiunitary elements are the *magnetic* (Shubnikov) groups: 1651 magnetic space groups in three dimensions, of four types (no  $\Theta$  at all;  $\Theta$  itself, the “grey” groups;  $\Theta$  only combined with point operations;  $\Theta$  combined with a translation). Their representation theory uses Wigner’s *corepresentations*, whose degeneracies are again decided by tests of the type above. The Bilbao server tabulates their irreducible corepresentations and band representations (magnetic topological quantum chemistry).

## Exercises and solutions

Exercise 6.1. Show that  $(\vec{E} \times \vec{P})_z = E_x P_y - E_y P_x$  is invariant under  $C_4$  but changes sign under the mirrors of  $C_{4v}$ . To which irrep of  $C_{4v}$  does it belong?

*Solution.* Under  $C_4^+$ ,  $(E_x, E_y) \rightarrow (-E_y, E_x)$  and likewise for  $\vec{P}$ :  $E'_x P'_y - E'_y P'_x = (-E_y)P_x - E_x(-P_y) = E_x P_y - E_y P_x$ . Under the mirror  $x \rightarrow -x$ :  $(-E_x)P_y - E_y(-P_x) = -(E_x P_y - E_y P_x)$ . So it is invariant under the rotations and odd under the mirrors:  $A_2$  of  $C_{4v}$ , like  $R_z$ .

Exercise 6.2. For the hexagonal lattice  $\vec{a}_1 = a(1, 0)$ ,  $\vec{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$ : (a) find  $\vec{b}_1, \vec{b}_2$ ; (b) show that  $K = \frac{2}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2$  is a corner of the Brillouin zone with  $|K| = 4\pi/(3a)$ , and that  $M = \frac{1}{2}\vec{b}_1$  is the midpoint of an edge; (c) show that the six corners fall into two classes of three equivalent points,  $K$  and  $K' \equiv -K$ , and that  $K \neq K'$ .

*Solution.* (a)  $\vec{b}_1 = \frac{2\pi}{a}(1, -\frac{1}{\sqrt{3}})$ ,  $\vec{b}_2 = \frac{2\pi}{a}(0, \frac{2}{\sqrt{3}})$ ,  $|\vec{b}_i| = \frac{4\pi}{\sqrt{3}a}$ . (b)  $K = \frac{2}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2 = (\frac{4\pi}{3a}, 0)$ . The zone is a hexagon with inradius  $|\vec{b}|/2 = \frac{2\pi}{\sqrt{3}a}$  and circumradius  $\frac{2\pi}{\sqrt{3}a}/\cos 30^\circ = \frac{4\pi}{3a} = |K|$ :  $K$  is a corner.  $M = \vec{b}_1/2$  has  $|M|$  equal to the inradius and lies on the edge bisector: the midpoint of an edge. (c)  $C_3^+ K - K = \frac{4\pi}{3a}(-\frac{3}{2}, \frac{\sqrt{3}}{2}) = -\vec{b}_1$ , so  $C_3 K \equiv K$ ; the corners at  $0^\circ, 120^\circ, 240^\circ$  are equivalent, and so are those at  $60^\circ, 180^\circ, 300^\circ$  (equivalent to  $K' = -K$ ). But  $K' - K = -2K = (-\frac{8\pi}{3a}, 0)$  would need  $m_1 = -\frac{4}{3}$  in  $m_1 \vec{b}_1 + m_2 \vec{b}_2$ : not a reciprocal vector, so  $K \neq K'$ .

Exercise 6.3. Prove the composition and inverse rules (6.6). Show that  $g\{E|\vec{t}\}g^{-1} = \{E|R\vec{t}\}$  for  $g = \{R|\vec{v}\}$ , so that  $\mathcal{T}$  is a normal subgroup.

*Solution.*  $\{R_1|\vec{v}_1\}\{R_2|\vec{v}_2\}\vec{r} = R_1(R_2\vec{r} + \vec{v}_2) + \vec{v}_1 = R_1R_2\vec{r} + \vec{v}_1 + R_1\vec{v}_2$ .  $\{R^{-1}|-R^{-1}\vec{v}\}\{R|\vec{v}\}\vec{r} = R^{-1}(R\vec{r} + \vec{v}) - R^{-1}\vec{v} = \vec{r}$ . Finally  $\{R|\vec{v}\}\{E|\vec{t}\}\{R^{-1}|-R^{-1}\vec{v}\} = \{R|\vec{v} + R\vec{t}\}\{R^{-1}|-R^{-1}\vec{v}\} = \{E|R\vec{t}\}$ , and  $R\vec{t}$  is a lattice vector because  $R$  maps the lattice onto itself.

Exercise 6.4. Verify from (6.4) that  $\hat{T}_{\vec{t}}\Phi_{\alpha\vec{k}} = e^{-i\vec{k}\cdot\vec{t}}\Phi_{\alpha\vec{k}}$  and  $\Phi_{\alpha,\vec{k}+\vec{G}} = \Phi_{\alpha\vec{k}}$ . What changes if the phase  $e^{i\vec{k}\cdot(\vec{t}+\vec{q}_\alpha)}$  is used instead?

*Solution.*  $\hat{T}_{\vec{t}} \Phi_{\alpha \vec{k}}(\vec{r}) = \sum_{\vec{t}'} e^{i\vec{k} \cdot \vec{t}} \varphi_{\alpha}(\vec{r} - \vec{t}' - \vec{t} - \vec{q}_{\alpha}) = \sum_{\vec{s}} e^{i\vec{k} \cdot (\vec{s} - \vec{t}')} \varphi_{\alpha}(\vec{r} - \vec{s} - \vec{q}_{\alpha}) = e^{-i\vec{k} \cdot \vec{t}'} \Phi_{\alpha \vec{k}}(\vec{r})$  with  $\vec{s} = \vec{t} + \vec{t}'$ . Since  $e^{i\vec{G} \cdot \vec{t}} = 1$ ,  $\Phi_{\alpha, \vec{k} + \vec{G}} = \Phi_{\alpha \vec{k}}$ . With the phase  $e^{i\vec{k} \cdot (\vec{t} + \vec{q}_{\alpha})}$  each Bloch sum is multiplied by  $e^{i\vec{k} \cdot \vec{q}_{\alpha}}$ : the translation property is unchanged, but now  $\Phi_{\alpha, \vec{k} + \vec{G}} = e^{i\vec{G} \cdot \vec{q}_{\alpha}} \Phi_{\alpha \vec{k}}$ , so  $\mathbf{H}(\vec{k})$  is periodic only up to a diagonal unitary transformation. Both conventions are in use. The one with  $\vec{q}_{\alpha}$  makes  $\mathbf{H}(\vec{k})$  reflect the positions of the orbitals (useful for Berry phases); the other makes it periodic.

**Exercise 6.5 (a glide line).** The plane group  $pg$  (rectangular lattice  $a\vec{e}_x, b\vec{e}_y$ ) is generated by translations and the glide  $g = \{m_x | \frac{b}{2}\vec{e}_y\}$ , where  $m_x: (x, y) \mapsto (-x, y)$ . (a) Show that  $g^2 = \{E | b\vec{e}_y\}$ . (b) On the line  $k_x = 0$ , Bloch states can be chosen as eigenstates of  $\hat{O}_g$ ; find the possible eigenvalues as functions of  $k_y$ . (c) Follow an eigenvalue continuously from  $k_y = -\pi/b$  to  $k_y = \pi/b$  and show that the two eigenvalue branches are exchanged: bands on this line come in connected pairs.

*Solution.* (a)  $g^2 = \{m_x^2 | \frac{b}{2}\vec{e}_y + m_x \frac{b}{2}\vec{e}_y\} = \{E | b\vec{e}_y\}$ , since  $m_x$  leaves  $\vec{e}_y$  unchanged. (b) On  $k_x = 0$ ,  $m_x \vec{k} = \vec{k}$ , so  $g$  is in the little group and  $\hat{O}_g$  can be diagonalised together with  $\hat{H}$ . Its eigenvalues satisfy  $\lambda^2 = e^{-ik_y b}$  (the eigenvalue of  $\hat{T}_{b\vec{e}_y}$ ), so  $\lambda_{\pm}(k_y) = \pm e^{-ik_y b/2}$ . (c)  $\lambda_+(-\pi/b) = e^{i\pi/2} = i$  and  $\lambda_+(\pi/b) = e^{-i\pi/2} = -i = \lambda_-(-\pi/b)$ . Since  $k_y = \pi/b$  and  $k_y = -\pi/b$  are the same point, a band that starts in the branch  $\lambda_+$  at  $-\pi/b$  arrives in the branch  $\lambda_-$ : it cannot close on itself and must connect to a partner. Bands along this line come in connected pairs, a consequence of the fractional translation.



## Chapter 7

# The group theory of electron bands

### Video: Lecture 7

Generators, graphically [L7@3:16](#) · tight binding for graphene [L7@9:40](#) · the point group and  $\sigma_h$  parity:  $\sigma$  and  $\pi$  bands [L7@18:03](#) · little groups [L7@24:36](#) · high-symmetry points and lines; why  $K$  has  $C_{3v}$  [L7@28:48](#) · the site-symmetry group [L7@35:04](#) · orbital mixing and distorted orbitals [L7@40:21](#) · the character formula for band states [L7@52:54](#) ·  $\Gamma$  point [L7@59:11](#) ·  $K$  point and its phases [L7@64:27](#) · compatibility relations [L7@73:01](#) · what symmetry cannot decide [L7@86:31](#)

This chapter applies the machinery to one concrete crystal. We label every  $\pi$  band of graphene by small irreps, prove that the Dirac point is enforced by symmetry, and introduce compatibility relations. They are the bridge to topology in [Chapter 8](#).

## 7.1 Generators

A set of elements *generates*  $G$  if every element is a product of them and their inverses.  $C_n$  is generated by  $C_n^+$ ;  $C_{nv}$  by  $C_n^+$  and one mirror;  $T_d$  by  $S_4$  and  $C_3$  (or  $S_4$  and a mirror that does not contain the  $S_4$  axis). To check that a set of functions transforms correctly under a representation, or that a matrix commutes with a representation, it is enough to check the generators.

The lecture opens with a graphical check: mark a generic point on a stereographic projection, apply the candidate generators repeatedly, and count the distinct images. If their number equals  $|G|$ , the set generates the group, and each element can be read off as a word in the generators.

## 7.2 A tight-binding model for graphene

Carbon has the configuration  $1s^2 2s^2 2p^2$ . The  $1s$  electrons are inert, so a minimal tight-binding model uses the four valence orbitals  $2s, 2p_x, 2p_y, 2p_z$  on each of the two atoms of the cell:  $8 \times 8$  matrices  $\mathbf{H}(\vec{k})$  and eight bands. With nearest- and next-nearest-neighbour hopping such a model already reproduces first-principles bands well near the Fermi level. Adding empty  $3s$  orbitals improves the upper bands.

The point group. With the origin at a hexagon centre (Fig. 6.1), the honeycomb is invariant under rotations by multiples of  $60^\circ$  and under six vertical mirrors. They fall into two classes (Fig. 7.1):

- $\sigma_v$ : the three mirrors that *contain* C–C bonds (lines at  $30^\circ, 90^\circ, 150^\circ$  with our axes); they map each sublattice onto itself;
- $\sigma_d$ : the three mirrors that *bisect* C–C bonds (lines at  $0^\circ, 60^\circ, 120^\circ$ ); they exchange sublattices A and B.

The in-plane point group is  $C_{6v}$  (plane group  $p6mm$ ). Graphene is also invariant under the reflection  $\sigma_h$  in its own plane, which doubles the group to  $D_{6h}$ . Since  $\sigma_h$  commutes with everything, we can use it once and for all:  $2s, 2p_x, 2p_y$  are even under  $\sigma_h$  and  $2p_z$  is odd. The Hamiltonian cannot couple states of opposite  $\sigma_h$  parity, so the  $8 \times 8$  problem splits into the six  $\sigma$  bands (bonding) and the two  $\pi$  bands from the  $p_z$  orbitals, which contain the Fermi level and the Dirac point. From now on we study the  $\pi$  bands with the group  $C_{6v}$ .

| $C_{6v}$ | $E$ | $C_2$ | $2C_3$ | $2C_6$ | $3\sigma_v$ | $3\sigma_d$ |
|----------|-----|-------|--------|--------|-------------|-------------|
| $A_1$    | 1   | 1     | 1      | 1      | 1           | 1           |
| $A_2$    | 1   | 1     | 1      | 1      | -1          | -1          |
| $B_1$    | 1   | -1    | 1      | -1     | 1           | -1          |
| $B_2$    | 1   | -1    | 1      | -1     | -1          | 1           |
| $E_1$    | 2   | -2    | -1     | 1      | 0           | 0           |
| $E_2$    | 2   | 2     | -1     | -1     | 0           | 0           |

(7.1)

Which of the two B irreps is called  $B_1$  depends on which mirrors are called  $\sigma_v$ ; with our choice  $B_1$  is even under the bond-containing mirrors.

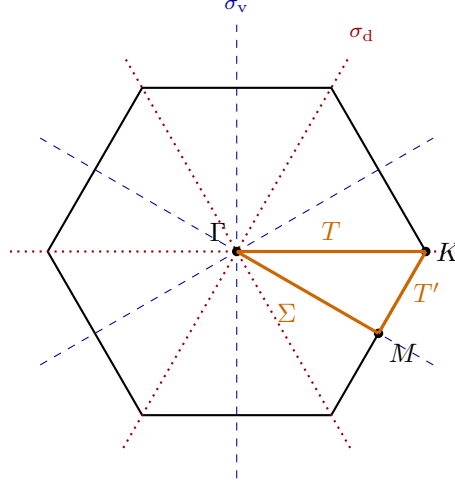


Figure 7.1: The Brillouin zone of graphene with its high-symmetry points and lines. Blue dashed: mirrors  $\sigma_v$  (containing C–C bonds in real space). Red dotted: mirrors  $\sigma_d$  (bisecting bonds). With  $\vec{a}_1 = a(1,0)$ , the point  $K = \frac{2}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2 = (\frac{4\pi}{3a}, 0)$  and  $M = \frac{1}{2}\vec{b}_1$ .

### 7.3 Little groups of graphene

The little co-group of each point or line is the set of point operations that map it to an equivalent point:

| $\vec{k}$              | $\bar{G}_{\vec{k}}$ | elements                                          |       |
|------------------------|---------------------|---------------------------------------------------|-------|
| $\Gamma$               | $C_{6v}$            | all 12                                            |       |
| $T$ ( $\Gamma K$ )     | $C_s$               | $E, \sigma_d(0^\circ)$                            |       |
| $K$                    | $C_{3v}$            | $E, C_3^\pm, 3\sigma_d$                           | (7.2) |
| $T'$ ( $KM$ )          | $C_s$               | $E, \sigma_d(60^\circ)$                           |       |
| $M$                    | $C_{2v}$            | $E, C_2, \sigma_v(150^\circ), \sigma_d(60^\circ)$ |       |
| $\Sigma$ ( $M\Gamma$ ) | $C_s$               | $E, \sigma_v(150^\circ)$                          |       |

Two remarks.

- At  $K$  the rotation  $C_3$  carries  $K$  to another corner of the hexagon. The two corners differ by a reciprocal-lattice vector, so they label the *same* irrep of  $\mathcal{T}$  and  $C_3 \in \bar{G}_K$ . For the same reason the six corners are only two inequivalent points,  $K$  and  $K' = -K$ . This is why graphene has two Dirac points, not six.
- The little co-group of a line is a subgroup of the little co-groups of its end points. This is what makes compatibility relations possible (Section 7.7).

## 7.4 The site-symmetry group

Definition 7.1. The *site-symmetry group*  $G_{\vec{q}}$  of a point  $\vec{q}$  is the set of space-group elements that leave  $\vec{q}$  fixed:  $g\vec{q} = \vec{q}$  exactly (not up to a lattice vector). It is always isomorphic to a point group. Points related by the space group form an *orbit*. Orbits whose site-symmetry groups are conjugate form a *Wyckoff position*.

The crystal as a whole is invariant under  $C_{6v}$  about a hexagon centre, but an atom does not see six-fold symmetry: along one direction it has a nearest neighbour, along the opposite direction a third-nearest one. The site-symmetry group of a carbon atom is  $C_{3v}$ , generated by the three-fold rotation about the atom and the three  $\sigma_v$  mirrors through it. For  $p6mm$  the Wyckoff positions with non-trivial site symmetry are

$$1a \text{ (hexagon centres, } C_{6v}), \quad 2b \text{ (the atoms, } C_{3v}), \quad 3c \text{ (bond centres, } C_{2v}).$$

Atomic orbitals become irreps of the site group. In the crystal the atomic orbitals are no longer those of a free atom; they transform under irreps of  $G_{\vec{q}}$ , obtained by subduction from  $O(3)$  (Section 5.3). At a  $C_{3v}$  site,  $s \rightarrow a_1$  and  $p \rightarrow a_1 \oplus e$  with  $p_z \in a_1$  and  $(p_x, p_y) \in e$ . The “ $p_z$  orbital” of a tight-binding model of graphene is really an  $a_1$  orbital of  $C_{3v}$ : a  $p_z$  orbital distorted by the crystal field and mixed with other  $a_1$  orbitals.

This has practical consequences. A model built as if the orbitals were spherically symmetric can miss couplings that are forbidden for free-atom orbitals but allowed by the true, lower site symmetry. For example, the hoppings to two neighbours at equal distance but in inequivalent directions need not be equal. Such an over-restricted model may display spurious degeneracies or miss phases. Knowing the site-symmetry irreps guarantees that the model is the most general one allowed by symmetry.

## 7.5 Characters of band states

Place one  $a_1$  orbital at each atom and form the two Bloch sums  $\Phi_{A\vec{k}}, \Phi_{B\vec{k}}$  of (6.4). At each  $\vec{k}$  they span a two-dimensional representation of the little group, and  $\mathbf{H}(\vec{k})$  is a  $2 \times 2$  matrix commuting with it. To find its small-irrep content we need its character.

Let  $g = \{R | \vec{0}\}$  be in the little group of  $\vec{k}$ . If  $g$  maps the atom at  $\vec{q}_\alpha$  onto an atom of the same sublattice,  $g\vec{q}_\alpha = \vec{q}_\alpha + \vec{t}_\alpha$  with  $\vec{t}_\alpha \in \mathcal{T}$ , the Bloch sum is mapped onto itself up to a phase. Otherwise it is mapped onto the other sublattice and does not contribute to the trace. A short computation with (6.4) gives

Character of a band representation at  $\vec{k}$

$$\chi_{\vec{k}}(g) = \sum_{\alpha: g\vec{q}_{\alpha} \equiv \vec{q}_{\alpha}} e^{-i\vec{k} \cdot \vec{t}_{\alpha}} \chi_{\rho}(g_{\alpha}), \quad \vec{t}_{\alpha} = g\vec{q}_{\alpha} - \vec{q}_{\alpha}, \quad (7.3)$$

where the sum runs over the atoms of the cell left invariant up to a lattice translation,  $g_{\alpha} = \{E | -\vec{t}_{\alpha}\}g \in G_{\vec{q}_{\alpha}}$  is the corresponding site-symmetry operation, and  $\chi_{\rho}$  is the character of the orbital irrep  $\rho$  of the site group.

This is the crystal version of  $\chi_M = N_{\text{inv}}\chi_V$  (Section 2.8): count invariant atoms, weight each by the orbital character, and add the Bloch phase. For our  $a_1$  orbitals  $\chi_{\rho} = 1$  and only the phases remain.

### 7.5.1 The $\Gamma$ point

At  $\vec{k} = 0$  all phases are 1 and  $\chi_{\Gamma}(g)$  is the number of sublattices mapped onto themselves: 2 for  $E, C_3, \sigma_v$ ; 0 for  $C_2, C_6, \sigma_d$  (these exchange A and B). From (7.1),

$$\chi_{\Gamma} = (2, 0, 2, 0, 2, 0) \implies \Gamma = A_1 \oplus B_1$$

(in the Bilbao labels used in the lecture,  $\Gamma_1 \oplus \Gamma_3$ ). Two non-degenerate levels: the bonding combination  $\Phi_A + \Phi_B$  ( $A_1$ ) and the antibonding one  $\Phi_A - \Phi_B$  ( $B_1$ , odd under the operations that exchange sublattices).

### 7.5.2 The $K$ point

Now phases matter. Only  $E$  and  $C_3^{\pm}$  keep the sublattices; the three  $\sigma_d$  in  $\bar{G}_K$  exchange them. Take  $C_3^+$  about the origin. It maps  $\vec{q}_A = \frac{1}{3}(\vec{a}_1 + \vec{a}_2)$  to  $\vec{q}_A - \vec{a}_1$  and  $\vec{q}_B = \frac{2}{3}(\vec{a}_1 + \vec{a}_2)$  to  $\vec{q}_B - 2\vec{a}_1$ . With  $K = \frac{2}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2$  we have  $K \cdot \vec{a}_1 = \frac{4\pi}{3}$ , so the phases are

$$e^{iK \cdot \vec{a}_1} = e^{4\pi i/3} = \omega^*, \quad e^{2iK \cdot \vec{a}_1} = e^{8\pi i/3} = \omega, \quad \omega = e^{2\pi i/3},$$

and  $\chi_K(C_3) = \omega + \omega^* = -1$ . Physically: under the rotation, the Bloch sum on sublattice A is multiplied by  $\omega^*$ , the one on B by  $\omega$ . Thus

$$\chi_K = (2, -1, 0) \text{ on } (E, 2C_3, 3\sigma_d) \implies K = E \quad (K_3).$$

The Dirac point is enforced by symmetry

At  $K$  the two  $\pi$  states form the two-dimensional irrep E of  $C_{3v}$ , so they are degenerate for *every* Hamiltonian with the symmetry of graphene. Whether the dispersion near the degeneracy is linear is a further question, answered in Section 7.6: it is.

### 7.5.3 The $M$ point

$\bar{G}_M = C_{2v} = \{E, C_2, \sigma_v, \sigma_d\}$ . The  $C_2$  and  $\sigma_d$  exchange sublattices, and  $\sigma_v(150^\circ)$  keeps them with trivial phases, so  $\chi_M = (2, 0, 2, 0)$  and  $M = A_1 \oplus B_1$ , where in  $C_{2v}$  we call  $B_1$  the irrep that is odd under  $C_2$  and even under  $\sigma_v$ .

## 7.6 $\vec{k} \cdot \vec{p}$ Hamiltonians and the method of invariants ★

Group theory tells us which bands must be degenerate at a high-symmetry point. The *method of invariants* tells us how they disperse nearby: it constructs the most general effective Hamiltonian allowed by symmetry, order by order in the distance from the point. It is the same idea as the invariant polarizability of [Section 3.3](#) and the invariant dynamical matrix of [Section 3.5](#).

### 7.6.1 The effective Hamiltonian

Let  $\vec{k}_0$  be a high-symmetry point and let  $\{|n\rangle\}$ ,  $n = 1, \dots, d$ , be the states of the (possibly degenerate) band multiplet of interest at  $\vec{k}_0$ , carrying a representation  $\mathbf{D}(g)$  of the little co-group  $\bar{G}_{\vec{k}_0}$  (usually one small irrep). For  $\vec{k} = \vec{k}_0 + \vec{q}$ , projecting the Hamiltonian onto these states (Löwdin partitioning, which includes the other bands perturbatively) gives a  $d \times d$  matrix  $\mathbf{H}(\vec{q})$ , expanded in powers of  $\vec{q}$ :

$$\mathbf{H}(\vec{q}) = \mathbf{H}_0 + \sum_i q_i \mathbf{H}_i + \sum_{ij} q_i q_j \mathbf{H}_{ij} + \dots$$

At first order  $\mathbf{H}_i \propto \langle n | \hat{p}_i | n' \rangle$ , hence the name  $\vec{k} \cdot \vec{p}$ .

### 7.6.2 The symmetry constraint

For every  $g$  in the little co-group, with rotational part  $R_g$ ,

$$\mathbf{D}(g) \mathbf{H}(R_g^{-1} \vec{q}) \mathbf{D}(g)^{-1} = \mathbf{H}(\vec{q}). \quad (7.4)$$

Expand  $\mathbf{H}(\vec{q}) = \sum_a c_a \mathbf{X}_a f_a(\vec{q})$ , where the hermitian matrices  $\mathbf{X}_a$  span the  $d^2$ -dimensional space of  $d \times d$  hermitian matrices and  $f_a$  are polynomials in  $\vec{q}$ . Both sets carry representations of the little co-group: the matrices transform as  $\mathbf{D} \otimes \mathbf{D}^*$  (by conjugation), the polynomials of degree  $n$  as the symmetric power  $[\Gamma_V^n]$  restricted to the relevant components of  $\vec{q}$ . Then (7.4) says that  $\mathbf{H}$  is an invariant, bilinear in matrices and polynomials. Hence

### Method of invariants

Decompose the matrices and the polynomials of each degree into irreps (with matched bases). The most general  $\mathbf{H}$  is a sum of one invariant, with a free real coefficient, for each pair (matrix irrep, polynomial irrep) related as in [Section 4.2](#). The number of free parameters at order  $n$  equals the multiplicity of  $A_1$  in  $(\mathbf{D} \otimes \mathbf{D}^*) \otimes [\Gamma_V^n]$ .

Additional antiunitary symmetries (time reversal, or time reversal combined with a spatial operation that maps  $\vec{k}_0$  to itself) add further constraints of the same type.

#### 7.6.3 Example: the Dirac cone of graphene

At  $K$  the  $\pi$  states carry the two-dimensional irrep  $E$  of  $C_{3v}$  ([Section 7.5.2](#)). Choose the band basis so that the matrices of  $E$  are those of in-plane vectors,  $\mathbf{D}(C_3^+) =$  rotation by  $120^\circ$  and  $\mathbf{D}(\sigma_d) = \text{diag}(1, -1)$  for the mirror that reflects  $q_y \rightarrow -q_y$ . The four hermitian  $2 \times 2$  matrices decompose as

$$\mathbf{D} \otimes \mathbf{D}^* = E \otimes E = A_1 \oplus A_2 \oplus E : \quad \mathbf{1} (A_1), \quad \sigma_y (A_2), \quad (\sigma_z, \sigma_x) (E).$$

Conjugation by a rotation by  $\varphi$  rotates the pair  $(\sigma_z, \sigma_x)$  by  $2\varphi$ . For  $\varphi = \pm 120^\circ$  this is the same as rotating by  $\mp 120^\circ$ , so  $(\sigma_z, -\sigma_x)$  transforms exactly like  $(q_x, q_y)$ . The polynomials decompose as: degree 0:  $\mathbf{1} (A_1)$ ; degree 1:  $(q_x, q_y) (E)$ ; degree 2:  $q_x^2 + q_y^2 (A_1)$  and  $(q_x^2 - q_y^2, 2q_x q_y) (E, \text{rotating by } 2\varphi \text{ like } (\sigma_z, \sigma_x))$ . Pairing equal irreps gives, up to second order,

$$\mathbf{H}(\vec{q}) = \varepsilon_0 \mathbf{1} + v(q_x \sigma_z - q_y \sigma_x) + \alpha q^2 \mathbf{1} + \beta[(q_x^2 - q_y^2)\sigma_z + 2q_x q_y \sigma_x] + O(q^3). \quad (7.5)$$

Consequences, all from symmetry:

- No constant  $\sigma_y$  term: the mass term transforms as  $A_2$  and there is no  $A_2$  polynomial of degree 0. The degeneracy at  $K$  cannot be lifted without breaking the symmetry.
- One linear invariant: the dispersion is linear and isotropic,  $E - \varepsilon_0 = \pm v|\vec{q}| + O(q^2)$ , a Dirac cone. The matrix  $q_x \sigma_z - q_y \sigma_x$  is unitarily equivalent to the familiar  $q_x \sigma_x + q_y \sigma_y$ .
- At second order, the  $\beta$  term produces the three-fold *trigonal warping* of the cone.
- If the sublattices become inequivalent (as in hexagonal boron nitride), the relevant irrep at  $K$  splits and a constant mass term, and a gap, become allowed.

The linear term exists because  $E \otimes E$  contains the irrep of  $\vec{q}$ . When that fails (for other groups or irreps), the crossing is quadratic or of higher order. This is exactly

the question asked in [Lecture 6](#). Complete catalogues of such  $\vec{k} \cdot \vec{p}$  models for all space and magnetic groups have been built with this method.

Further reading. G.L. Bir and G.E. Pikus, *Symmetry and Strain-Induced Effects in Semiconductors* (Wiley, 1974), the classic source of the method; R. Winkler, *Spin–Orbit Coupling Effects in Two-Dimensional Electron and Hole Systems* (Springer, 2003), with tables of invariants; Dresselhaus, Dresselhaus and Jorio, the chapter on  $\vec{k} \cdot \vec{p}$  perturbation theory.

## 7.7 Compatibility relations

Along a line  $\vec{k}$  varies continuously, and the little co-group is a subgroup of those of the end points. A state at  $\Gamma$  carrying an irrep  $\Gamma^{(\mu)}$  evolves continuously into the line, where it must carry the subduced representation  $\Gamma^{(\mu)} \downarrow \bar{G}_{\text{line}}$ . These subductions are the *compatibility relations*. On every line  $\bar{G} = C_s$  with irreps  $A'$  (even under the mirror) and  $A''$  (odd). Reading the character of the relevant mirror:

| line (mirror)           | from $\Gamma$                             | from $K$                      | from $M$                                  |
|-------------------------|-------------------------------------------|-------------------------------|-------------------------------------------|
| $T$ ( $\sigma_d$ )      | $A_1 \rightarrow A', B_1 \rightarrow A''$ | $E \rightarrow A' \oplus A''$ |                                           |
| $\Sigma$ ( $\sigma_v$ ) | $A_1 \rightarrow A', B_1 \rightarrow A'$  |                               | $A_1 \rightarrow A', B_1 \rightarrow A'$  |
| $T'$ ( $\sigma_d$ )     |                                           | $E \rightarrow A' \oplus A''$ | $A_1 \rightarrow A', B_1 \rightarrow A''$ |

Each line can be reached from both ends, and the two subductions must agree. On  $T$ : from  $\Gamma$ ,  $A' \oplus A''$ ; from  $K$ ,  $A' \oplus A''$ . The same holds on  $\Sigma$  and  $T'$ . Such consistency checks catch almost every mistake. In practice one never computes characters along lines; subduction from the end points gives them for free.

## 7.8 What symmetry decides and what it does not

Symmetry fixes the *set* of irreps at each point:  $\{A_1, B_1\}$  at  $\Gamma$ ,  $\{E\}$  at  $K$ ,  $\{A_1, B_1\}$  at  $M$ . It does not fix their energetic order. In [Fig. 7.2](#) the  $A_1$  state is below at  $\Gamma$  but above at  $M$ . The model produced this, not symmetry, and different hoppings could reorder them. The compatibility relations then decide how the labels are connected. The order of the irreps at high-symmetry points, combined with the compatibility relations, is exactly the information that topological quantum chemistry uses.

Irrep labels are also what selection rules need: optical transitions between bands at  $\vec{k}$  are allowed or forbidden according to the small irreps involved ([Section 4.5](#)).

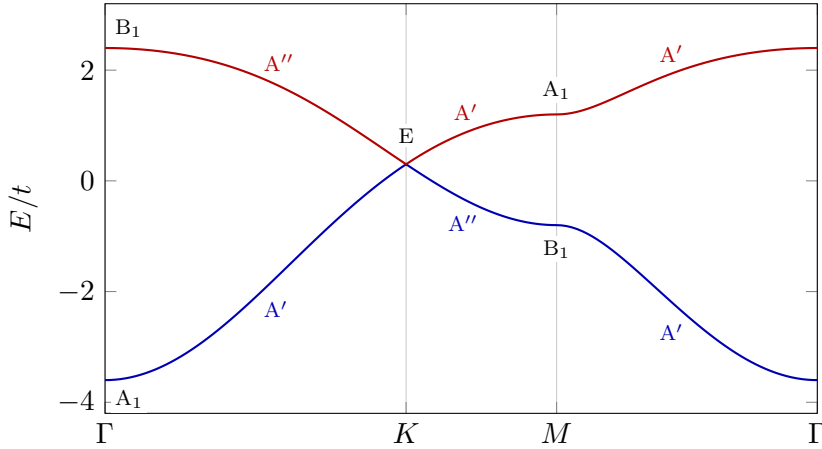


Figure 7.2: The  $\pi$  bands of graphene in a tight-binding model with nearest- ( $t$ ) and next-nearest-neighbour ( $t' = 0.1t$ ) hopping, labelled by small irreps of  $C_{6v}$  and its subgroups. The labels at  $\Gamma$  and  $M$  were obtained from the eigenvectors; symmetry alone fixes only which pair of labels appears at each point.

## Exercises and solutions

Exercise 7.1. Verify the little co-groups (7.2). In particular show that  $\sigma_d(60^\circ)$  maps  $K$  to an equivalent point and that  $\sigma_v(150^\circ)$  fixes  $M$ .

*Solution.*  $\sigma_d(60^\circ)$  reflects the direction  $0^\circ$  of  $K$  to  $120^\circ$ , i.e. to  $C_3^+ K$ , which is equivalent to  $K$  (Exercise 6.2).  $M$  lies at  $-30^\circ$ , on the mirror line at  $150^\circ \equiv -30^\circ$ , so  $\sigma_v(150^\circ)$  fixes it exactly. By contrast  $\sigma_v(30^\circ)$  sends  $M$  to  $90^\circ$ , i.e. to  $\vec{b}_2/2$ , and  $\vec{b}_2/2 - \vec{b}_1/2$  is not a reciprocal vector. The lines inherit the mirror that contains them:  $\sigma_d(0^\circ)$  for  $T$ ,  $\sigma_v(150^\circ)$  for  $\Sigma$ , and  $\sigma_d(60^\circ)$  for  $T'$  (it maps the segment  $KM$  onto an equivalent one).

Exercise 7.2 (*the  $\sigma$  bands*). The  $\sigma$  bands come from  $s$  ( $a_1$ ) and  $(p_x, p_y)$  ( $e$ ) orbitals at the  $C_{3v}$  sites. Use (7.3) to show that the six  $\sigma$  states decompose as

$$\Gamma : A_1 \oplus B_1 \oplus E_1 \oplus E_2, \quad K : A_1 \oplus A_2 \oplus 2E, \quad M : 2A_1 \oplus A_2 \oplus 2B_1 \oplus B_2.$$

How many symmetry-enforced degeneracies are there at  $\Gamma$  and at  $K$ ?

*Solution.* Each atom carries  $a_1 \oplus e$  with site characters  $\chi_\rho(E) = 3$ ,  $\chi_\rho(C_3) = 1 - 1 = 0$ ,  $\chi_\rho(\sigma_v) = 1 + 0 = 1$ .  $\Gamma$ : sublattices kept by  $E, C_3, \sigma_v$  with trivial phases,  $\chi_\Gamma = (6, 0, 0, 0, 2, 0)$  on  $(E, C_2, 2C_3, 2C_6, 3\sigma_v, 3\sigma_d)$ . The reduction gives  $m_{A_1} = m_{B_1} = \frac{1}{12}(6 + 6) = 1$ ,  $m_{E_1} = m_{E_2} = \frac{1}{12}12 = 1$ , others 0.  $K$ :  $\chi_K(C_3) = (\omega + \omega^*) \cdot 0 = 0$ ,  $\sigma_d$  exchanges sublattices, so  $\chi_K = (6, 0, 0)$  and  $K = A_1 \oplus A_2 \oplus 2E$ .  $M$ :  $\chi_M = (6, 0, 2, 0)$  on  $(E, C_2, \sigma_v, \sigma_d)$ , giving  $2A_1 \oplus A_2 \oplus 2B_1 \oplus B_2$ . Enforced degeneracies: two doubly degenerate levels at  $\Gamma$  ( $E_1, E_2$ ) and two at  $K$  ( $2E$ ); none at  $M$ .

Exercise 7.3 (*nearest-neighbour model*). With only nearest-neighbour hopping  $t$ ,  $\mathbf{H}(\vec{k}) = -t \begin{pmatrix} 0 & f \\ f^* & 0 \end{pmatrix}$  with  $f(\vec{k}) = 1 + e^{-i\vec{k} \cdot \vec{a}_1} + e^{-i\vec{k} \cdot \vec{a}_2}$  (Bloch sums (6.4)).

Compute the energies at  $\Gamma$ ,  $K$  and  $M$  and identify the irrep of each eigenvector by applying  $C_2$ .

*Solution.*  $\Gamma$ :  $f = 3$ ,  $E = \mp 3t$ . The lower state  $(1, 1)/\sqrt{2}$  is invariant under all operations:  $A_1$ ; the upper one  $(1, -1)/\sqrt{2}$  is odd under the operations exchanging sublattices:  $B_1$ .  $K$ :  $K \cdot \vec{a}_1 = \frac{4\pi}{3}$ ,  $K \cdot \vec{a}_2 = \frac{2\pi}{3}$ , so  $f = 1 + \omega + \omega^* = 0$ :  $E = 0$ , doubly degenerate ( $E$ ).  $M$ :  $M \cdot \vec{a}_1 = \pi$ ,  $M \cdot \vec{a}_2 = 0$ ,  $f = 1 - 1 + 1 = 1$ ,  $E = \mp t$ .  $C_2$  maps  $\vec{q}_A \rightarrow -\vec{q}_A = \vec{q}_B - (\vec{a}_1 + \vec{a}_2)$ , so  $\hat{O}_{C_2}\Phi_A = e^{iM \cdot (\vec{a}_1 + \vec{a}_2)}\Phi_B = -\Phi_B$  and likewise  $\hat{O}_{C_2}\Phi_B = -\Phi_A$ . The lower state  $(1, 1)$  has  $C_2$  eigenvalue  $-1$  ( $B_1$ ), the upper  $(1, -1)$  has  $+1$  ( $A_1$ ), as in Fig. 7.2.

Exercise 7.4. Write the compatibility relations for the  $\sigma$  bands of Exercise 7.2 along the line  $T$ , and check that the subductions from  $\Gamma$  and from  $K$  agree.

*Solution.* On  $T$  the mirror is  $\sigma_d$ . From  $\Gamma$ :  $A_1 \rightarrow A'$ ,  $B_1 \rightarrow A''$ ,  $E_1 \rightarrow A' \oplus A''$ ,  $E_2 \rightarrow A' \oplus A''$ : in total  $3A' \oplus 3A''$ . From  $K$ :  $A_1 \rightarrow A'$ ,  $A_2 \rightarrow A''$ ,  $2E \rightarrow 2(A' \oplus A'')$ : again  $3A' \oplus 3A''$ . The two subductions agree.

## Chapter 8

# Group theory and topology

Video: Lecture 8 (no captions; times read off the slides)

Band topology L8 @ ≈10:00 · Wannier representability L8 @ ≈18:20 · Wannier representability and TQC L8 @ ≈25:00 · classification of band representations L8 @ ≈35:00 · EBRs from Wyckoff positions L8 @ ≈45:00 · irreps from EBRs L8 @ ≈55:00 · double-valued EBRs of  $p6mm$  L8 @ ≈65:00 · spinful  $p_z$  orbitals in graphene L8 @ ≈73:20 · EBR analysis L8 @ ≈80:00 · what TQC can and cannot diagnose L8 @ ≈85:50

Topological quantum chemistry (TQC) connects the symmetry labels of [Chapter 7](#) with the global, topological properties of bands. Its central idea fits in one sentence: *a set of bands is topologically trivial exactly when it can be described by localised, symmetric orbitals in real space; the irreps at high-symmetry points often reveal when this is impossible.*

### 8.1 Local versus global

A cylinder and a Möbius strip are made by gluing two opposite edges of a rectangle, with or without a twist. Every small piece of either looks like a piece of rectangle: locally they are identical. Globally they differ, and no continuous deformation turns one into the other. Topology studies such global properties.

Bands carry the same kind of structure. The Bloch Hamiltonian is periodic in reciprocal space,  $\mathbf{H}(\vec{k} + \vec{G}) = \mathbf{H}(\vec{k})$ , so the Brillouin zone is a torus. At each  $\vec{k}$  an isolated set of  $n$  bands defines an  $n$ -dimensional subspace of states. As  $\vec{k}$  runs over the torus these subspaces form a *vector bundle*. Locally one can always choose smooth basis vectors (a gauge). Whether one can choose them smoothly and periodically over the *whole* zone is a global question, and the answer can be no. The Chern number of a two-dimensional band is the simplest obstruction.

## 8.2 Wannier functions

From Bloch states  $\psi_{n\vec{k}}$  of an isolated set of bands one builds *Wannier functions*

$$w_{n\vec{t}}(\vec{r}) = \frac{V_{\text{cell}}}{(2\pi)^d} \int_{\text{BZ}} d^d k e^{-i\vec{k}\cdot\vec{t}} \sum_m U_{mn}(\vec{k}) \psi_{m\vec{k}}(\vec{r}), \quad (8.1)$$

labelled by a band index and a lattice vector, and related by translations,  $w_{n\vec{t}}(\vec{r}) = w_{n\vec{0}}(\vec{r} - \vec{t})$ . The unitary matrix  $U(\vec{k})$  is the gauge freedom. Wannier functions are the real-space, orbital-like description of the bands, and their localisation depends on the gauge: a smooth, periodic gauge gives exponentially localised functions.

### Wannier representability

The Wannier functions of an isolated set of bands can be chosen *exponentially localised and transforming among themselves under all the symmetries of the crystal* if and only if the bands are topologically trivial. Such a set of bands is called *Wannier representable*.

In TQC this equivalence *defines* “trivial”. For two-dimensional insulators without extra symmetry it contains the theorem that exponentially localised Wannier functions exist precisely when all Chern numbers vanish. The global property in reciprocal space (topology) is thus tied to a local property in real space (localisation of symmetric orbitals).

## 8.3 Band representations

### 8.3.1 Definition

Take a Wyckoff position with sites  $\vec{q}_1, \dots, \vec{q}_n$  in the cell, site-symmetry group  $G_{\vec{q}_1}$ , and a set of orbitals at  $\vec{q}_1$  transforming as a representation  $\rho$  of  $G_{\vec{q}_1}$ . Copy them to the other sites with the space-group operations (coset representatives: the cosets of  $G_{\vec{q}_1}$  in  $G$  label the sites of the orbit, [Section 1.4](#)) and to all cells with translations. The infinite set of orbitals so obtained is mapped onto itself by the space group: it carries a representation of  $G$ , the *band representation*

$$\rho \uparrow G \quad (\text{the representation induced from } \rho). \quad (8.2)$$

In the Bilbao notation,  $\rho@w$  ( $w$  being the Wyckoff label); for instance  $A_1@2b$  is our graphene  $p_z$  example. Fourier transforming to Bloch sums, the band representation becomes, at each  $\vec{k}$ , a finite representation of the little group, of dimension  $n \dim \rho$ , whose character is exactly (7.3). Its small-irrep content at the high-symmetry points is the “irrep vector” of the band representation.

A set of bands is Wannier representable if and only if it carries (is equivalent to) a band representation. The Wannier functions are then the orbitals from which the band representation was induced.

### 8.3.2 Elementary band representations

Band representations can be added. A band representation that cannot be written as a sum of other band representations is *elementary* (EBR). EBRs are the building blocks of all atomic-limit band structures.

- Every EBR is induced from an irrep of the site-symmetry group of a *maximal* Wyckoff position (one whose site-symmetry group is not contained in that of another position). Orbitals at non-maximal positions can be moved continuously to a maximal one without breaking symmetry, so they give composite band representations. (A few irreps at maximal positions also give composite band representations; these exceptions are listed in the tables.)
- For each space group the number of EBRs is finite. They are tabulated, with their irrep vectors, in the **BANDREP** application of the Bilbao Crystallographic Server, for all 230 space groups (with and without spin-orbit coupling) and for the magnetic groups.

For  $p6mm$  the maximal positions are  $1a$  ( $C_{6v}$ ),  $2b$  ( $C_{3v}$ ) and  $3c$  ( $C_{2v}$ ). Evaluating (7.3) for every irrep of each site group gives the spinless EBRs:

| $\rho@w$ | bands | $\Gamma$ ( $C_{6v}$ ) | $K$ ( $C_{3v}$ )          | $M$ ( $C_{2v}$ )                       |
|----------|-------|-----------------------|---------------------------|----------------------------------------|
| $A_1@1a$ | 1     | $A_1$                 | $A_1$                     | $A_1$                                  |
| $A_2@1a$ | 1     | $A_2$                 | $A_2$                     | $A_2$                                  |
| $B_1@1a$ | 1     | $B_1$                 | $A_2$                     | $B_1$                                  |
| $B_2@1a$ | 1     | $B_2$                 | $A_1$                     | $B_2$                                  |
| $E_1@1a$ | 2     | $E_1$                 | $E$                       | $B_1 \oplus B_2$                       |
| $E_2@1a$ | 2     | $E_2$                 | $E$                       | $A_1 \oplus A_2$                       |
| $A_1@2b$ | 2     | $A_1 \oplus B_1$      | $E$                       | $A_1 \oplus B_1$                       |
| $A_2@2b$ | 2     | $A_2 \oplus B_2$      | $E$                       | $A_2 \oplus B_2$                       |
| $E@2b$   | 4     | $E_1 \oplus E_2$      | $A_1 \oplus A_2 \oplus E$ | $A_1 \oplus A_2 \oplus B_1 \oplus B_2$ |
| $A_1@3c$ | 3     | $A_1 \oplus E_2$      | $A_1 \oplus E$            | $A_1 \oplus B_1 \oplus B_2$            |
| $A_2@3c$ | 3     | $A_2 \oplus E_2$      | $A_2 \oplus E$            | $A_2 \oplus B_1 \oplus B_2$            |
| $B_1@3c$ | 3     | $B_1 \oplus E_1$      | $A_2 \oplus E$            | $A_1 \oplus A_2 \oplus B_1$            |
| $B_2@3c$ | 3     | $B_2 \oplus E_1$      | $A_1 \oplus E$            | $A_1 \oplus A_2 \oplus B_2$            |

(8.3)

Conventions: the  $C_{6v}$  labels are those of (7.1). At  $1a$  the site irrep  $B_1$  is even under the bond-containing mirrors  $\sigma_v$ . At  $3c$  the site irrep  $B_1$  is even under the

mirror containing the bond, i.e. it is a  $p$  orbital pointing along the bond. At  $M$ ,  $B_1$  is odd under  $C_2$  and even under  $\sigma_v$ . The row  $A_1@2b$  is exactly the  $\pi$ -band result of Chapter 7. In the Bilbao labels the same information appears as vectors of multiplicities, one entry per small irrep; always check the conventions of the table you use.

## 8.4 The TQC criterion

Every band representation has an irrep vector that is a sum of EBR vectors with non-negative integer coefficients. Consequently:

### Topological quantum chemistry

Take an isolated set of bands (separated by gaps from the rest) and list its small irreps at the maximal  $\vec{k}$  points.

1. If this irrep vector *cannot* be written as a sum of EBR vectors with non-negative integer coefficients, the set is *necessarily* topological: no false positives.
2. If it can be written only with some negative coefficients, the set is *fragile* topological: it becomes trivial when certain trivial bands are added.
3. If it can be written with non-negative coefficients, the set *may* still be topological: false negatives exist, because not all topology is *symmetry indicated*. Two sets of bands can have identical irreps everywhere and different topology (a Chern insulator and a trivial insulator, for example).

A second, purely structural statement is often decisive.

**Theorem 8.1 (Disconnected EBRs).** *If the bands of an EBR split into two or more groups separated by gaps, at least one of the groups is topological.*

Indeed, if every group were Wannier representable, each would carry a band representation, and the EBR would be their sum, contradicting its elementarity.

The compatibility relations of Section 7.7 are what make band structures connected or disconnected. Along each line the subduced irreps must match. If there is no way of connecting the irreps at the high-symmetry points into two separated groups that satisfy all compatibility relations, the EBR is forced to be connected. A disconnection, when allowed, is a signal of topology.

## 8.5 Graphene without spin-orbit coupling

For the spinless  $\pi$  bands,  $A_1@2b$  has the two-dimensional irrep  $E$  at  $K$ . The two bands are forced to touch there, so the EBR cannot split: graphene is a semimetal with symmetry-enforced Dirac points. No gap, no topological insulator.

## 8.6 Graphene with spin-orbit coupling

With spin, each  $p_z$  orbital becomes a Kramers doublet. At the  $C_{3v}$  site,  $a_1 \otimes D_{1/2} = \bar{E}_1$  (Chapter 5), so the four spinful  $\pi$  bands carry the double-valued EBR  $\bar{E}_1@2b$ . Using the double-group characters of  $C_{6v}$ ,  $C_{3v}$  and  $C_{2v}$  (Exercise 8.4) one finds

$$\bar{E}_1@2b: \quad \Gamma: \bar{E}_1 \oplus \bar{E}_2, \quad K: \bar{E}_1 \oplus {}^1\bar{E} \oplus {}^2\bar{E}, \quad M: 2\bar{E}. \quad (8.4)$$

In the Bilbao labels shown in the lecture these are  $\Gamma_7 \oplus \Gamma_8$ ,  $K_4 \oplus K_5 \oplus K_6$  and  $2M_5$ , with  $K_4$  the two-dimensional irrep at  $K$ .

With spin-orbit coupling the degeneracy at  $K$  need not be fourfold: the four bands can separate into two isolated pairs. In the lecture's example (a Kane-Mele-type model),

- one pair carries  $\{\Gamma_7 | K_4 | M_5\}$ . This is exactly the irrep vector of the EBR  $\bar{E}_1@1a$  (spinful orbitals at the hexagon centres). By irreps alone it *could* be trivial: an “obstructed atomic limit” with Wannier centres in the empty hexagons rather than on the atoms;
- the other pair carries  $\{\Gamma_8 | K_5 \oplus K_6 | M_5\}$ . The two-band EBRs of this group are  $\bar{E}_1@1a = \{\Gamma_7 | K_4 | M_5\}$ ,  $\bar{E}_2@1a = \{\Gamma_8 | K_4 | M_5\}$  and  $\bar{E}_3@1a = \{\Gamma_9 | K_5 \oplus K_6 | M_5\}$ . None of them matches, so this pair is necessarily topological.

This is consistent with the theorem on disconnected EBRs. In the Kane-Mele model the topological pair is the  $\mathbb{Z}_2$  quantum spin Hall insulator. There, in fact, *both* pairs are  $\mathbb{Z}_2$  non-trivial. The pair whose irreps mimic an atomic limit is a false negative of the irrep criterion, a concrete instance of point 3 above.

### Beyond the lectures: where to go next

- The BANDREP, Check Topological Mat. and Topological Materials Database tools of the Bilbao server apply this analysis to first-principles band structures; programs such as IrRep extract the irreps from DFT output.
- Symmetry indicators (Po, Vishwanath, Watanabe) are the group of irrep vectors modulo band representations; they classify what irreps can diagnose.
- Magnetic topological quantum chemistry extends everything to the 1651 magnetic space groups, where time reversal combines with spatial operations (Section 6.8.2).
- Review: J. Cano and B. Bradlyn, Annu. Rev. Condens. Matter Phys. 12, 225 (2021).

## Exercises and solutions

Exercise 8.1. Derive the rows  $A_1@1a$  and  $E_1@1a$  of (8.3). For orbitals at the origin all phases in (7.3) are trivial. Why does  $E_1@1a$  split at  $M$ ?

*Solution.* An orbital at the origin is mapped onto itself by every point operation, with  $\vec{t}_\alpha = 0$ , so  $\chi_{\vec{k}}(g) = \chi_\rho(g)$ : the band representation at  $\vec{k}$  is simply  $\rho$  subduced to  $\bar{G}_{\vec{k}}$ . For  $A_1$  this is the identity irrep everywhere. For  $E_1$ : at  $\Gamma$ ,  $E_1$ ; at  $K$  the characters  $(2, -1, 0)$  on  $(E, C_3, \sigma_d)$  give  $E$ ; at  $M$  the characters  $(2, -2, 0, 0)$  on  $(E, C_2, \sigma_v, \sigma_d)$  give  $B_1 \oplus B_2$ . It splits because  $C_{2v}$  has only one-dimensional irreps.

Exercise 8.2. Show that  $A_1 @ 2b$  is not the sum of EBRs induced from  $1a$ . Then explain why the two spinless  $\pi$  bands can never be separated into two isolated bands without breaking a symmetry of  $p6mm$ .

*Solution.* One-band EBRs from  $1a$  have one-dimensional irreps at  $K$ , so any sum of two of them has two one-dimensional irreps at  $K$ , not  $E$ . The two-band EBRs from  $1a$  have  $E_1$  or  $E_2$  at  $\Gamma$ , not  $A_1 \oplus B_1$ . Hence  $A_1 @ 2b$  is not a sum of  $1a$  EBRs (the Wannier functions of the  $\pi$  bands sit on the atoms). The two  $\pi$  bands can never be separated: at  $K$  they form the two-dimensional irrep  $E$ , so they touch there for every symmetric Hamiltonian.

Exercise 8.3. A hypothetical isolated band of a  $p6mm$  crystal has irreps  $B_1$  at  $\Gamma$ ,  $A_2$  at  $K$  and  $B_1$  at  $M$ . Is it Wannier representable according to the irrep criterion? Where would its Wannier centres be?

*Solution.*  $\{B_1 | A_2 | B_1\}$  is exactly the irrep vector of  $B_1 @ 1a$  (8.3). By the irrep criterion the band is compatible with a Wannier function centred at the hexagon centres ( $1a$ ), with the symmetry of an  $f$ -like orbital (odd under the bond-bisecting mirrors). If the atoms sit at  $2b$ , such a band is an *obstructed atomic limit*: trivial, but with Wannier centres away from the atoms. Irreps cannot exclude hidden topology (false negatives), but nothing in them requires it.

Exercise 8.4 (*spinful  $p_z$  orbitals*). The double group of  $C_{6v}$  has three two-dimensional double-valued irreps with characters (classes  $E, \bar{E}, C_2, C_3, \bar{C}_3, C_6, \bar{C}_6, \sigma_v, \sigma_d$ )

|             | $E$ | $\bar{E}$ | $C_2$ | $C_3$ | $\bar{C}_3$ | $C_6$       | $\bar{C}_6$ | $\sigma_v$ | $\sigma_d$ |
|-------------|-----|-----------|-------|-------|-------------|-------------|-------------|------------|------------|
| $\bar{E}_1$ | 2   | -2        | 0     | 1     | -1          | $\sqrt{3}$  | $-\sqrt{3}$ | 0          | 0          |
| $\bar{E}_2$ | 2   | -2        | 0     | 1     | -1          | $-\sqrt{3}$ | $\sqrt{3}$  | 0          | 0          |
| $\bar{E}_3$ | 2   | -2        | 0     | -2    | 2           | 0           | 0           | 0          | 0          |

(class sizes 1, 1, 2, 2, 2, 2, 2, 6, 6;  $C_2$  and  $\bar{C}_2$ , and each mirror and its barred partner, fall in the same class). Using (7.3) with the spin character  $\chi_{1/2}(\theta) = 2 \cos(\theta/2)$  for rotations and 0 for mirrors, derive (8.4).

*Solution.* Each site carries  $a_1 \otimes D_{1/2}$ ; its site character is the spin character. At  $\Gamma$ :  $E \rightarrow 4, \bar{E} \rightarrow -4; C_2, C_6, \sigma_d$  exchange sublattices  $\rightarrow 0; C_3$  keeps them with trivial phases,  $\rightarrow 2\chi_{1/2}(120^\circ) = 2$ , and  $\bar{C}_3 \rightarrow -2; \sigma_v \rightarrow 2 \cdot 0 = 0$ . The reduction in the double group of  $C_{6v}$  (order 24, class sizes 1, 1, 2, 2, 2, 2, 2, 6, 6) gives  $m_{\bar{E}_1} = m_{\bar{E}_2} = \frac{1}{24}(8 + 8 + 4 + 4) = 1$  and  $m_{\bar{E}_3} = \frac{1}{24}(8 + 8 - 8 - 8) = 0$ . At  $K$ :  $C_3$  gives

$(\omega + \omega^*) \cdot 1 = -1$ ,  $\bar{C}_3$  gives +1, the mirrors 0, so  $\chi_K = (4, -4, -1, 1, 0, 0) = \chi_{D_{3/2}}$ ,  
 i.e.  $\bar{E}_1 \oplus {}^1\bar{E} \oplus {}^2\bar{E}$  by (5.9). At  $M$ :  $(4, -4, 0, 0, 0)$  on the five classes of the double  
 group of  $C_{2v}$  (order 8), whose only double-valued irrep  $\bar{E}$  has  $(2, -2, 0, 0, 0)$ :  $2\bar{E}$ .